



CIM-BASED NETWORK MODEL MANAGEMENT IMPLEMENTATION GUIDE

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Revision History

Table 1: Revision History Table

Version	Release	Date	Comments
1	0	2026-07-02	First version drafted by CIM WG members and other ENTSO-E members, with the review of NMM vendors. ICTC approved for publication on 2 July 2026.

Executive Summary

The Common Information Model (CIM) -Based Network Model Management (NMM) Implementation Guide provides Transmission System Operators (TSOs) with a structured, practical, and standards-aligned approach for establishing a single source of network model data. It enables TSOs to modernize their modelling landscape, enhance data quality, and ensure interoperability across consuming processes such as operational, planning, and coordination processes.

Today, network model data is typically fragmented across multiple tools, formats, and organizational silos. This leads to duplicated modelling efforts, inconsistent data semantics, and increased operational risk. A CIM-based NMM addresses these challenges by consolidating all model-related processes into a unified ecosystem built on the IEC CIM and ENTSO-E application profiles (CGMES & Network Code Profiles).

The purpose of the Guide is to support TSOs in transitioning from their current heterogeneous modelling landscape to a CIM-compliant NMM ecosystem by:

- Defining foundational concepts such as NMM and CIM.
- Presenting a clear and reusable implementation methodology.
- Providing harmonisation, profiling, serialisation, and validation guidance.
- Outlining tool requirements and procurement-ready use-cases.
- Offering governance insights and practical lessons learned.

The document is vendor-agnostic and tool-neutral, ensuring TSOs can apply its methodology regardless of their specific modelling environment.

The CIM-based NMM vision is to become the single source of truth for network modelling data across all TSO business processes. It will:

- Aggregate data from diverse network modelling sources.
- Enforce semantic consistency via CIM.
- Enable controlled versioning, approvals, and project-based changes.
- Support multi-context modelling across different processes.
- Deliver validated, reproducible datasets to all consumers.
- Strengthen interoperability with external stakeholders (RCCs, DSOs, ACER, ENTSO-E).

This transformation improves model transparency, reduces manual reconciliation, enhances traceability, and accelerates the propagation of validated model changes.

The Implementation Approach introduces a six-step iterative process to establish and maintain a CIM-based NMM:

- **Network Models Harmonisation:** Consolidate heterogeneous model sources into a Totally Harmonised Model (THM), using either a CGMES/NCP-based or canonical CIM-based path.
- **Profiling and Schema Generation:** Create conceptual CIM-aligned representations of the THM and, if needed, model extensions. Generate machine-readable schemas (RDFS, SHACL, XSD) to support interoperability and validation.
- **Data Serialisation:** Convert harmonised data into CIM-based instance files (e.g., CIMXML), ideally through automated ETL pipelines.
- **Data Validation:** Apply multi-layered validation—from syntactic checks to topological and load-flow validation—leveraging existing CGMES/NCP and QoCDC rules where applicable.
- **Data Loading:** Deliver structured data products (datasets, schemas, validation results) to NMM tools and downstream consumers.
- **Iterative Improvement:** Repeat the cycle as new models, extensions, or process requirements emerge.

This methodology applies across Pre-NMM, Shadowrun, and Operational phases.

NMM Tool Requirements and Use-Cases translate the NMM concept into concrete capabilities that commercial NMM tools must support, including:

- Reliable network model storage and versioning.
- Object registry and ID consistency management.
- Multi-user, multi-context workspaces.
- Controlled branching, project-based updates, and approval workflows.
- Import/export of CIM profiles and extensions.
- Layered validation (syntax → schema → topology → power flow).
- Integration with diverse internal and external systems.

Six generic, testable use-cases allow TSOs to evaluate tools during procurement, covering roundtrip import/export, incremental updates, historical models, project management, hierarchical dependencies, and boundary adjustments.

Successful NMM implementation requires not only technical readiness but also organisational alignment and process maturity. Best practices emphasise:

- Clear role definitions and ownership of network modelling processes.
- Structured governance aligned with frameworks such as ISO/IEC 38500.
- A progressive go-live strategy, starting with internal offline processes.
- Strong change management to ensure adoption.
- Early wins enabled by harmonisation and CIM-based workflows.

Key practical challenges—and strategies to mitigate them—are documented, including ID consistency, managing calculated values, and coordinating modelling responsibilities across teams.

By adopting the approach described in this guide, TSOs can transition toward a future-proof, standards-driven, interoperable NMM environment. This enables higher model quality, stronger cross-border coordination, reduced operational risks, and more efficient integration of new tools, data consumers, and grid development processes.

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1 Introduction

This Implementation Guide aims to support TSOs in achieving a Common Information Model (CIM)-based Network Model Management, i.e. a single-source of the network model. This document provides a step-by-step approach to implement a reliable and interoperable Network Model Management (NMM), following best practices and on-the-ground experiences available to date.

The first section of the document defines the notion of NMM and the benefits it brings. A general introduction to the Common Information Model is provided, from the canonical CIM to the ENTSO-E extensions – Common Grid Model Exchange Specification and Network Code Profiles, followed by TSO NMM use cases and CIM application for NMM.

The second section details the process step by step to realise a Totally Harmonised Model (THM): Network Models Harmonisation – unifying source models into a THM; Profiling and Schema Creation – defining structure and constraints; Data Serialisation – producing CIM datasets; Data Validation – ensuring data quality; Data Loading – serving data products.

The fourth chapter is dedicated to NMM tools capabilities. General principles an NMM tool should follow are proposed along six generic use cases: CIM import and export roundtrip, CIM import of an increment and model assembly, CIM incremental and historical model export, Power System Projects and exporting the future network model, creating and managing power system hierarchies, revisiting and expanding the model and its boundary parts.

Chapter five brings guidance to govern pre- and post-NMM phases, making a clear distinction between building and maturing a CIM-based NMM (pre-NMM) and, on the other hand, the operational use of the NMM to serve consuming needs (post-NMM).

Eventually, chapter six gathers organisational challenges, best practices and on-the-ground experiences from TSO when building an NMM.

1.1 In Scope

This Implementation Guide explains NMM implementation in the sense of preparatory work performed before an NMM becomes operational as a single source of the network model. The focus is on building and maturing the foundations of a CIM-based NMM rather than on its operational exploitation.

The document covers key concepts and principles underlying CIM-based NMM and provides guidance for TSOs on preparing network model data accordingly. This includes harmonising existing or newly available network models into a THM, aligning proprietary data with CIM – including CGMES and Network Code Profiles (NCP) where applicable – and applying profiling, schema creation, serialisation, and validation as core implementation steps.

In addition, this guide describes a generic and tool-agnostic implementation approach applicable to the pre-NMM phase, focusing on data preparation, standardisation, and readiness. Typical implementation challenges, good practices, and high-level NMM capabilities are outlined to

support informed design and procurement choices, while remaining vendor-neutral and standards-driven.

1.2 Out of Scope

On the other hand, this Implementation Guide does not cover the operational use of an NMM once it is established as a production system.

Activities related to the post-NMM phase are outside the scope of this document, including day-to-day use of the NMM by consuming processes, as well as long-term operation, maintenance, and support.

A distinction should be made considering the authority of the NMM on model data and the real-time decisions happening within the control room. In the fully operational phase, the NMM is the designated authority for all model data, while the real-time decision making remains out of scope for the NMM. We acknowledge that the CIM-based NMM is a foundational element of the future control room environments, but we also keep the focus on the principal NMM capabilities within this document.

Furthermore, detailed system integration aspects – such as IT/OT (Information-/Operation-Technology) architectures, interfaces, APIs, or service orchestration – as well as detailed governance models, organisational arrangements, and tool-specific configuration are not addressed. Topics related to post-implementation model lifecycle operations, including continuous maintenance or workflow automation, may be covered in future iterations of this Implementation Guide.

2 Background Information

In this chapter, a general introduction to NMM is included to clarify how the network model management fits into the electrical utility business contexts and technical solutions. This chapter explains *what* the NMM is and then provides insight into *how* the NMM becomes the single source of network models. The prospective benefits of NMM adoption and the procedural aspects of NMM implementation are also explored.

2.1 Introduction to Network Model Management (NMM) Concept

The Network Model Management (NMM) refers to a set of consistent practices dealing with the network model - a digital representation of the actual power system network [1]. The NMM relates to creating, maintaining, and exchanging the network model with relevant business processes consistently [2]. The Network Model Manager is a software tool / platform that implements the harmonised management and exchange of network models [3]. In this document, we consider the NMM a set of practices implemented via a software tool or platform - the network model *manager*.

The high-level core NMM functions are:

- Model creation and maintenance: acquiring the relevant data from the relevant sources, data validation, versioning and handling multiple models or model parts simultaneously.
- Data exchange: consistent data exchange with the original data sources and with the consumers of the network model (both internal within the TSO, and external), in line with the requirements of the model consumers.
- Model change management: management and propagation of updates, so appropriate actions can be triggered on model update.

The handling of different model instances or versions simultaneously relates directly to the different business contexts where the network model should be utilized. As an example, long term expansion planning requires a different dataset, compared to a dynamic security analysis. Any analysis, however, requires consistent network model data.

Additionally, the correctness network-related data may impact a wider business context, well beyond the tasks directly related to network modelling. The quality of network models directly and indirectly impacts other systems in the electrical utility. Geographic information systems or SCADA systems and even general business systems can be influenced. For this reason, the benefits of a well implemented NMM can reach a wider business context. However, to reap its benefits, it must be adopted consistently, within the network model handling procedures across business contexts.

NMM is critical for TSOs to ensure accurate and consistent development and maintenance of the network models representations across processes such as operational and planning processes within the TSO and beyond the TSO (e.g. for Regional Coordination Processes). NMM serves therefore as the backbone for network models for internal and external data exchange, enabling the critical processes such as static and dynamic security analysis in a control room to utilise the

most consistent and the most relevant data. NMM promotes interoperable systems and organisations, reduces complexity by eliminating proprietary silos, and enhances data quality and consistency for the TSO processes.

Figure 1 illustrates this vision of NMM becoming the source of network model(s). The NMM is the aggregation point of diverse network data sources, and it holds and maintains the Master Source Model, itself becoming the point of reference (or the source of truth) for all the model consuming processes. The master source model can be seen as a union of different aspects and contexts of network modelling. Please note that the model consuming processes can also be the same as the original network model data sources, especially in the transition period to full-scale NMM implementation.

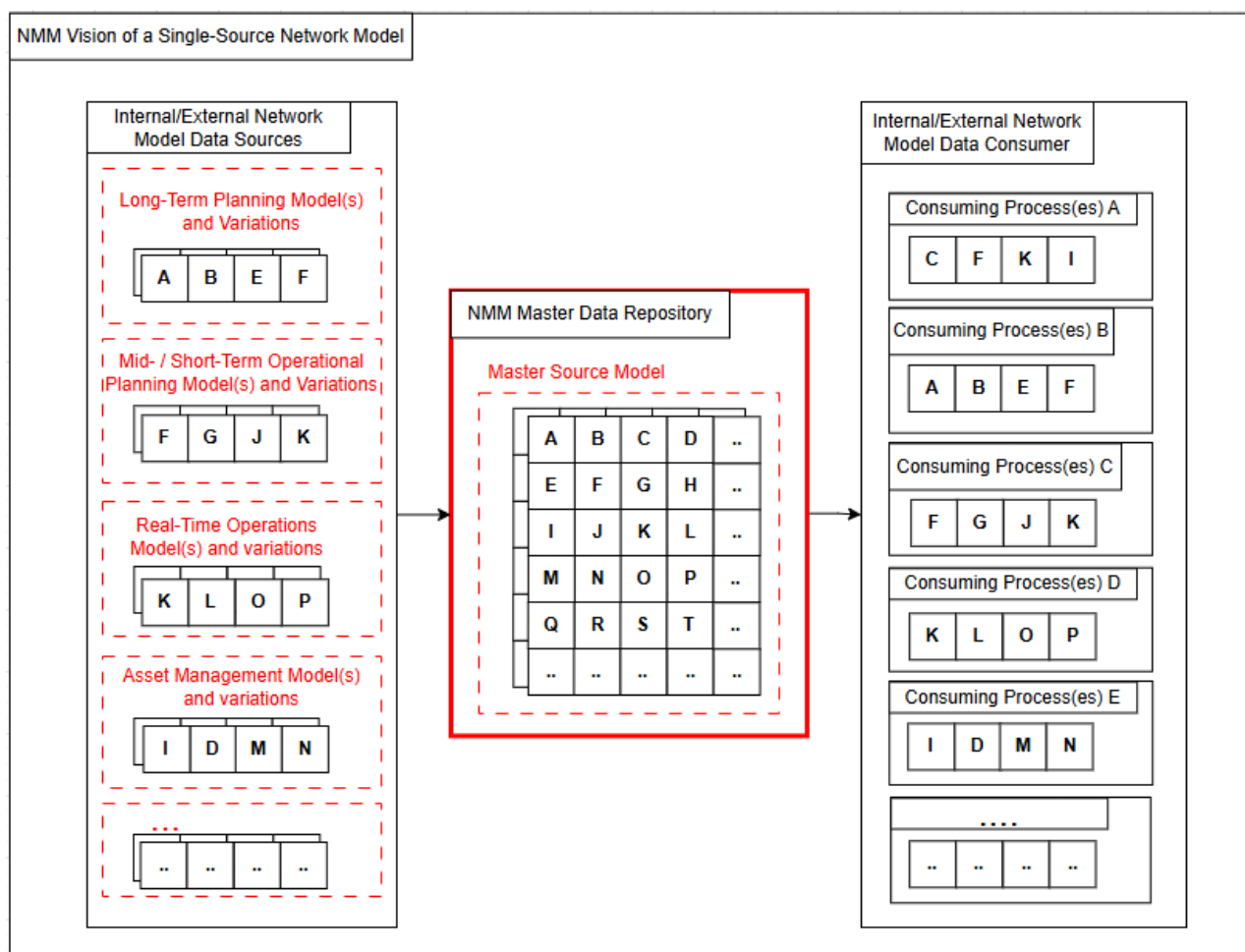


Figure 1: NMM Vision of a Single-Source Network Model. Adapted from [2, 4].

As described earlier, different various model consuming processes typically utilise only a subset of the master source model depending on their context. The NMM vision considers *all* these consumers retrieve their network model data *from the NMM* and do not go directly to the model data originators anymore – the NMM then becomes *the* origin where the network model data should be consumed. In this context it is easy to envision the case where previous model data

originators become integrated with the NMM and, in fact, become *consumers*, relying on the NMM for their input data after the full NMM implementation.

This way, the NMM becomes *the* relevant source of modelling data for the consuming processes – and it is the NMM where the network model is handled. This approach ensures the effect of proprietary silos holding the modelling data is reduced, as the NMM becomes responsible for handling the data. Consequently, the model consistency across different consuming processes is improved as each of the processes can reach correct data by interfacing to the NMM only. This must, however, be followed up by the appropriate procedural updates in the organization.

The interfacing aspects of the NMM may suggest its similarity to an enterprise service bus, but the NMM is not merely an intermediary agnostic to the data payload being exchanged. The NMM is a full-fledged network model source of truth instead. A fully implemented NMM becomes *the* single source of consistent network data (i.e. connectivity, topology, physical parameters, ratings), by assembling and validating the data, composing the master source model from multiple external data sources, and then providing different subsets of that master model to the model data consumers.

These tasks consider more than just providing a communication channel between the model data providers and consumers. The NMM:

- provides the communication channel, and;
- composes the master model;
- handles the model changes and updates in a tractable and accountable manner, and;
- ensures consistency and traceability of the model data.

This directly and indirectly improves the results quality of all processes dealing with network model data. For the readers versed in computer source code control, the NMM becomes similar to the source code management system and a central repository. A well-implemented source code management enables traceable builds, automatic creation of software bill of materials and other benefits beyond just the collaborative development.

Besides the technical aspect, the procedural aspect of the NMM implementation should not be underestimated. A very significant share of business processes is affected by the NMM integration and a well-defined NMM integration roadmap takes into account the current business processes in the organization implementing the NMM, ranks the features and affected roles, and clearly implements the change management in the processes affected so that insight into changes and impacts as well as the implementation benefits is clear to everyone involved [5] [6]. Furthermore, a well-designed NMM implementation process would first assess the situation within the company implementing the NMM and select the most viable path that reaches the modelling benefits quicker, with a gradual introduction of other business contexts.

2.1.1 Benefits of NMM

The benefits of the NMM come from its explicit ownership network of network model handling tasks. More precisely, these benefits come from technical and procedural improvements in the model handling, compared to the situation prior to the NMM implementation.

Reducing the work duplication and improvements in documentation and process ownership

Without NMM, the modelling work is often duplicated and inconsistent as different consumers would get their required model data from the originating contexts and assemble their “master model subset” manually and independently so this error-prone process is essentially duplicated for each of the model data consumers.

This duplicated work is often necessary: different consumers require different types of data. A steady-state analysis would need a bus-branch model, while a dynamic security analysis calls for a node-breaker model. Each process needs a network model in a suitable form. As the NMM is designed to store all the required data for these processes, the NMM can reduce work duplication by preparing the exact required data for all the consuming processes. The model consuming processes are then relieved of the need to source the data and manually assemble their own models.

Refer to Figure 1 for a visual illustration: Consuming process 1 would have to reach the four data originators, assemble the model, and same with Consuming process 2 and so on.

The modelling then becomes an error-prone process as it requires process-specific (manual) composition of data from different sources. Tracking and documenting the original data sources is a laborious and intensive task. With an NMM handling the master model and its different aspects, this task can be automated more easily as the NMM handles the data provenance automatically and consistently, and it is the sole reference point where the model data is taken from.

Updating and ensuring that the most recent and the most relevant data is used in the modelling context without an NMM is often complex, in some cases even with manual update tracking with external tools. The NMM approach makes this process less convoluted and more manageable.

The NMM provides the data update and provenance management directly. The model quality control and sanity checks are easier to implement in the NMM directly. Different model validations relevant for different business context should also be applied in parallel. The reductions in manual interventions result in shorter business response time to the modelling tasks and thus improve the modelling quality. The NMM approach makes the documentation of processes easier, eliminates the duplicate processes for model management, defines the better modelling practices and eases the automation, and reduces the process-related root causes for the serious operating or planning errors coming from the models using outdated and erroneous data.

Respecting and complementing the original data sources

Assembling the network models from several sources becomes the task of an NMM – and the NMM then takes over as the single source of network model data.

NMM makes the ownership of the model handling tasks explicit, and ensures all model-related processes benefit from the consistent model management. Any newly introduced processes would become a new consuming process for the NMM, and it would consume the most relevant model data from the NMM and *only* from the NMM. The eventual modelling updates would also feed back into the NMM exclusively, and NMM would handle the brokerage and approval of the updates, so relevant consumers are always served with correct and up-to-date network model data.

As illustrated above, while the NMM becomes the reference point to refer to for all business processes consuming the network modelling data, and its data handling capabilities complement other tools such as long-term expansion planning or dynamic network simulation tools. Once these tasks call for updates to the network model, it is the NMM responsibility to consolidate the updates. The NMM subsequently provides consolidated data with the updates to all consumers in all business contexts requiring these data. In other words, with the NMM in place, model updates are propagated only through the NMM. The NMM thus respects the most recent network models from different sources when consolidating and assembling the master network model, while having the explicit role of the model data handler.

Consistency and relevance improvements

This way, the models in all business context always contain the best-known information, and the provenance of the modelling information is kept explicit by the NMM. All the consuming processes then use the most reliable and the most relevant modelling data they require.

The existing processes are then complemented by making the data handling tasks explicit and obvious. All model updates and case creation is generated from the NMM as the single source. The model data is well-organized and clearly identifiable and traceable, which makes modelling replicable and modelling case assembly processes can be automated and scriptable. The model content is then consistent across different consumer applications and business context, which may in many cases eliminates the manual processes required for the comparison and settlement when the models in different business contexts diverge one from another.

Faster and consistent propagation of model amendments

Eventual model corrections are available for all the business contexts immediately, significantly reducing the time required for manual comparisons and duplication of entering the model changes in different business context. This improves the modelling accuracy over time in all business contexts, as the NMM makes the validated and official models available as the modelling basis for e.g. the network-model based configuration of operational technology solutions or the network simulation studies. All cases then use the updated data.

The NMM reduces the manual steps in the model update process (Figure 2): upon creating a model change proposal, the model validation post-correction can be triggered automatically. The

requests for approval are immediately channelled to the correct roles as well. The NMM should also allow manual inspection of changes at all steps, so the changes can also be rejected and the model rolled back to the previous state. Once updated model is approved and becomes valid, the model consuming processes update can be triggered automatically. This makes all model handling processes quicker and clearer, and the time to propagate model updates and correction reduces significantly.

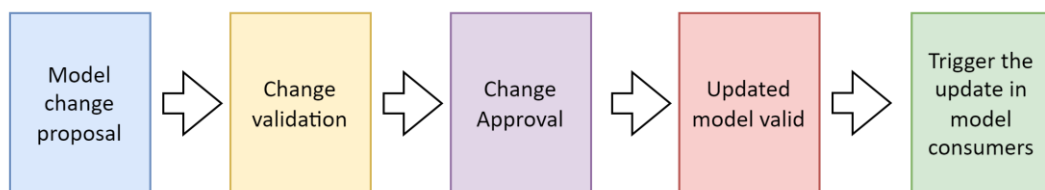


Figure 2: Model update process

As an example: a correction in a parameter for a line due to dynamic rating can automatically trigger a re-creation of the dynamic simulation modelling cases affected by this value update. This approach improves the quality of analyses significantly as corrections are propagated automatically and the reduction in manual interventions directly results in less process-related errors.

Improvements in quantitative model validation

With an NMM, the model handling roles and responsibilities are clearer and data entry points well-defined so a clear set of layered model validation rules can be set up. For a physical network model, data are validated at multiple points: upon entering in the physical network model data editor, internally within the NMM when the collection and composition of data from other sources is completed, as well as externally in other modelling tools once the modelling case is assembled by the NMM. Model validations and tests can be triggered automatically, analogously to continuous integration in software development where a new code committed to the code repository can trigger the automatic tests.

Improvements in model maintenance workflows

Any modelling updates, including significant ones such as commissioning or derating can be communicated across different business processes quickly and efficiently. As the needs for network modelling data had to have been identified in the project lifecycle earlier, the affected processes update can be triggered automatically which reduces the workload required to manually update the models and study result notably.

Streamlined and clearer responsibilities for the (standardised) model data flows

The implementation of NMM requires the processes to have clear requirements for their modelling data subsets. With a CIM-based NMM interfaces, this also lessens the dependency on a proprietary vendor-specific tools. This also provides a much easier means for the dry-run implementation of new modelling tools and enables the utility to implement new processes and tools much easier.

Keeping track records, keeping the history and improvements in documentation

For each business process consuming the data from the NMM, the NMM keeps the version of the model provided, enabling traceability and replicability of the studies. Cases can then be fully reconstructed from the model source data at a later point in time, which provides an audit trail of case assembly process and enables the users to re-create the cases if the model changes. This significantly improves the ability of post-event analysis, reducing the required documentation workload at the same time.

2.2 Introduction to Common Information Model (CIM) Concept

The Common Information Model (CIM), standardised under the International Electrotechnical Commission (IEC) TC57 series (notably IEC 61970, IEC 61968, and IEC 62325), provides a semantically consistent, object-oriented information model for representing power system components, their attributes, and relationships [7]. At a high level, CIM serves as the canonical information model enabling interoperability across heterogeneous systems within Transmission System Operators (TSOs), Distribution System Operators (DSOs), and market participants.

The CIM implements a layered architecture that separates the data from the format in which it is serialized. Figure 3 represents CIM as a layered reference architecture, showing how an abstract information model is progressively specialised and then encoded into exchangeable data.

- **Semantic Layer:** contains the most generic meaning of the domain. This is where the canonical CIM lives, defined in UML in the IEC 61970-300 series and extended for distribution by IEC 61968-11. In this layer, the focus is not yet on file formats or specific exchanges, but on the shared conceptual meaning of power-system concepts, their attributes, and relationships.
- **Context Layer:** contains profiles which are a constrained subset of the canonical CIM chosen for a particular business context, use case, or exchange scenario. The figure's note "Derived from (According to 61970-401)" means that these profiles are systematically derived from the canonical CIM using formal profiling rules. So, rather than using the full CIM everywhere, a specific exchange uses only the classes, attributes, and associations needed for that context. The IEC 61970-401 [8] defines how to create such subsets from the canonical UML model.
- **Syntax Layer:** shows how profiles are rendered into machine-processable artefacts. A schema is generated, for example in RDF Schema (RDFS), XSD, or other syntactic representations. Then actual datasets are produced that conform to that schema, such as CIM RDF/XML exchange files. The note "Described by (According to 61970-501)" indicates that in the context of utilising CIM, the schema/vocabulary for this machine-readable representation is done in RDFS and is defined by IEC 61970-501 [9], while practical model exchange in CIMXML is specified in IEC 61970-552 [10].

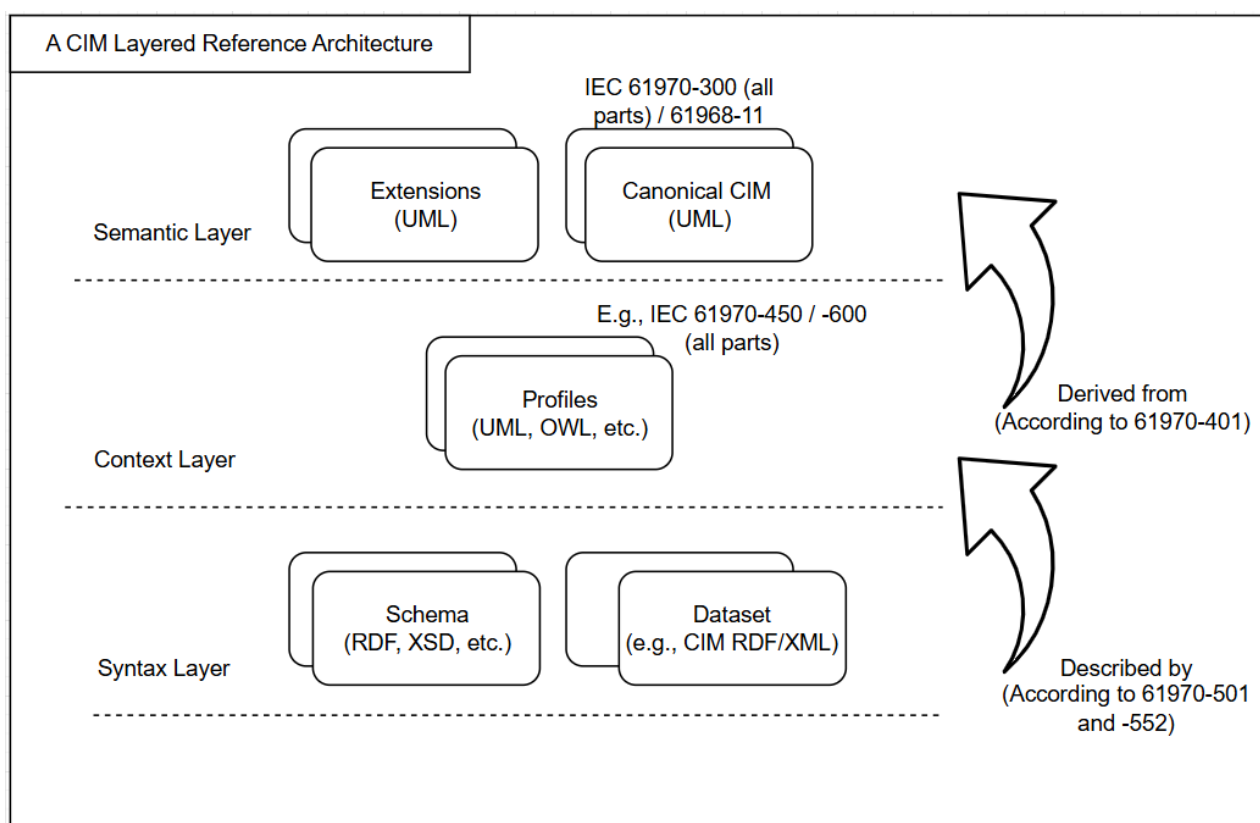


Figure 3: A CIM Layered Reference Architecture. Adapted from [11, 12, 13, 8]

For further information and more insight into the interaction of the above layers, as well as the available tooling landscape, please refer to the ENTSO-E RDF Syntax Guide [14].

With reference to Figure 4, in SGAM (Smart Grid Architecture Model) [15], Canonical CIM sits in the information/enterprise interoperability layers. Its main role is to make data from various systems interoperable across the other systems within and among organizations such as EMS of TSOs and DSOs. In SGAM terms, the concept of CIM supports information exchange across the identified domains and zones, especially where operational, enterprise, and market systems need to share consistent models and business context. Based on the SGAM (Smart Grid Architecture Model) layering concept [15], the positioning of CIM can be summarized as follows in Table 2:

Table 2: Role of CIM in Smart Grids

SGAM Layer Role of CIM

<i>Business</i>	Enables harmonised interpretation of regulatory and operational processes
<i>Function</i>	Supports interoperable applications through shared data semantics

Information | **Primary layer** – defines data models, semantics, and structure

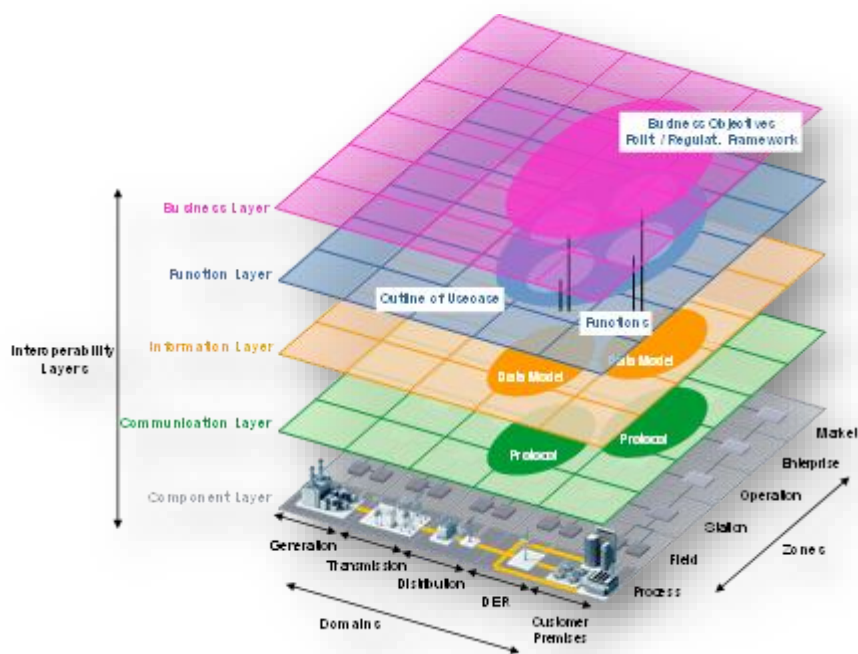


Figure 4: Smart energy Grid Architecture Model (SGAM) is a three-dimensional architectural framework [15].

2.2.1 Canonical CIM

With reference to Figure 5, the Canonical CIM is the IEC's vendor-neutral, technology-independent semantic information model for the electricity sector. In formal IEC terms, CIM is an abstract model that represents the major objects in an electric-utility enterprise and their relationships. In practice, it provides the common vocabulary and structure for describing things such as substations, lines, transformers, terminals, measurements, schedules, markets, assets, and dynamic behaviour, so that different applications can exchange data with the same meaning. The core canonical base is standardised in IEC 61970-301; dynamic behaviour is extended in IEC 61970-302. Related CIM families then extend that same semantic foundation into other business areas, notably IEC 61968 for distribution/asset-oriented utility integration and IEC 62325 for electricity market communications.

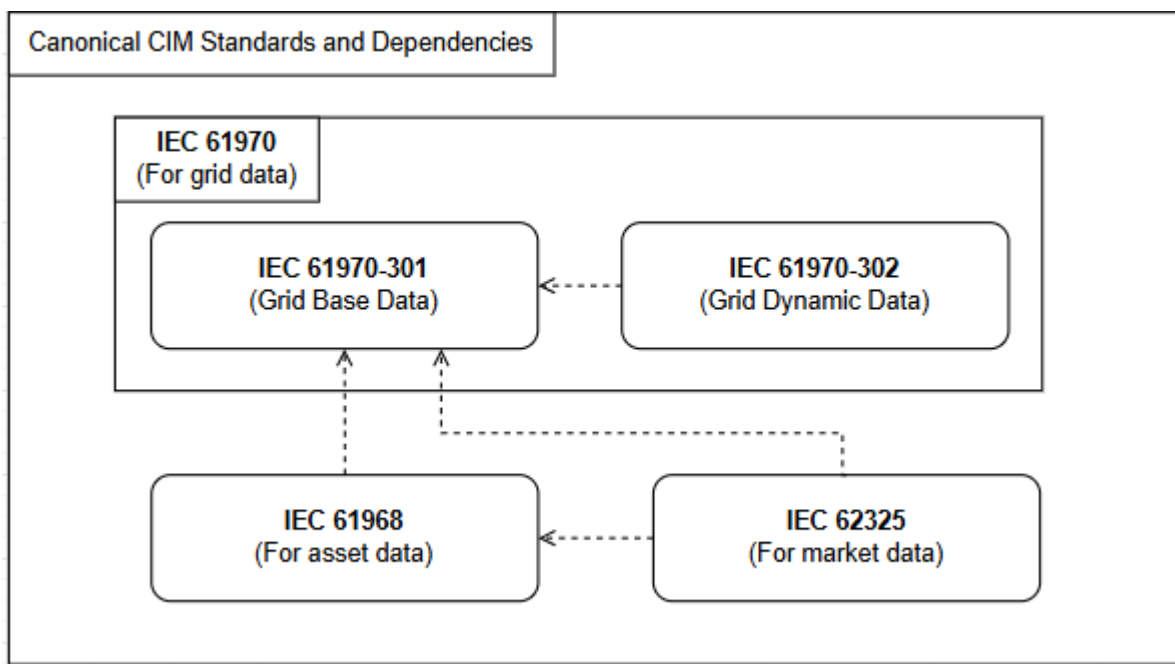


Figure 5: Canonical CIM Standards and Dependencies

2.2.2 Profiling and Schema Generation

This section provides a brief introduction to CIM profiling and schema generation and points the readers to sources where they can gain more in-depth knowledge about these topics.

The Common Information Model (CIM) provides a rich and comprehensive semantic foundation for representing power system and market-related information. However, practical data exchange rarely requires the full canonical CIM. Instead, implementations rely on profiles—well-defined, purpose-specific subsets of CIM that capture exactly the classes, attributes, and associations needed for a particular business use case. Example of these profiles are the ones defined in CGMES [16]. The act of creating such constrained subsets is known as profiling, and it ensures consistency, interoperability, and clarity by using a controlled fragment of the broader CIM while maintaining semantic alignment with the canonical model. Figure 6 shows a hypothetical example of how profiling looks like. As seen, some canonical CIM classes are extended and used as basis for a constrained profile. As seen, profiling include:

- **Extended canonical CIM with classes and attributes:** in this case *ExtCIM Class A* contacting *attribute ExtA* is created.
- **Selecting classes and attributes and leaving others:** in this case selecting *Profile Class A* and *Profile Class B* based on *CIM Class A* and *CIM Class B* with the respective attributes: *attribute A*, *attribute B*, *attribute ExtA*, *attribute E*, and *attribute F*.
- **Restricting cardinalities:** in this case restricting *attribute A* in *Profile Class A* and *attribute F* in *Profile Class B* as well as the association between the two classes, meaning that *Profile Class B* must have exactly one association to *Profile Class A*. Note that *attribute ExtA* is originally restricted in the source class *ExtCIM Class A*.

- **Specifying direction of reference:** in this case mandating the direction of association between *Profile Class A* and *Profile Class B*, where *Profile Class B* must reference *Profile Class A*.

The foundational rules for creating CIM profiles are formally described in IEC 61970-401 [8], which defines how profile specifications are structured, how they relate to the canonical CIM, and the principles for selecting and constraining CIM elements for data exchange. Complementing this, the energy-market-specific profiling methodology that is defined in IEC 62325-450 [17], which specifies the profiling and context-modelling rules for market communication scenarios and describes how business requirements translate into contextual and assembly models derived from CIM.

While profiling determines what information is included, schema generation determines how that information is represented syntactically for message exchange. CIM profiles can be serialised into different schema forms, most commonly RDF Schema (RDFS) or XML Schema (XSD), depending on the interoperability requirements of the domain. Example of these schemas are the ones associated with CGMES [16]. The rules for producing a machine-readable CIM schema in RDF format are standardised in IEC 61970-501 [9], which defines how the CIM is expressed in an RDF vocabulary suitable for data access, validation, and exchange. For XML-based interoperability (i.e., for XSD), the authoritative specification is IEC 62361-100 [18], which provides the formal mapping from CIM profiles to W3C XML Schema, ensuring that syntax generation follows prescribed naming and design rules and that the resulting schema remains semantically consistent with the originating CIM profile.

Together, these standards form the methodological backbone for moving from the full CIM to a well-defined profile and finally to an implementation-ready schema suitable for systems integration. Readers who wish to explore the underlying profiling rules, modelling constraints, and schema-mapping processes in detail should consult the standards mentioned above—IEC 61970-401, IEC 62325-450, IEC 61970-501, and IEC 62361-100—as they describe the full frameworks, constraints, and design principles needed to create interoperable and standards-compliant profiles and schemas. Additionally, readers could consult UCAIug CIM Modeling Guide [19] and CGMES Profiling User Guide [20] to gain more knowledge and experience on the profiling topic.

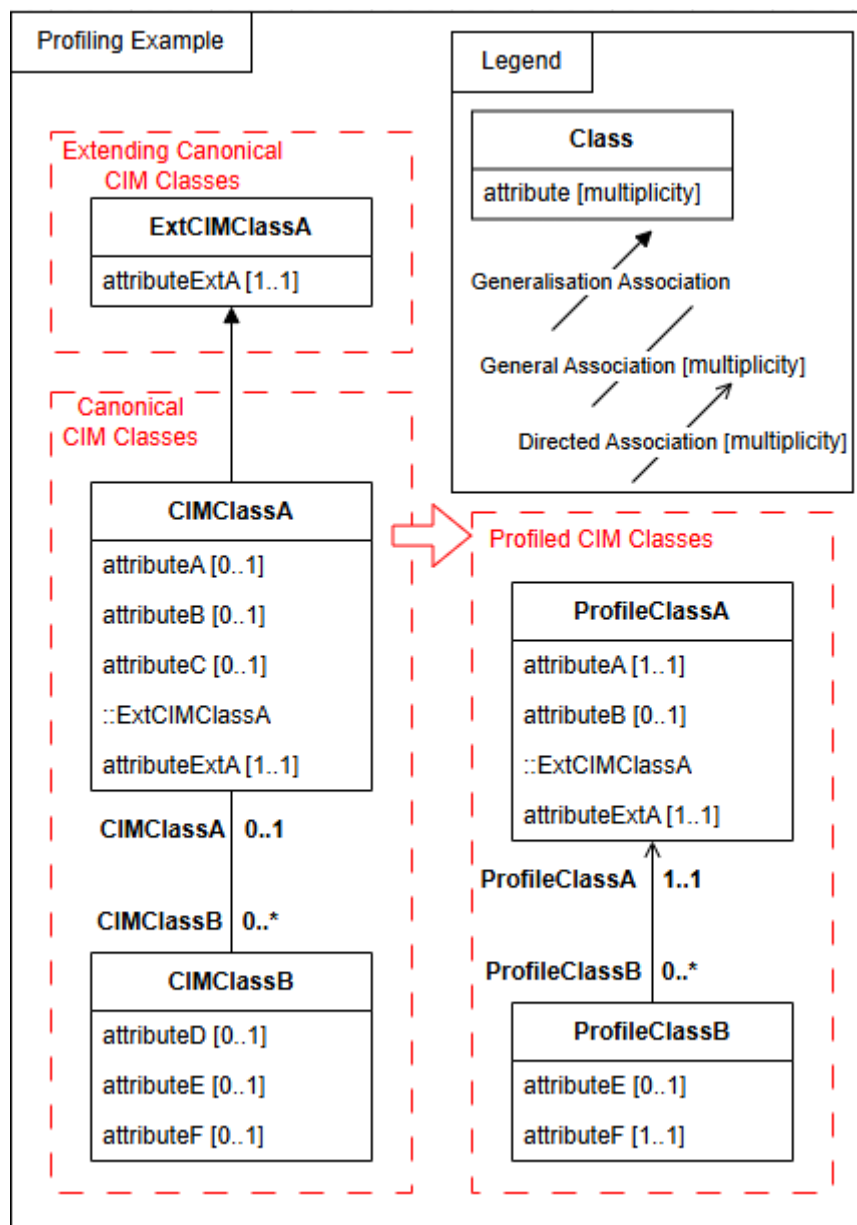


Figure 6: Profiling Example. Adapted from [8]

2.2.3 ENTSO-E Application Profiles

The ENTSO-E application profiles [21] are standardised artefacts designed to ensure interoperability and consistency in data exchange for e.g. Regional Coordination Processes (RCP) [22]. The profiles are organised into packages distinguishing between:

- The Common Grid Model Exchange Standard (CGMES), as defined in the IEC 61970-450 / -600 (all parts) as illustrated in Figure 3 and the IEC 61968-13 [23] and

- Network Code Profiles (NCP) specialised for the implementation of European Network Codes used in system operation, coordination and security analysis.

The application profiles are published as artefacts in a publicly accessible ENTSO-E GitHub repository and consist of RDFS vocabulary schemas and SHACL constraints (derived from the underlying UML information model). The profiles are published under the Apache 2.0 open-source license. ENTSO-E uses these artefacts in standard-vetting interoperability tests [24] [25] to test the conformity and effectiveness of the profiles in real-world scenarios.

Figure 7 depicts a high-level dependency view of CGMES/CIM-related standards and profiles, with a specific emphasis on which parts are relevant for in the NMM context (high-lighted in blue); where it should be possible for TSOs to leverage the CGMES-based data export capabilities in their utilised systems for transferring the network models into the Master Source Model, and/or to serve consuming processes such as for RCP.

- **Equipment profile (EQ)** for static models and physical connectivity;
- **Dynamic (DY)** for dynamic models;
- **Geographical Location (GL)** for geographical data;
- **Short Circuit profile (SC)** for short-circuit studies;
- **Diagram Layout (DL)** for single line diagram representations;
- **Operation (OP)** for real-time operations data;
- **Steady-State Hypothesis (SSH)** for scheduled operations;
- **Topology (TP)** for the definition of electrical topological connections and;
- **State Values (SV)** for the values resulting from the power flow.

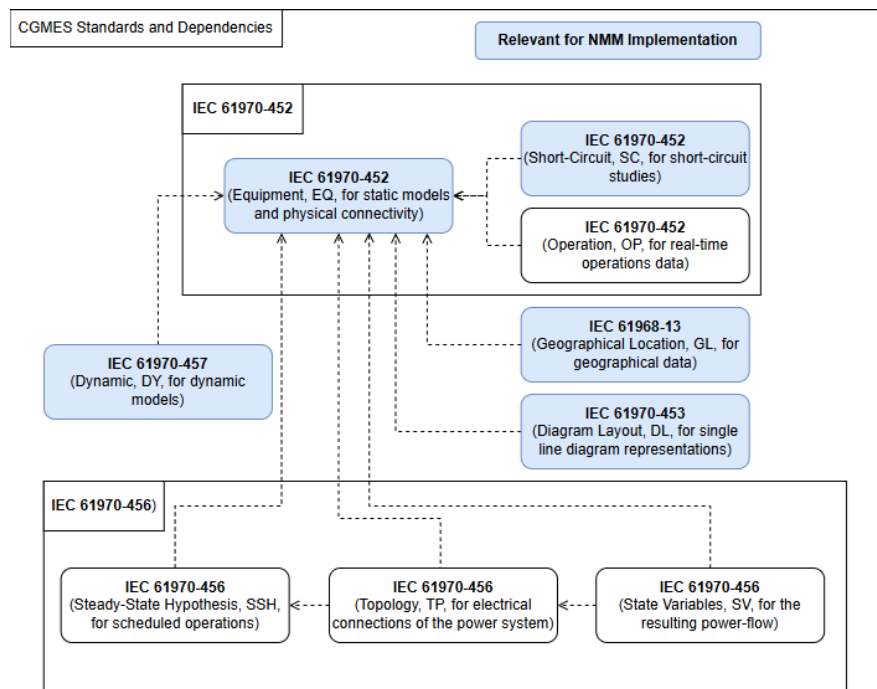


Figure 7: CGMES Standards and Dependencies. Coloured in blue are relevant to NMM.

Figure 9 indicates the Network Code Profiles (NCP) and their dependencies, as well as the dependencies on the relevant CGMES profiles. The NCP is used for system operation, coordination, and security analysis, where in the context of an NMM, the structural data (highlighted in blue) is applicable. Just like can in CGMES, it should be possible for TSOs to leverage the NCP-based data export capabilities in their utilised systems for transferring the network models into the Master Source Model, and/or to serve consuming processes such as for RCP.

- **Equipment Reliability (ER):** a major central element of the NCP, which acts as the anchor for information such as equipment reliability status, limits, areas, corridors, overlapping zones, and related network-code concepts.
- **Power Schedule (PS):** to exchange planned power schedules. Its classes cover scheduled power behaviour of producing/consuming units, DC corridors, and collections of units allocated to control areas. In practical terms, it contains time-series schedule values, typically planned injections, withdrawals, exchanges, or dispatch-related schedule quantities
- **Availability Schedule (AS):** to exchange availability-related information, with time points indicating when an event is valid and whether it is active. In practice, it contains time-dependent availability or outage windows, such as when equipment, a remedial action capability, or some operational resource is available or unavailable.
- **Monitoring Area (MA):** for the area model used by the regional process: control/monitoring areas, their boundaries, and the assignment of monitored objects or

processes to those areas. This is how the data exchange knows which geographical or operational area a given object belongs to for monitoring purposes.

- **Object Registry (OR):** contains identifiers, naming authorities, code systems, aliases, and object-type references so that different actors can refer to the same object consistently across exchanges.
- **Assessment Element (AE):** contains identifiers, naming authorities, code systems, aliases, and object-type references so that different actors can refer to the same object consistently across exchanges.
- **Remedial Action (RA):** contains the catalogue of actions that can be used, such as topology changes, PST changes, redispatch, etc.
- **Contingency (CO):** contains the outage/event scenarios to be analysed, for example the loss of a line, transformer, generator, or a combination thereof.
- **Grid Disturbance (GD):** covering already occurred grid disturbances, faults, outages, and interruptions, including switching triggered automatically, unintentionally, or manually due to faults. So this profile contains event-based disturbance records, not future plans.
- **Steady State Hypothesis Schedule (SHS):** enables exchange of schedules of the operating point, that is, information over time. Conceptually, this is the scheduled version of SSH-like operating assumptions. It contains planned values or statuses for the future horizon, such as scheduled operating states that are needed as analysis input across multiple time intervals.
- **Impact Assessment Matrix (IAM):** to exchange impact assessment matrices needed within the process. In practical terms, it captures the structured assessment of whether some object, action, or scenario has an impact on another object or process outcome.
- **Remedial Action Schedule (RAS):** contains when, with what status, and with what intensity/parameter values a remedial action is to be applied.
- **Sensitivity Matrix (SM):** contains numerical sensitivity coefficients showing how a change in one variable or action affects another monitored quantity, which is central for e.g. Coordinated Security Analysis (CSA) and optimisation.
- **Security Analysis Result (SAR):** contains the assessment outputs: violations, calculated monitored values, margins, and other results produced by e.g. CSA.
- **State Instruction Schedule (SIS):** contains future time-series instructions or overrides to structural data, but at schedule granularity rather than per individual MTU (Market Time Unit) precision.
- **Steady State Instruction (SSI):** provides the most exact MTU-level updates and can override both structural data and scheduled data for a specific time unit. It is therefore used for

precise operational instructions or per-time-unit operating-point values, such as unit dispatch instructions, shift keys, or other detailed state-setting inputs.

Finally, Figure 8 visually maps the CGMES profiles across the domains (mainly transmission and distribution) and zones (mainly operational, enterprise, and market) where it is applied the most. NCP profiles, for implementing the network codes specific requirements, are visualised as supporting subsets alongside CGMES.

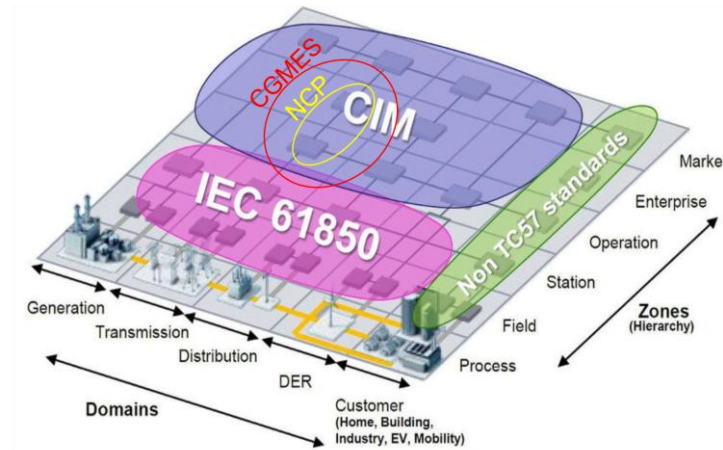


Figure 8: CGMES / NCP for Smart Grids Adapted from [26, 15]. The Plane (Component Layer in the SGAM) is for Illustration only.

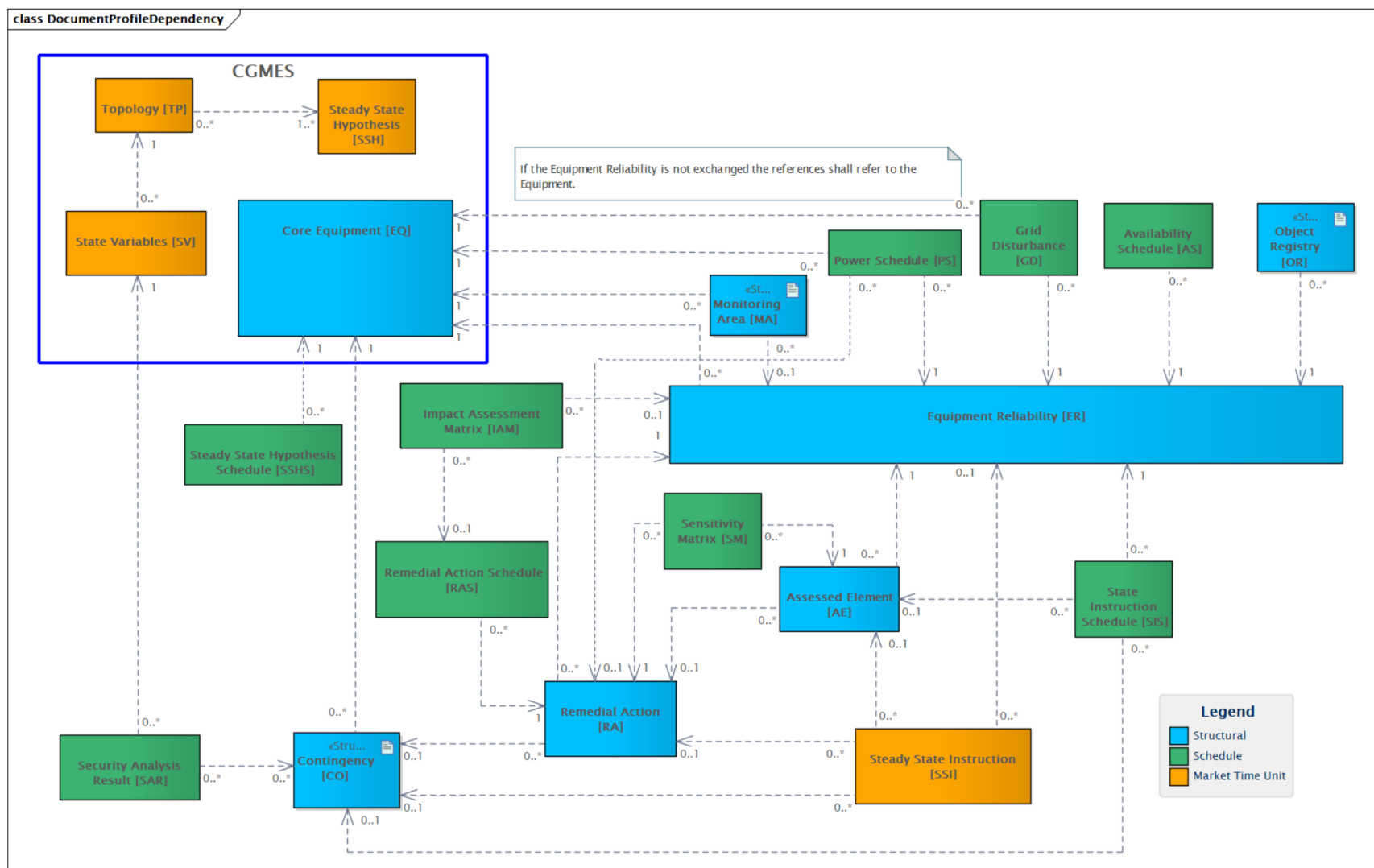


Figure 9: ENTSO-E Network Code Profiles and Dependencies (incl. dependency with CGMES).

2.2.4 Serialisation and Validation

The CIM model operates in different levels of abstraction and relies on technologies originally envisioned for semantic web, in particular the RDF (Resource Description Framework). RDF is the foundational standard data model for representing the information on semantic web as triples, structured as *subject-predicate-object*. RDF standard provides a basic syntax to express the statements about resources and URIs (Uniform Resource Identifiers) to identify and refer to resources uniquely. It is a graph data format, defining the instance data in these triple structures. RDF can be serialised in different formats. The chosen serialisation of the data does not impact the actual RDF content: an RDF serialised in RDF/XML format can also be serialised in Turtle or JSON-LD formats. This means that there is a distinction between the data and the syntax – the data and its serialization format are actually separable.

The RDF/XML serialisation is the most common serialisation format. The CGMES standard defines the specific CIMXML serialisation format, which is defined in the IEC 61970-552 standard [10]. It is based on RDF XML serialisation specification restricted to the processing of instance data, with additional changes and special assumptions, which means there are items in CIM that not correct by the RDF/XML standard: it does use the RDF graph-based standards (as defined by the W3C) to describe the model and the data. CIMXML has been published in 2003, while the W3C only completed the RDF/XML 1.0 specification in 2004. This results in challenges with general purpose RDF tooling that cannot be used directly on CIMXML.

Figure 10 shows how the data format, serialization, basic ontological schema definition and rule checking interact in terms of level of abstraction. The syntax, context and semantic layer refer to the layers in Figure 3.

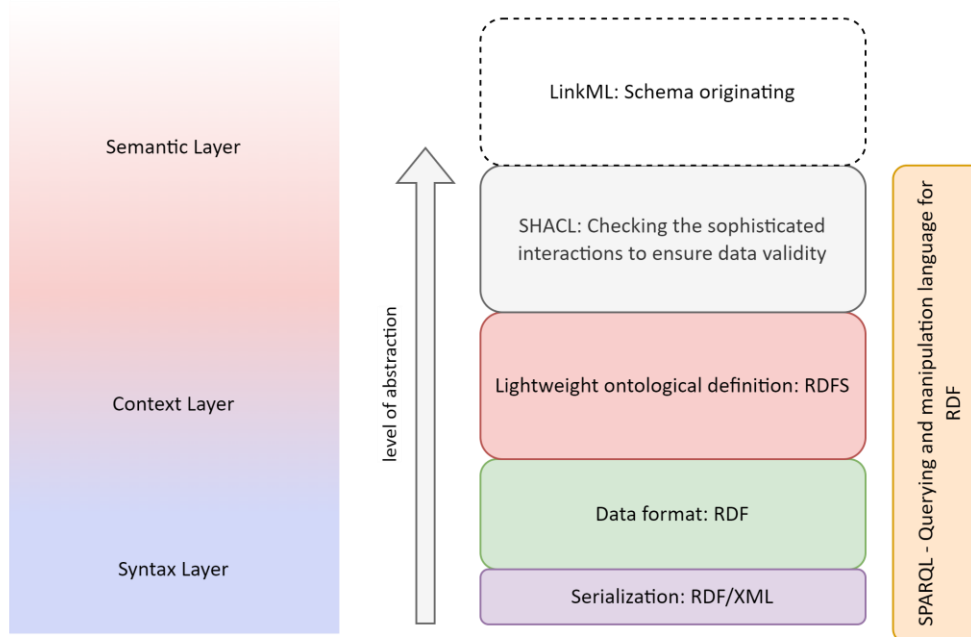


Figure 10: The relation between CIM serialization, data format, ontological (schema) definition, and shape validation [14]

The RDF format defines a data model. Its serialisation sits below the RDF in terms of data abstraction. The RDFS (RDF schema) only provides lightweight ontological restrictions and defines the vocabulary, i.e. what types of things should exist. RDFS is not going beyond these definitions. A more elaborated shapes constraint language (SHACL) is used to provide data validation and enforce structure beyond simple enumeration of classes and properties. The SHACL rules provide more complex and interacting validation capabilities, so SHACL is primarily designed to solve the data quality and conformance checking. The SPARQL language also indicated on the image is somewhat orthogonal to the above as it is a query language to retrieve and manipulate RDF data.

Finally, there are recent efforts to use LinkML at yet higher level of abstraction and use the LinkML generators to automatically create and maintain the code such as SHACL validation rules.

Finally, there are other ways to define the validation rules, including directly writing them in general purpose language such as C# or Python and this is also a common approach given the challenges in general purpose tooling and the tooling performance. This paragraph only serves as an illustration of the scope that a CIM-compliant NMM should be aware of and support.

For a deeper insight into the data formats, serialisation possibilities, challenges in schema and shapes validation and tooling landscape please refer to the ENTSO-E RDF Syntax Guide [14], where the topic is addressed in more detail. Supporting the standards indicated above is needed to support the functionalities of the CIM-based NMM and keeping that support up to date.

2.3 Introduction to TSOs Network Models and Use-Cases

Typically, Transmission System Operators (TSOs) develop and maintain different types of network models, composing of internal (e.g., network model of TSO A) and external (e.g., network model of neighbouring TSOs of TSO A) models. These network models are used for different purposes (e.g., N-1 contingency analysis) across different timeframes (e.g., Day-ahead) using tools such as PowerFactory, PSS/E, and PSCAD. These different power systems modelling tools are used for their simulation capabilities and are often employed for managing the data of the network models. The use of such tools therefore often results in vendor lock-in where proprietary data format is used for storing and maintaining data. The fragmentation of network models across different tools complicates data exchange interoperability and increases costs for integration and maintenance of supporting solutions.

In Figure 11, a high-level overview of the various internal and external models which together form a complete network model used for simulations and analysis. As seen, the various models are categorised as internal and external model parts and can be split into national and international parts. Furthermore, and regardless of whether a model is internal or external, the modelling perspective or representations of the network models for power systems are commonly recognised as follows and shown in Figure 12:

- **Bus-Branch Model:** Used for quick assessments for real-time operations and short-/mid-term operational planning. It is simplified representation of the network, focusing on buses and branches for fast load-/power-flow calculations.

- **Node-Breaker Model:** Used for more detailed and accurate representation for mid-/long-term planning and dynamic studies. It includes switching devices and their connectivity details, enabling more precise simulations and analysis.
- **Hybrid or Proprietary Variants:** Often tailored to specific tools or vendor solutions, combining elements of both Bus-Branch and Node-Breaker models.

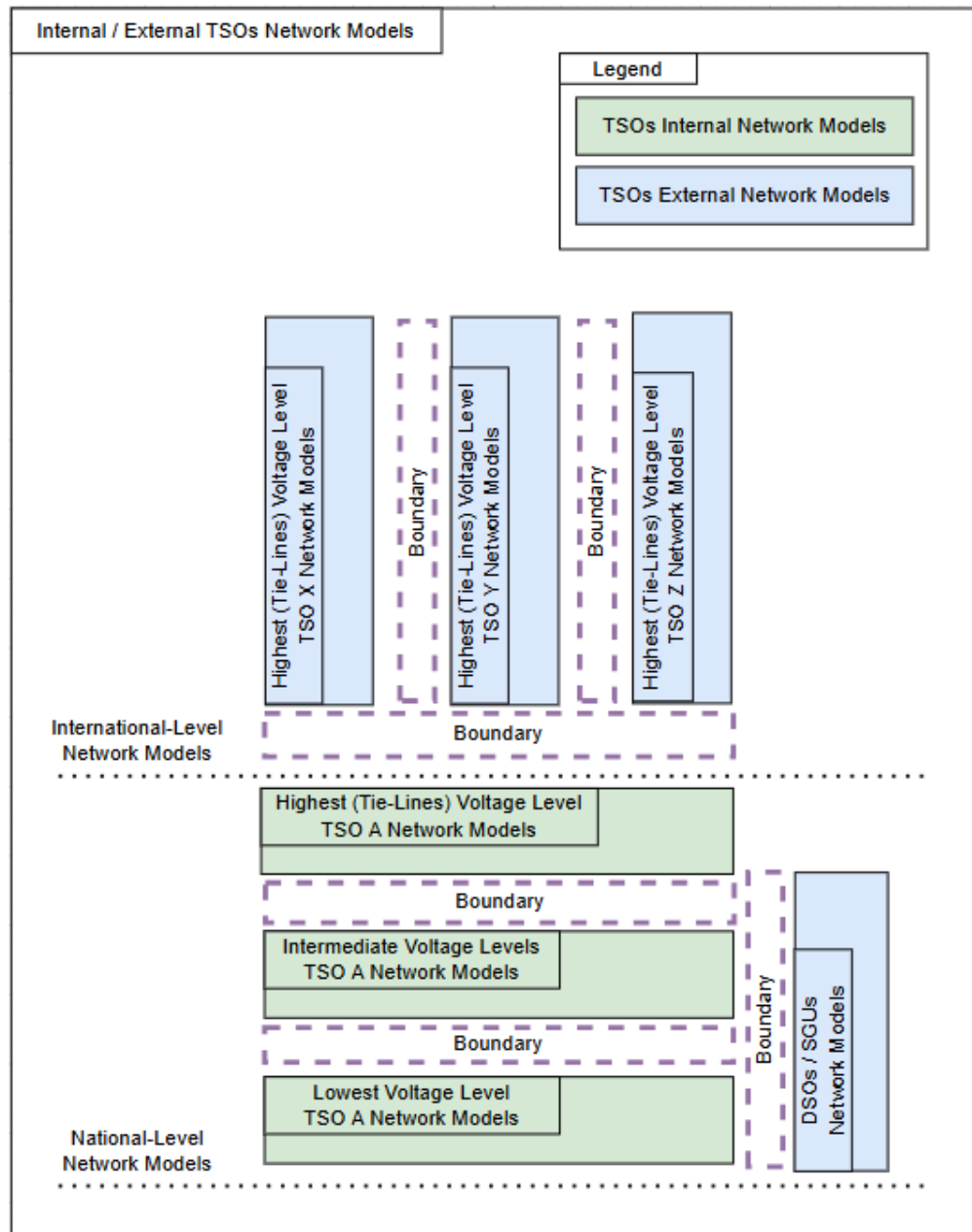


Figure 11: High-Level Composition of Network Models TSOs Deal with. Adapted form [3].

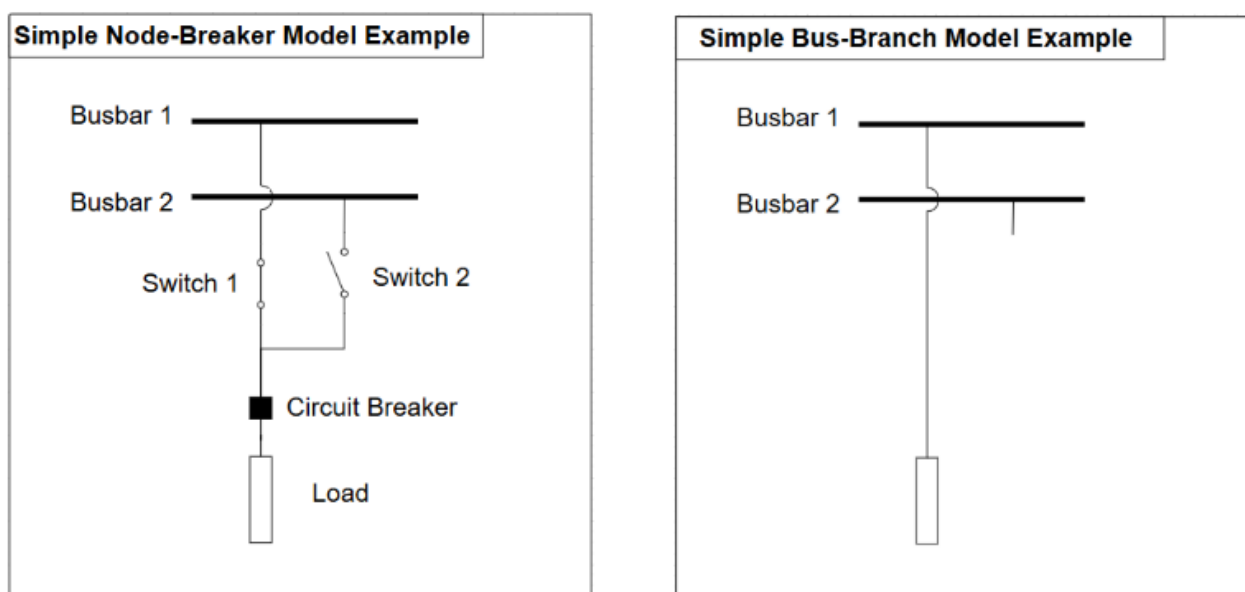


Figure 12: Simple Node-Breaker and Bus-Branch Examples.

Moreover, the different timeframes across which the various network models are used are given below and shown in Figure 13:

- **Real-Time Operations:** Immediate decision-making and intraday security analysis.
- **Short-/Mid-Term Operational Planning:** Day-ahead to (multi-)year-ahead studies.
- **Mid-/Long-Term Planning:** Multi-year-ahead dynamic stability and harmonics analysis.

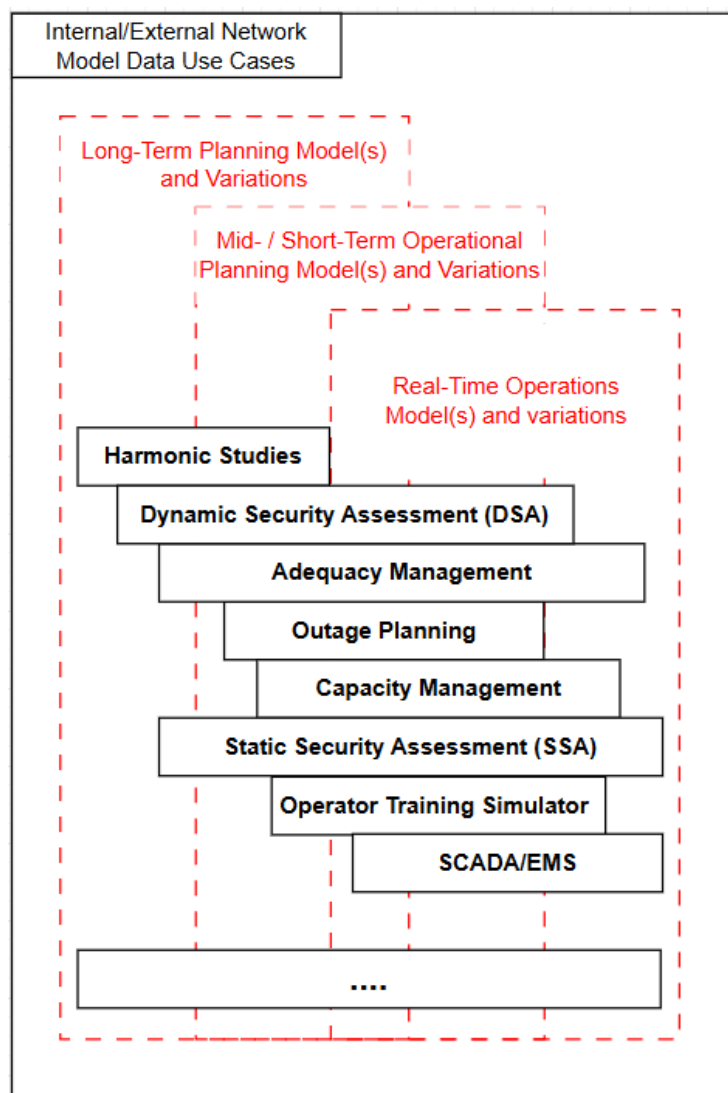


Figure 13: Common Internal/External Use-Cases of Network Models.

Table 3: Common Use-Cases of Network Models.

Common Network Models Use-Cases.	Timeframe of Use-Cases	Explanation
Harmonic Studies	Long-term studies	
Dynamic Security Assessment (DSA)	Long-term and down to real-time studies	DSA has a number of interfaces to other modules within a future control room. For example, static and dynamic grid models are imported from the NMM. Further, for analysis of real-time grid behavior, the DSA may use the results of Dynamic State Estimation for more accurate initialization of the dynamic RMS or EMT models.
Static Security Assessment (SSA)	Mid-term and down to real-time studies	
Adequacy Management	Long-term and down to short-term studies	
Capacity Management	Mid-term and down to short-term studies	
Outage Planning	Mid-term and down to short-term studies	
Operator Training Simulator	N/A	

SCADA/EMS	Real-time operations and studies	
TYNDP [27, 28]		

2.4 Using CIM in NMM

CIM can be used in Network Model Management (NMM) as the common semantic model that standardises how grid assets, connectivity, topology, operating state, and exchange datasets are represented across tools and organisational boundaries. Specific applications could have dedicated profiles and schemas that are derived from the canonical CIM, and the serialisation and validation could be done by leveraging CIM supporting technologies such as CIM RDF/XML and SHACL. Studies such as [4] shows the many advantages of adopting a CIM-Based NMM. Additionally, leading practices demonstrates effectiveness of such an CIM-based NMM implementation to avoid model management fragmentation – each functional group maintains its own model, leading to duplicated (manual) work, inconsistencies, errors, and slow processes [3]. The next chapter provides an actionable approach for realising a CIM-Based NMM for TSOs.

3 An Approach for Realising CIM-Based NMM

This chapter describes an approach for TSOs to realise an NMM ecosystem and serve CIM-based data to internal and external consuming processes. This approach is shown in Figure 14 and is generic by design, making it as a common process for:

- **Establishing a CIM-based NMM:** refers to a *pre-NMM Phase* during which the TSO is developing the necessary building blocks for an NMM ecosystem. During this phase, the proposal process in Figure 14 serves as an iterative process not only for the development and realisation of the NMM ecosystem but also for maintaining an up-to-date network model(s) representation.
- **Shadowrun CIM-based NMM:** refers to a *post-NMM Phase* during which the TSO has realised a functional CIM-based NMM and is ready to service consuming processes as *To-Be Situation* in parallel to the *As-Is Situation*. During this phase, the integration of the NMM is being tested and accepted in the operational business processes. Therefore, during this phase, the process proposed in Figure 14 can be deployed again for maintaining an up-to-date network model(s) representation in the NMM ecosystem.
- **Operational CIM-based NMM:** refers to a *post-NMM Phase* during which the realised CIM-based NMM is fully tested, accepted, and operational as the single-source of network models for (all) consuming process(es). Therefore, during this phase, the process proposed in Figure 14 can be leveraged for maintaining an up-to-date network model(s) representation in the NMM ecosystem.

The description provided in this chapter focuses on deploying the proposed process in Figure 14 during the *pre-NMM Phase*. Therefore, it is out of the scope of this chapter to describe how the proposed process is deployed in the *post-NMM Phase*. A flowchart of the pre-/post-NMM phases is given and described in chapter 5 for more info.

Furthermore, the proposed process in Figure 14 is triggered during the *pre-NMM Phase* by the initiation of the NMM project and/or due to the availability of more recent network model(s) data.

- **Network Models Harmonisation:** A step for harmonising the already existing or newly available network models into a Totally Harmonised Model (THM). After this step is either *Profiling and Schema Generation* and/or *Data Serialisation*. A more detailed description is provided in section 3.1. In the case of *post-NMM Phase*, this step could be taken for the

harmonisation or synchronisation of newly available network model(s) data from internal/external sources.¹

- **Profiling and Schema Creation:** A step for creating profile(s) as semantical representation of the THM where the data structure, relationships, and specifications are modelled and visualised in term of UML representation(s). Profiling is dependent on *Network Models Harmonisation* for defining what exactly to be profiled. Additionally, this profiling supports generating schema as machine-readable artefacts for machines/systems interpretation and validation of basic UML specifications. This step advances to the next step, *Data Serialisation*. A more detail description is provided in section 3.2. In the case of *post-NMM Phase*, this step could be taken for the profile and schema artefacts creation to incorporate the representation and interpretation of the newly introduced network model(s) data from internal/external sources.
- **Data Serialisation:** A step for serialising the proprietary data of (parts of) the THM in a CIM-based format. Serialisation is dependent on the *Network Models Harmonisation* for defining what exactly to be serialised. Furthermore, while serialisation is not necessary dependent on the *Profiling and Schema Generation*, it can benefit from it as a reference and a mean for visualisation. This step advances to the next step; either *Data Validation* or *Data Loading*. A more detail description is provided in section 3.3. In the case of *post-NMM Phase*, this step could be taken for the serialisation of the newly available proprietary network model(s) data from internal/external sources.
- **Data Validation:** A step for validating the CIM-based THM serialised data. Validation is dependent on *Data Serialisation*. Note that TSO may choose to do the validation on data consumer side after loading the data for the consuming process. This step advances to the next step, *Data Loading*. A more detail description is provided in section 3.4. In the case of *post-NMM Phase*, this step could be taken for the validation of the newly available CIM-based network model(s) data from internal/external sources.
- **Data Loading:** A step for preparing and making the data product available for consuming processes. Data Loading depends on the *Data Serialisation* for physical dataset(s) to be validated. Furthermore, while Data Loading is not necessary dependent on the Data Validation, it can benefit from it as a compliance proof before a data product is served to a consuming process, potentially alongside the validation report. This step concludes a full cycle, after which, a new cycle may be triggered. In the case of *post-NMM Phase*, this step

¹ The optimal sequence of harmonization steps will depend on the situation in the TSO: it is possible to harmonize the model even without profiles, e.g. by exporting the data in terms of CGMES and merging these data into a THM, and only then proceeding with the profiling steps iteratively. In an ideal case, it is better to work with profiles but the optimal approach for a given TSO will depend on their situation when starting the NMM implementation.

could be taken for preparing and making CIM-based data product(s) available for internal/external consuming processes.

Finally, as seen, every step in the cycle undergoes iterative loops for improving the results, before advancing towards a *post-NMM Phase*.

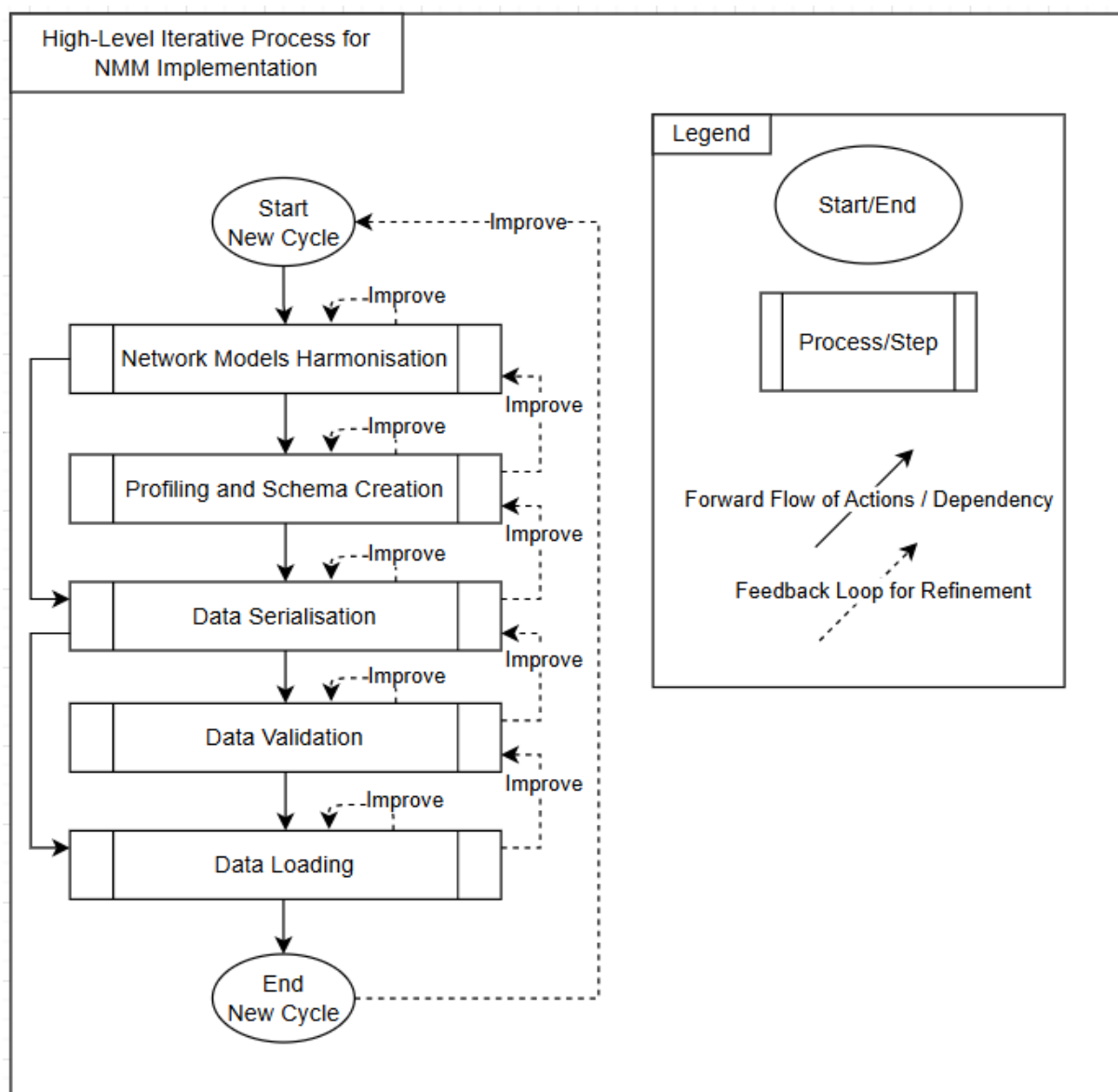


Figure 14: Proposed Process for Implementing a CIM-Based NMM.

3.1 Network Models Harmonisation

This section outlines a method for TSOs to harmonise the proprietary data of the various network models into a Totally Harmonised Model (THM), serving as a single source of network model. The *THM* term is used to refer to a part or all of network models harmonised in their native proprietary format, while the term *CIM-based THM* refers to the THM representation based on CIM, including ETNSO-E Application Profiles, in addition to any necessary TSO specific extensions. The CIM-based THM is in this sense the Master Source Model in the NMM. The proposed harmonisation options presented in this chapter strives to ensure alignment with CIM standards so that TSOs can more effectively achieve CIM-based THM that supports both internal processes and collaboration with external systems and organisations.

While establishing the THM, one of key steps is to identify the correct ownership and processes regarding the THM attributes as the NMM, once implemented, will support and respect these processes. This will ensure that the object IDs (and other attributes) in the THM are consistent, with a clear business ownership behind it.

Figure 16 shows how the THM acts as a merging of various network models. Examples of these models are the ones used for Dynamic Security Analysis (DSA), Static Security Analysis (SSA), Energy Management System (EMS), Asset Management (AM), etc. There are different situations regarding existing model data, and establishing a fully harmonized model does not provide a one-fits-all solution. The original data, in terms of its storage, formatting and representation can be found as:

- **Case 1 – Fully Homogenous Proprietary Data:** All models are represented/modelled using the same modelling tool, and therefore, they are all represented in the same proprietary format. In this case, as shown in , it is an option for the TSO to do the harmonisation of the network models in the proprietary format of the used tool. It is however (almost) never the case that all the various network models are modelled in the same tool/proprietary format, and this is one of the major reasons why there is a need for an CIM-based NMM. The harmonisation can be done by exporting the various models in the native language of the modelling tool, comparing them and defining the differences, and finally merge them to keep the unique data and remove the duplicate data. Next step would be the standardisation, where data (i.e., THM) is serialised in a CIM-based format. Data serialisation is described later in this document.
- **Case 2 – Partial Homogenous Proprietary Data:** Some models are represented/modelled using the same modelling tool, but others are not. In this case, as shown in , it is an option for the TSO to do the harmonisation of the network models of the similar proprietary format as an intermediate-partial step for the harmonisation. Following this step would be the standardisation of the various models, including those that are partially harmonised in proprietary format, and then perform a total harmonisation of all the models using a CIM-based format.
- **Case 3 – Fully Heterogeneous Proprietary Data:** All models are represented/modelled using different modelling tools, and therefore, they are all represented in different proprietary format. In this case, as shown in , TSO's (best and almost only) option is to standardise the data first and then do the harmonisation. Note that this approach is

applicable to the previous cases as well, where TSOs can choose to follow this approach regardless of the similarity in the proprietary data formats.

So far, the description establishes what a THM is and what are the TSO's options to do this based on the native proprietary format of the modelling tool(s).

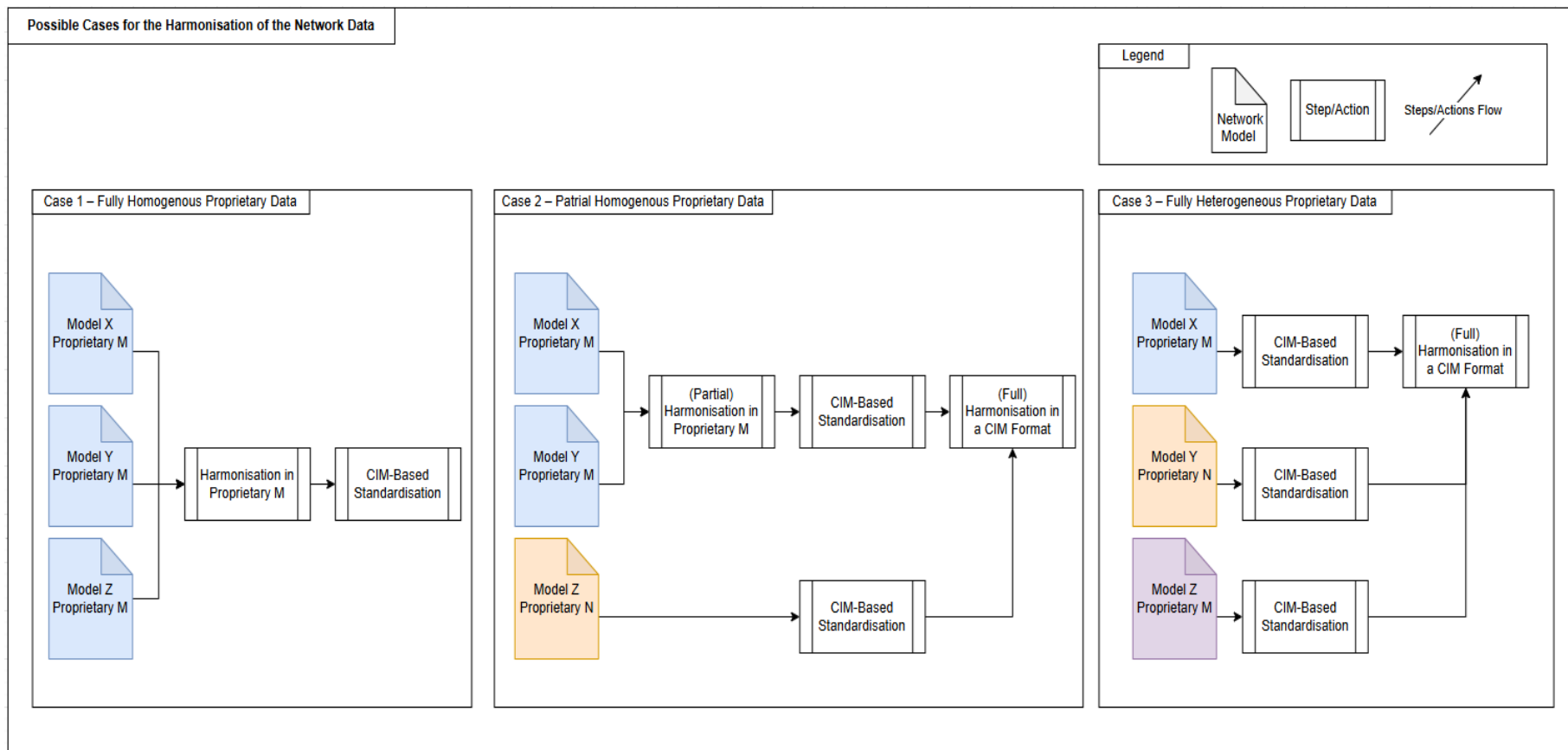


Figure 15: Possible Cases for the Harmonisation of the Network Data

With reference to Figure 16, as seen, the data of the THM needs to be standardised (i.e., transformed into CIM-based), and based on this, the data can be divided into two standardisation categories or parts:

- **Part 1 – Highlighted in Green:** A basic and bare minimum part of the THM that is needed for basic simulations and analysis. This part is high-lighted in green and is possible by most modern modelling tools to be exported as NCP and/or CGMES and in various CIM-based formats (e.g., CIMXML). This export of the THM forms a part of the representation in canonical CIM since CGMES/NCP are derived from canonical CIM.
- **Part 2 – Highlighted in Red:** A more detailed and complex part of the THM that is complementary but crucial part for other more specialised simulations and studies. This part of the THM is not part of CGMES/NCP, however, it is found partially in canonical CIM. What is not found in canonical CIM (e.g., asset owner) is the other part that could be extended.

To conclude this chapter, at this stage of the CIM-based NMM implementation, for the harmonised as well as standardisation of the various network models data and realising a CIM-based THM, TSOs have two overarching approaches to choose from as shown in Figure 16:

- **Choice 1 – CGMES-/NCP-Based Approach:** Leverage CGMES/NCP exporting functionalities to collect and merge the network models data of the various modelling tools as CGMES/NCP datasets in e.g. CIMXML format. This approach entitles also leveraging the already existing artefacts (e.g., profiles, schema, validation rules, etc.) made available by ENTOS-E. The representation of the rest of the data is either found in canonical CIM or should be extended and linked to the CGMES/NCP dataset(s).
- **Choice 2 – Canonical CIM-Based Approach:** To represent the various network models data based on canonical CIM; where part of the representation must cover at least CGMES/NCP, and part is to be extended to cover TSO's specific needs.

Furthermore, and regardless of the harmonisation approach chosen by the TSO, a mapping list is needed for various steps to realise a CIM-based NMM – mainly profiling, serialisation, and validation as described later in this document. Furthermore, as seen in Figure 16, (at least some) data must be mapped across CIM and the proprietary format to define what classes and attributes correspond to each other. The exact and the amount of data that needs to be mapped depends on the approach chosen.

- **In Approach 1 – CGMES-/NCP-Based,** the data to be mapped is all the data that forms the THM except that covered by CGMES/NCP; while;
- **In Approach 2 – Canonical CIM-Based,** all the data that forms the THM must be mapped including that covered by CGMES/NCP.

In either approach, the least data that requires mapping is the missing data in the canonical CIM representation which should be defined and mapped for extending it on top of CIM. The mapping list should among others define:

- Which classes/attributes in the THM are:
 - Found **in CGMES/NCP** and could be leveraged based on export capabilities of modelling tools and available artifacts by ENTSO-E (refer to [16] for artifacts).
 - Not found in CGMES/NCP but are found **in canonical CIM**.
 - Neither found in CGMES/NCP nor in canonical CIM, and thus the data **to be extended**.
- Additional rules such as restrictions, datatypes, enumeration tables, naming conventions, etc.

When the harmonisation approach is chosen by the TSO, and the mapping work is in progress or even done, the TSO can proceed with the proposed NMM implementation process. Suggested next steps are *Profiling and Schema Creation* and/or *Data Serialisation*. The following chapters describes the implementation process steps in more details.

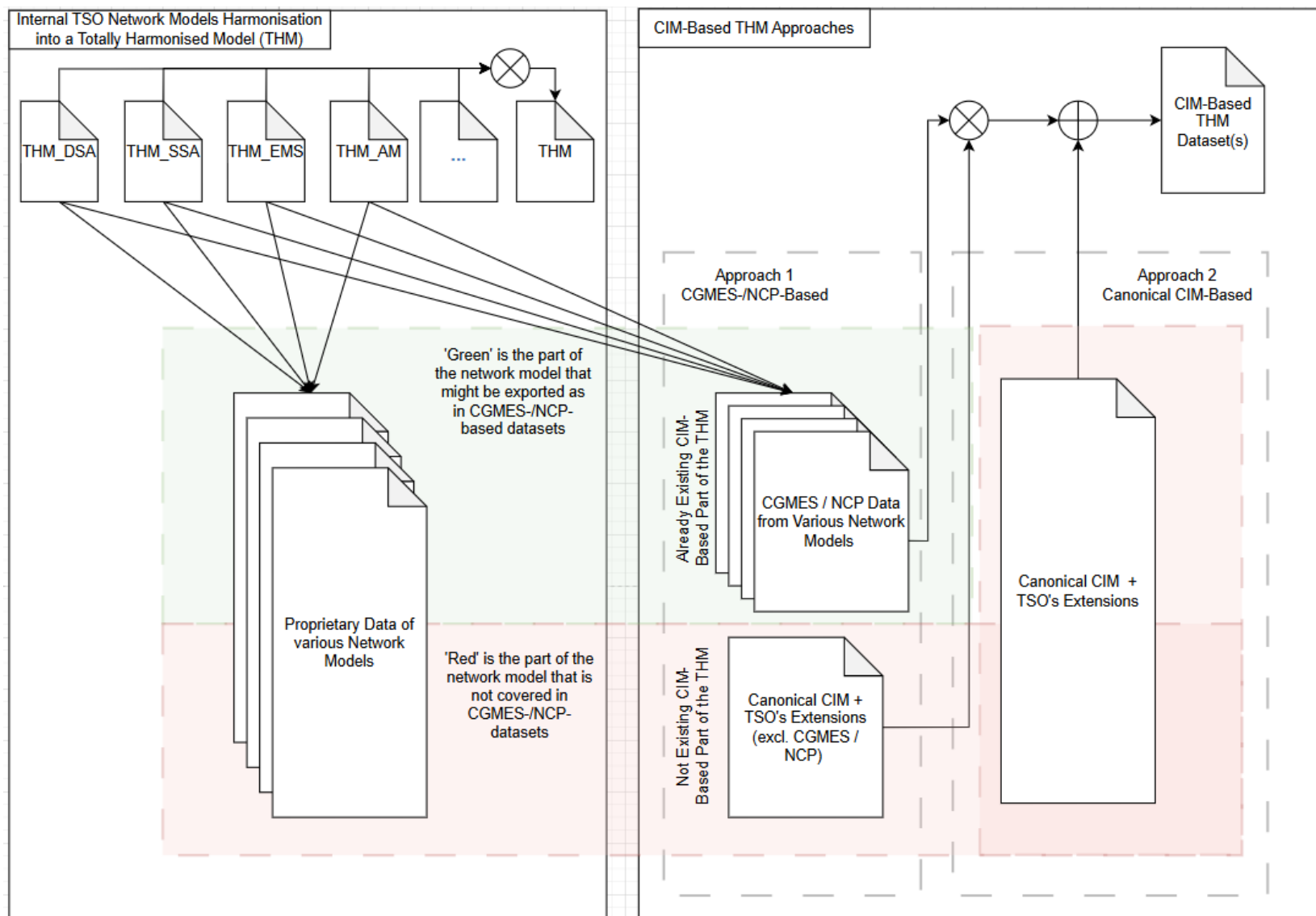


Figure 16: Network Models Harmonisation into a Totally Harmonised Model (THM) and Standardising the Data. Green/Red ratios are indicative.

3.2 Profiling and Schema Generation

This section describes how to establish a canonical semantic layer based on the IEC Canonical CIM model for the CIM-based Totally Harmonised Model (THM). Additionally, this chapter describes how this semantic layer, represented as UML (Unified Modelling Language) models, is used for visualising the data structure, relationships, and specifications of the derived profiles, and how these contribute to schema generation that could be used for the validation of standard UML restrictions (e.g., restricting reference direction and associated cardinalities). For an introduction to profiling and schema generation, refer to 2.2.2. Finally, this chapter describes how to leverage the ENTSO-E CGMES/NCP application profiles, saving man-hours and advancing the development of realising a CIM-based NMM.

As concluded in the previous chapter, TSOs have two overarching approaches to choose from for the harmonisation and standardisation of the various network models:

- **Choice 1 – CGMES-/NCP-Based Approach:** Leverage CGMES/NCP exporting functionalities (if applicable) to collect and merge the network models data of the various modelling tools as CGMES/NCP datasets. In this case, the already existing artifacts (i.e., profiles, validation rules, etc.) made available by ENTSO-E can also be leveraged. The profiling of the rest of the data is either found in canonical CIM or could be extended; and thus, the schema is generated accordingly.
- **Choice 2 – Canonical CIM-Based Approach:** To represent the various network models data in terms of canonical CIM, where part of the representation is what is already covered in CGMES/NCP and part is to be extended.

For either option, and as shown in Figure 17, there is a need for profiling so that the data structure, relationships, and specifications are conceptually modelled (a.k.a. logical model) and ready to be used for schema generation. Note that profiling is needed at least for the CIM extensions to cover the missing representation of the THM as well as the internal/external processes, and hence, the profiling of the THM itself and/or CGMES/NCP is not a blocker. However, the lack of a dedicated profile for the THM could result in undesired issues down the line:

- **THM Validation:** It is not possible to leverage a schema based on the THM profile for the basic UML validation. This still can be done using e.g. SHACL or other validation techniques.
- **THM Migration:** Transition from a particular NMM tool to another could be challenging, because the THM data structure, relationships, and restrictions are not conceptually represented (UML in this case), making difficult to determine criteria for a complete and successful migration.
- **THM Quality:** The quality of the THM could decrease over time since developers do not have a conceptual representation of the data (-structure, -relationships, and -restrictions), resulting in wrong assumption and uninformed-decisions.

Furthermore, below is a list of necessary activities for profiling and schema generation for the THM and/or internal and external consuming processes as well as associated extensions:

- **THM $\leftrightarrow$ CIM Mapping:** If not yet done, create a list of the classes and attributes from the THM and map them onto CIM to support the following steps. This list should define a.o. which classes and attributes are in CIM; which part is already part of CGMES/NCP; what part need to be extended; rule, etc.
- **CIM Semantic Model Reference:** Create, with the permission of the UCAIug [29], a reference (a.k.a. golden copy) UML model of the IEC CIM classes and European extensions to ensure semantic alignment across international standards. This reference model could be used for the profiling of the THM and/or modelling extensions.
- **Contextual Profiling:** Derive CIM subsets (i.e., profiles) from the reference model to represent the THM data structure, relationships, and restrictions. Contextual modelling is dependent on the mapping sheet to determine what classes and attributes should be profiled and what their specifications are. Furthermore, it is necessary to introduce the CGMES/NC profiles which are necessary as part of the data needs for internal as well as external processes (e.g., (Inter-)national Security Analysis).
- **Extending CIM:** Extend the reference CIM model to cover the missing data representation of the THM. Extending CIM might also include extended to the CGMES/NCP for internal TSO-specific use-cases. Modelling extensions is dependent on the mapping sheet to determine what classes and attributes should be extended and what their specifications are.
- **Schema Generation:** Create schema(s) for the THM model from the THM profile for machine interoperability and partial validation (i.e., basic UML validation). This schema also includes the modelled extension. Depending on the approach the TSO chooses, it might be applicable to use the ENTSO-E Application Profiles Library (RDFS/SHACL) [16] as the constraints set for machine validation and interoperability of CGMES/NC datasets.

Enterprise Architect (EA) by SPARX is the main tool used for the CIM UML models development and maintenance [19], and hence, it is recommended to use EA for NMM. Profiling. Schema can be created using tools like CIMTool or CimConteXtor, where the generated schemas serve as machine-readable artifacts to understand the associated datasets as well as for (partial) validation. The reader can refer to chapter 9 in IEC61970-401:2022 [8] for requirements for the profiling tools.

In the next chapter, serialisation of data is introduced where conceptual or logical models of data are translated into a physical data load. Serialisation results in the dataset containing the actual values. Later in the document, it is also described how schemas – based on the defined profiles – could be leveraged for basic UML rules validations. All this paves the way for establishing a validated CIM-based single-source for the network models for internal and external consuming processes of special needs.

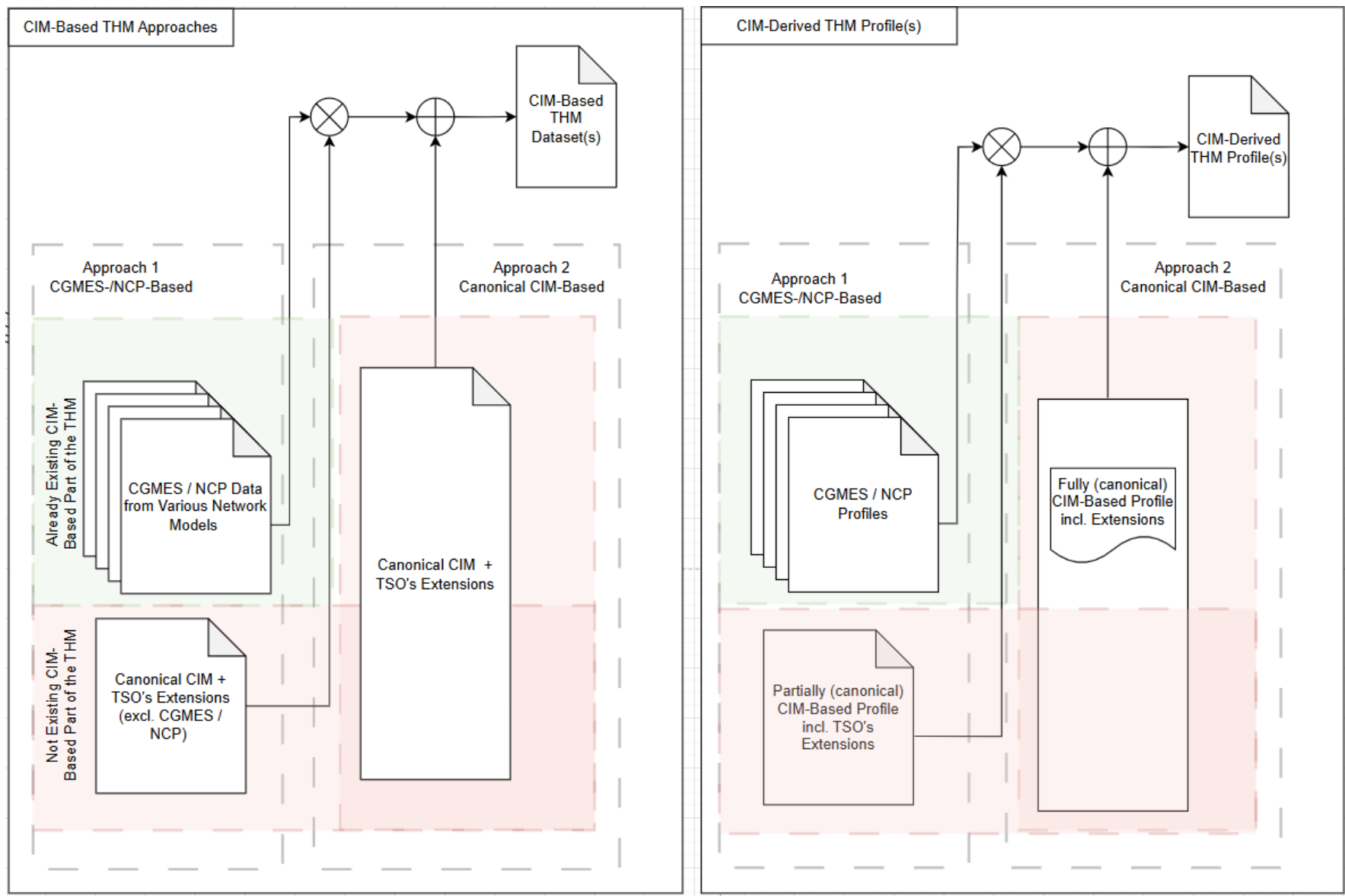


Figure 17: Profiling of the Standardised Totally Harmonised Model (THM). Green/Red ratios are indicative.

3.3 Data Serialisation

This section describes how serialisation is an essential step in the realisation of a CIM-based NMM. Data serialisation is the act of generating a dataset instance file in a specific format such as CIMXML, RDFXML, Turtle, JSON-LD, N-Triples, etc [11]. Each of these different serialisation formats has its own advantages and disadvantages. Refer to 0 for a more elaborate introduction to data serialisation. The recommended and adopted format in the context of CIM/CGMES is CIMXML in accordance with the IEC 61970-552 [10].

Serialisation of the various existing network models (i.e., the THM), as a step in the implementation process, can start right after the *Network Models Harmonisation* and can benefit from the *Profiling and Schema Generation*, if work has been done. A THM profile could support visualising the data structure, relationships, and specifications. Nevertheless, serialisation is a challenging task that would require many iterations before a stable point is reached. Serialisation should be supported by a repeatable and auditable transformation process. This process may be implemented using ETL tooling, NMM-native import/export functions, adapters, scripts or other automation mechanisms. The key requirement is not the specific technology used, but the repeatability, traceability, validation capability and ability to reproduce the same CIM-based data product from the same source inputs.

Nevertheless, this Implementation Guide recommends developing an Extract-Transform-Load (ETL) pipeline to automate the high-level process shown in Figure 18:

- **Extract:** Automate the process of extracting the proprietary data of the various existing network models in the modelling tool(s) (such as PowerFactory, PSS/e, PSCAD, etc.).
- **Transform:** Automate the data transformation (or serialisation) into CIMXML instances.
- **Load:** Automate the loading of the data instance files into its destination; staging area, validation tool (e.g., the tool used for the NMM), consumers, etc.. Furthermore, depending on the capabilities of the toolchain, it could also be advantageous to automate the validation and results reporting of the loaded data
- **Improvement:** Throughout the implementation process, improving the ETL process is needed, and hence, the feedback loops in Figure 18. The question is what needs to be improved in each new cycle. This question is answered by assessing, analysing, and reporting the results.

An input to the data serialisation process is the mapping sheets needed to allocate THM data to the serialised files. E.g., the actual resistance of transformer winding in CIMXML. Additional input for the serialisation, is the THM profile which could be used to visually and/or automatically validate the correctness of the serialised files.

Assessing and validating the serialised data can be done by comparing the serialised output data with the THM conceptual or semantical model. E.g., if the THM profile is derived and specifies that `cim:PowerTransformer.vectorGroup` is mandatory (i.e., multiplicity is set to [1..1]), the validator

should be able to find this attribute in every `cim:PowerTransformer` object. The automatic assessment and validation – at least for the basic UML restrictions – can be done in one of the deployed NMM toolchains using a generated schema (e.g., RDF Schema) from the derived THM profile. In case no tool in the toolchain is capable of validation, additional tool integration would be necessary for this purpose. Finally, a more elaborate description on the validation of the THM data is treated in the next chapter.

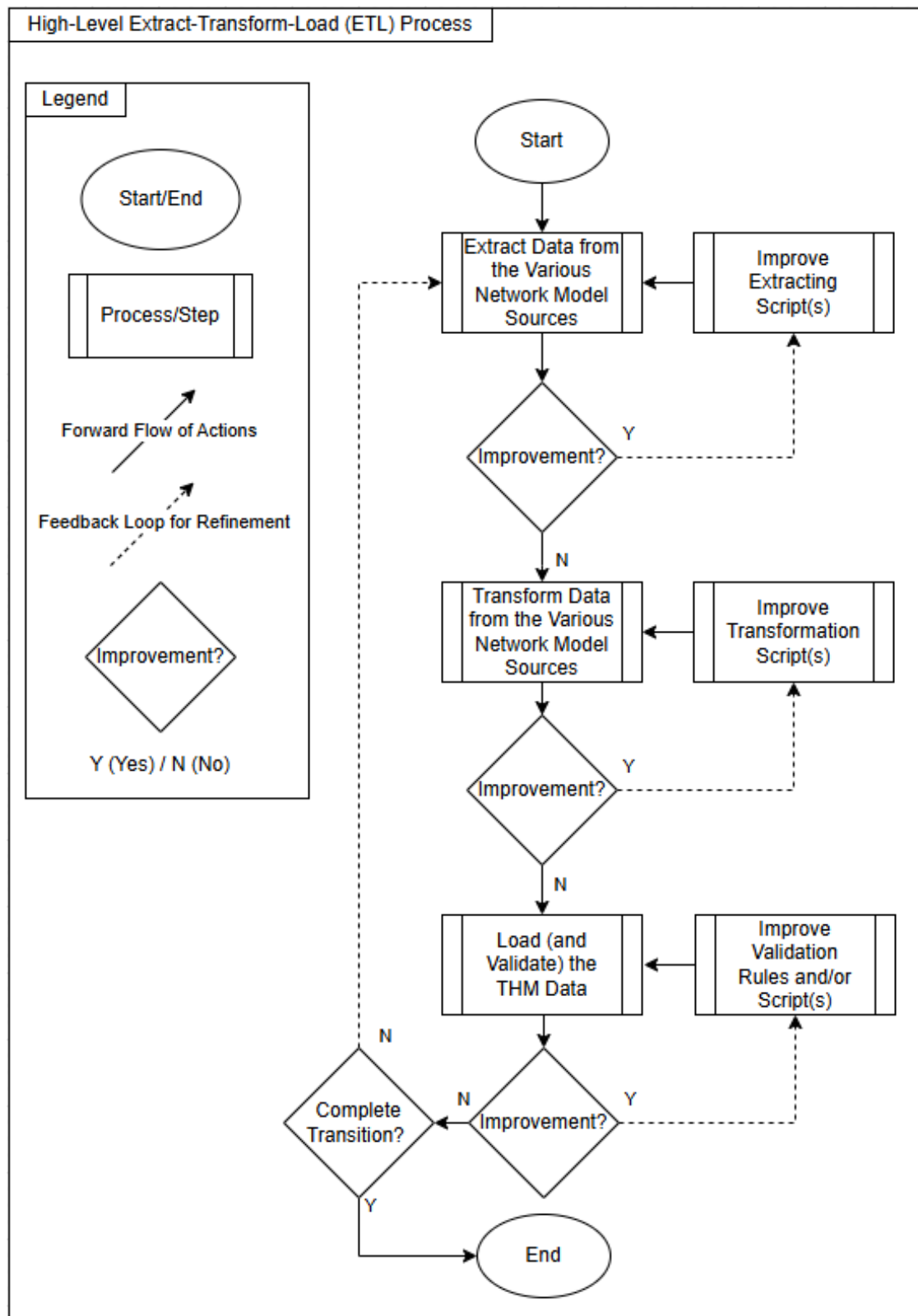


Figure 18: High-Level Extract-Transform-Load (ETL) Process.

3.4 Data Validation

This section describes how to validate the CIM-based Totally Harmonised Model (THM). This validation step depends therefore on the availability of the THM dataset(s), and hence, this step depends on the output from the *Data Serialisation* step. The validation also depends on the implementation approach chosen by the TSO. In the case that CGMES/NCP is leveraged by the TSO for the NMM implementation, the existing validation rules of CGMES/NCP [16] can be leveraged as well for validating the CGMES/NCP serialised part of the THM.

Figure 19 depicts a high-level process for the validation of CIM-based data (e.g., the CIM-based THM or CGMES-based IGMs). As proposed, the validation process should start with defining the validation rules, which might be represented using CSV, SHACL, OCL and others, depending on the toolchain in the NMM ecosystem. Further on in this section, an elaborate description for defining validation rules is given. Furthermore, referring to Figure 19, with the available validation rules, the dataset validation (e.g., THM CIMXML) may start, and the validation results may be documented (e.g., CSV) and made available for potential use-case.

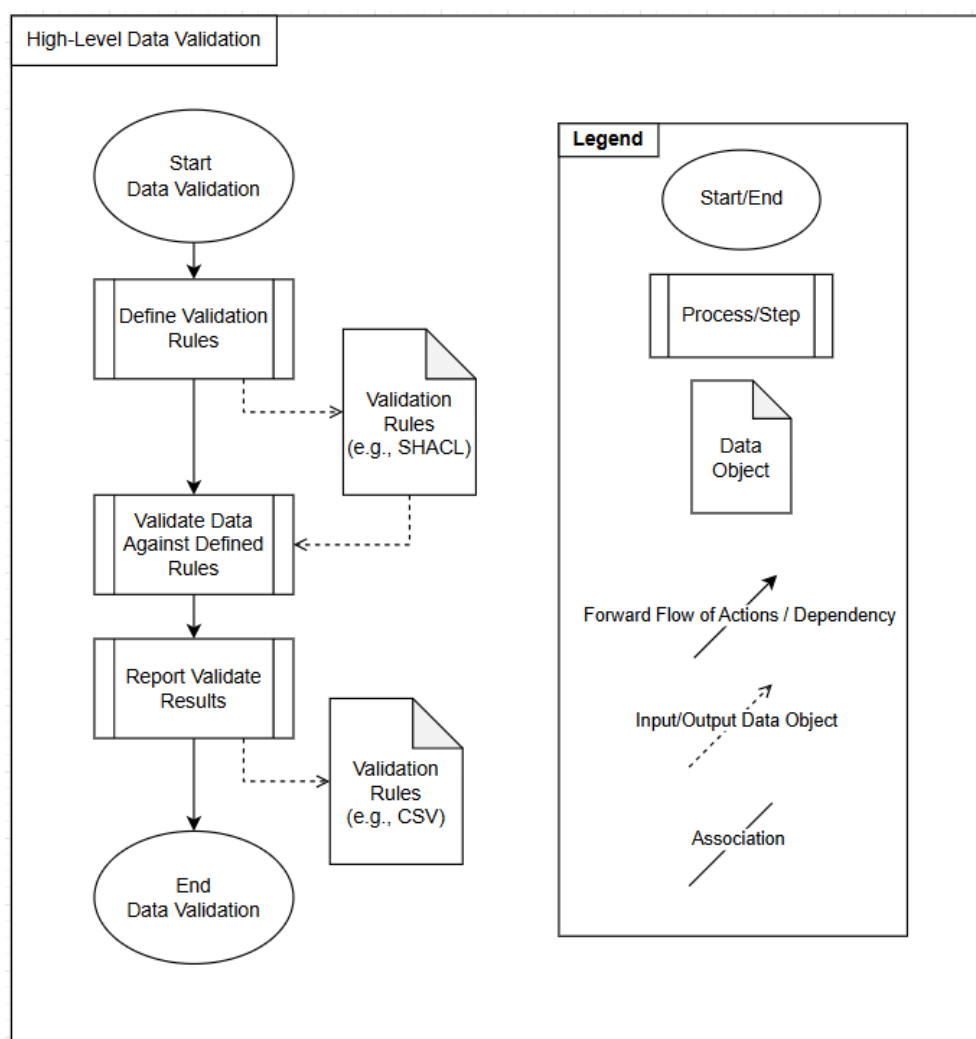


Figure 19: High-Level Process for Data Validation.

Potential Validation Aspects

Table 4 lists high-level potential validation aspects for the THM and provides a description and an example per aspect. This table refers to the simple UML (Unified Modeling Language) model in Figure 20 as a mean for visualisation and relating to the examples given in the table. In the case that CGMES/NCP is leveraged by the TSO for the NMM implementation, where CGMES/NCP is part of the CIM-based THM, the validation aspects are only applicable on the other remaining part of the CIM-based THM.

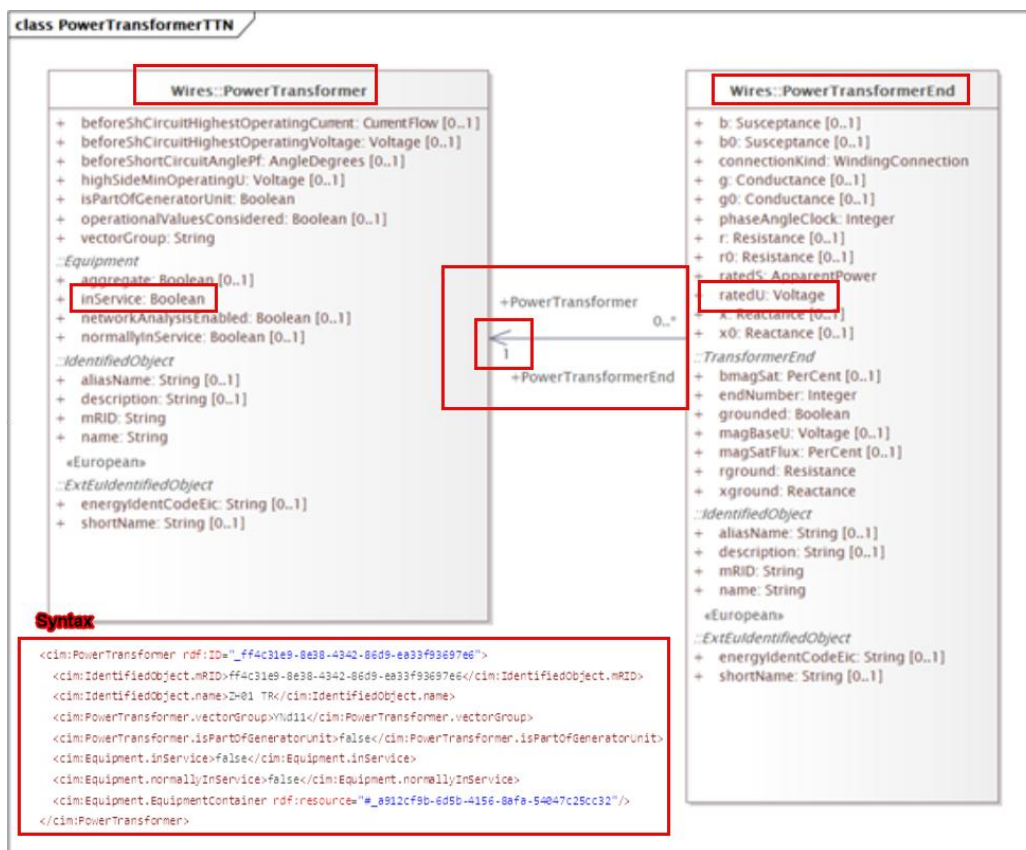


Figure 20: Simple UML model for illustration. Source: IEC 61970 CIM, adapted by ENTSO-E.

For more comprehensive validation rules, Appendix in A, provides a list of potential QoCDC (Quality of the CGMES Datasets and Calculations [16]) rules that can be leveraged for the validation of the serialized THM. Furthermore, refer to Appendix 0 for an example of utilising RDF Schema for (partial) validation; and refer to [14] for more info on validation technologies and tools.

Table 4: Validation Aspects of the CIM-Based Serialised THM Dataset.

Validation Aspect	Sub-Cat.	Description	Example
1. Classes and Attributes Completeness.	1.1. Class & Attributes Existence. 1.2. Class & Attributes Completeness. 1.3. Interdependency Completeness.	To validate the existence and completeness of all concrete classes and mandatory attributes, and possibly interdependency on datasheet(s).	Completeness of all PowerTransformer objects and mandatory attributes.
2. Association Completeness and Allocation Correctness.	2.1. Associations Completeness. 2.2. Associations Direction. 2.3. Associations Correctness.	To validate the correctness of 1) associations between classes and 2) equipment allocations.	Association between PowerTransformer objects and PowerTransformerEnd objects, where the right transformer are allocated to the right substation.
3. Values Correctness and Accuracy.	3.1. Numerical Values Accuracy. 3.2. Datatypes Correctness. 3.3. Multiplicity Correctness. 3.4. Duplicate Contradicting Definitions	To validate the correctness of values in terms of 1) numerical value and 2) datatype and multiplicity; and to avoid multiple contradicting definitions, e.g. true/false.	TransformerEnd.grounded is Boolean and should be false; or TransformerEnd.ratedS is float and should be in Volt. <namespace:ReferencingClass.ReferencedClass rdf:resource="#rdf:ID of ReferencedClass"/>

3.5 Data Loading

This section describes a high-level process of *Data Loading* as a step in the CIM-based NMM implementation. This process is shown in Figure 21, where not only the process is depicted but also a definition for terminology is given. Below is a description of this process and associated terminologies.

- **Preparing Data Product for Consuming Process:** the starting point where the data product in question is prepared for a consuming process. In this case, this step is for the preparation of the THM data product to be served to the NMM solution (or tool) as the consuming process.
- **Data Product:** a data product that includes the dataset (e.g., THM CIMXML), Schema (e.g., THM RDFS), and possibly, Validation Rules (e.g., THM SHACL) and Validation Results. The validation rules and results are outputs of *Data Validation*, and hence the dependency. The TSO may choose to not deploy a validation mechanism as a step prior to *Data Loading*. Instead, the TSO may for example choose to do the validation on the side of the consuming process; in this case, the NMM solution may be capable of validate the THM data.
- **Serving Data Product for Consuming Process:** the final action in *Data Loading* where the data product is *served* to the consuming process. *Served* may refer to the storage and make available of the data product on a particular ecosystem, loading into a particular system, etc. In this case of establishing a CIM-based NMM, *Served* could refer to the actual loading into the NMM solution. This might also include triggering the (internal) *Data Validation* in the NMM solution.

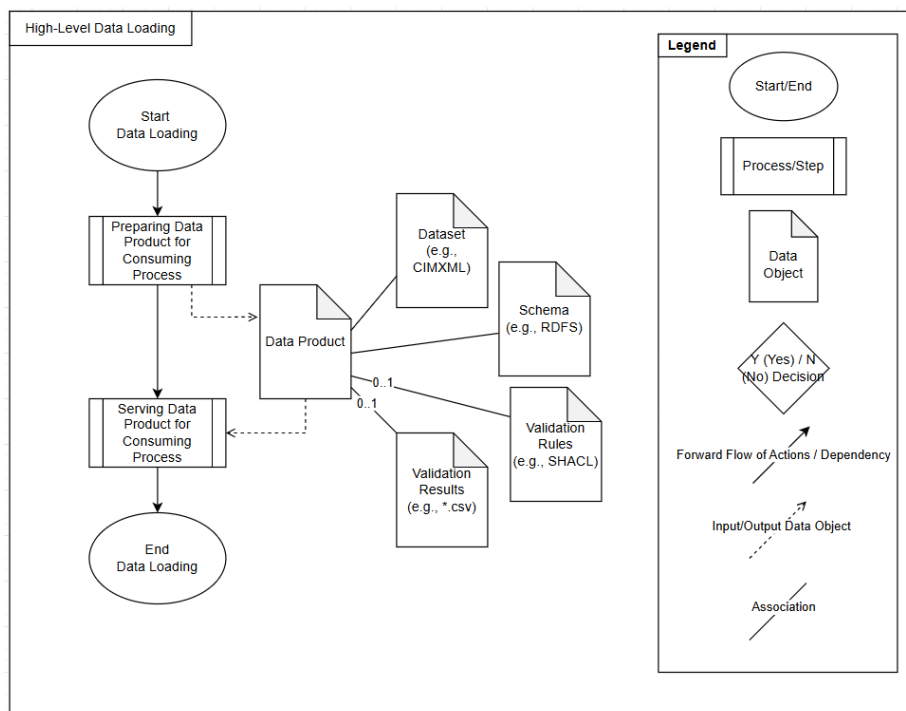


Figure 21: High-Level Data Loading Process.

4 NMM Solution Requirements

4.1 Introduction

In this chapter, the CIM-based NMM solution requirements are specified. This chapter is derived from the requirements as defined in [30] as well as the “Gilde TSOs Requirements for NMM solutions”. The latter is quoted in Appendix C: A multi-TSO proposal for Requirements for Network Model Management (NMM) Solutions enabled by CIM / CGMES [1]. Swissgrid and Statnet started an initiative in 2018 to collect the requirements for the NMM tools, as seen from the TSO perspective, and this work by Swissgrid, Statnet and Svenska kraftnät resulted in a list of requirements that assume CIM / CGMES as a primary vector for semantic and data exchange, and has been shared with the TSO and DSO community. It is essentially a list of principal requirements to illustrate the set of capabilities a commercial CIM-based NMM solution should have.

This chapter aim is to explain the reasoning behind specific requirements that the NMM must have to reach the proposed benefits. Upon the assessment of the situation in the TSO pondering the implementation of the NMM, using the such as [2] [30] and the document quoted in CC, the tendering documentation for the purchase and implementation of the NMM solution can be derived. With that in mind, to make this specification easier, the requirements below are formulated so that their importance reflects their relevance to the tendering process.

When a utility decides to implement the NMM, the path is not entirely straightforward, due to the impact the NMM has on the network model related *processes*, both internal and external. The NMM can only become a relevant source of network model truth if that is respected in the organizational sense. It is chiefly for this reason that the implementation will occur on a different pace each of the utilities - as the starting implementation points and process ownerships may differ. However, consistently and clearly involving the participants in the NMM process is an essential step to a successful NMM implementation and this step shouldn't be underestimated.

We expect the reader to have a solid base of understanding for the required NMM capabilities: for instance, why the *shall* requirements are linked to very significant functionalities of the NMM and not having these would notably reduce the potential benefits. On the other hand, there are features requiring a certain degree of organization readiness that might be deemed as “nice to have”.

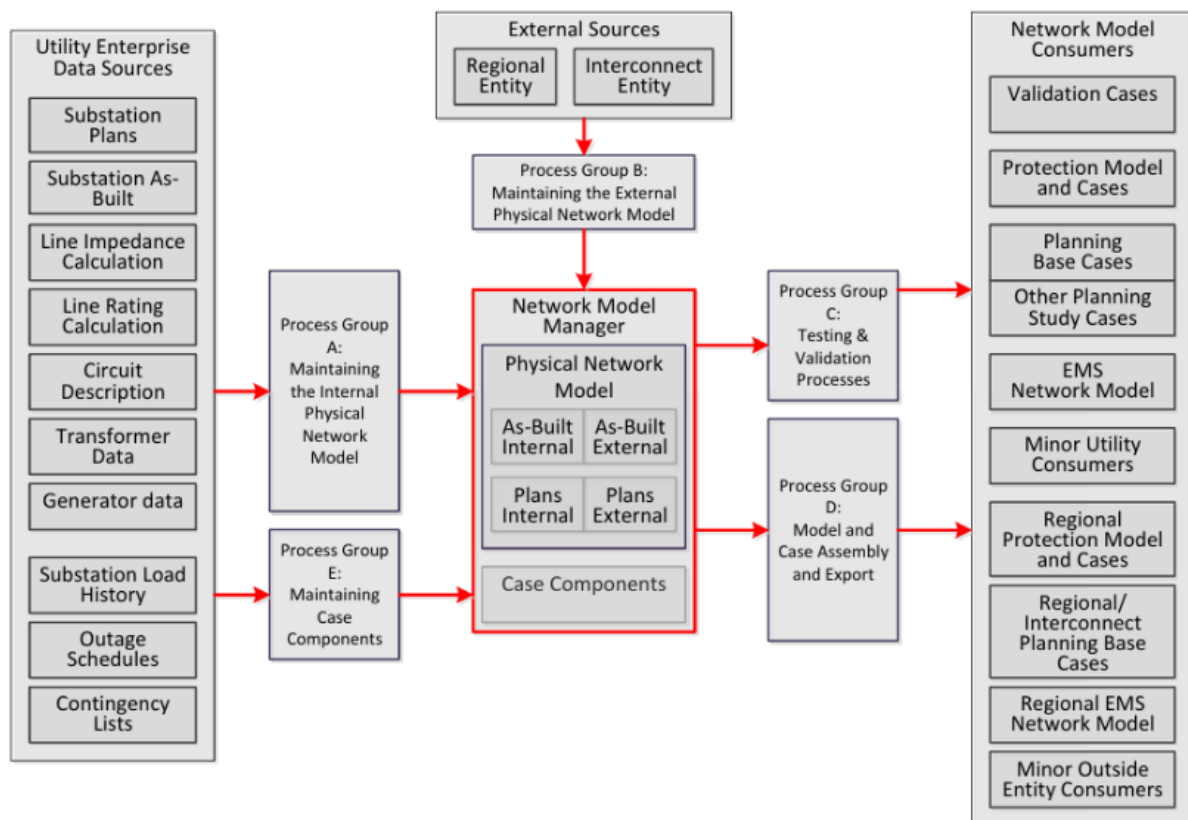


Figure 22: Role of the NMM Tool [30]

4.2 High-Level NMM Tools Capabilities

Here, a set of high-level functional requirements for the NMM tool is provided first. A commercial CIM-based NMM solution should capably address the following high-level requirements, coming from the business context of using the NMM.

Consistent, reliable and secure internal storage of physical network model

The NMM solution should function as a reliable storage of the physical network model. Since it is designed as *the* source of truth considering network model data, it should offer reliable features. If the NMM is implemented as the sole source of network model data, if it fails with no failover solution, all tasks requiring the NMM data will also fail.

Commercial NMM solutions take different approaches on model data storage, in most cases relying on commercial database solutions for model data handling. However, by enforcing proper access control, the NMM should *only* allow editing of the network model data via its model handling procedures. In other words, even if the underlying storage uses a commercial database, manual editing not respecting the model handling procedures should not be allowed.

Any amendments to the physical model should be traceable, and it should not invisibly change the model. For example, an edit to physical network model information should create a new version

of the network model explicitly, so it is clear for the network model consumers which version of the model has been used in the studies. This approach ensures traceability – it is possible to retrace the modelling artifacts for the consuming processes and replicate (replay) the process. This is a key value proposition of an NMM.

Handling the network model updates in a consistent and traceable manner

This requirement further reiterates that the NMM shall work as a trustworthy data storage for the physical network. For this reason, any updates to the model should be traceable, easily identifiable, and the NMM should allow the capability to trigger actions based on the updates.

This is required as the model changes should trigger updates to the models derived for the model consumers. For instance, derating of a line directly impacts any of the power flow and dynamic simulation studies involving that network element.

For instance, NMM automatic trigger could then signalize that the existing studies do not refer to the currently valid network model data anymore. While this automatic triggering can be seen as “nice to have” feature, handling the model updates in a traceable form for model consumers is a must-have feature. It must be implemented in a clear and consistent manner, not only for the NMM internally, but also for processes external to the NMM. It should be clear which model change triggered the model data update.

In other words, integrating the model update management is an essential requirement for the NMM tool.

Partial information provenance depending on the business context – Object Registry Requirements

The NMM solution shall support the object registry services so that the network modelling objects have consistent naming and data referencing across different business contexts. This means that the NMM should consistently and accurately handle the data semantics so that data related to a single canonical object is clearly identified in the object registry and named accordingly. This eliminates the erroneous or duplicate naming problems and the possible splitting of different partial sources of truth about the same object.

Here, a particularly challenging issue is to establish clear identifier governance: how exactly the identifiers (mRIDs and others) are issued, how the legacy identifiers that may be needed elsewhere are handled, how the naming authorities and object matching decisions are handled during harmonisation. The NMM should support traceable resolution of conflicting or missing identifiers to keep the traceability and auditability of the harmonized model.

Multi-user, multi-context capabilities, role-based access control

The NMM should not require the full model to be locked for editing as multiple business contexts happen concurrently. An NMM satisfying the concurrency requirement should therefore allow multiple concurrent views of the model. To control and track changes properly, this requires the

capability of partial locking of the model segments while these are being edited so editing conflicts and duplications are avoided and consistency. In turn, this requirement implicitly requires different contexts for different model data consumers and a close related requirement is the role-based access control and allowing the proper role-based assignment of privileges. The NMM system must support roles such as model users, model developers, model managers and others. When exchanging the data with external actors, the NMM should discern the data that can and should be shared from the data that the external actors should not have access to.

Explicit model branching – Different business contexts, model and case assembly capabilities, project management

Concurrent model usage that results in a single assembled model is generally not enough – the NMM should also support *explicit* “branching” of the model that ties into different model contexts. Similarly, the key NMM role is coordinating and composing the model updates from different stakeholders. To support creation, approval, revising and assembly of the network models for different contexts, the NMM should support explicit “branching” of its network model, i.e. it should be able to define what parts of the master model are accessible to which consuming internal or external business processes. In the CIM context this closely relates to CIM profiles.

The NMM should allow controlled evolution of the model via the project management features. The project is a semantic aggregation of the edits to the network model in a human-understandable semantic container - linked to a certain real-life project. A well-desired feature is to be able to do explicit comparisons of prospective model variants and scenarios, compared to the current state.

Explicit branching also permits the so-called *staging* of model updates. The same network model can exist in its official state and in the draft edited state prior to approval. Its main official state only becomes updated after the approval process. The NMM should permit the definition of the approval triggering process as well as match these to the actual roles in the role-based access control. To support this, the NMM should provide mechanisms for model exploration, browsing, comparison, visualization and navigation across model versions. Note that these functions should also cover the model versions failing the validation tests.

Flexible handling of temporal contexts and “implicit model branching”

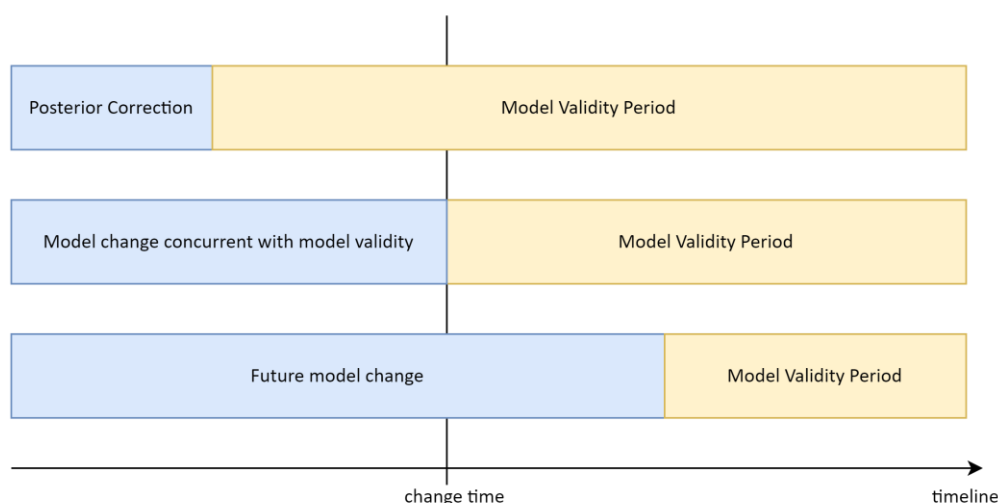


Figure 23: Model changes and model validity

The NMM should offer a mechanism to support all the following three cases for the model change temporal instant and model validity period:

- **Post-model-correction:** there is a model change that occurred in the past but went undetected so the awareness only happened later; this implies a period where the actual state and the modelled state diverged and this is from the actual change until the post-correction.
- **Concurrent change and model validity:** a model is valid immediately and right when committed change occurs.
- **Planned (future) model change:** this is typical for e.g. expansion planning context or scheduled maintenance; a model change that is planned for the future is submitted to the NMM and is inactive until the specified validity instant in the future.

In other words, supporting that the model editing and model validity do not have to coincide temporally is an essential feature of the NMM tool.

Modelling, Validation, Naming, Unique ID provisioning, Lifecycle Management

The utility managing the network normally relies on established modelling practices. This, for instance, can relate to naming schemes for network assets. With the NMM being central to managing the network model, it should allow the modelling users (in line with their assigned roles) to enforce these constraints in the validation phase.

This closely relates to universally unique ID provisioning: to ensure object uniqueness and avoid unwanted duplications, the NMM should be able to expose its unique object IDs to the external applications (to consumer processes) and include capabilities of mapping and matching of the external names to internal IDs.

Model validation functionality is crucial for the basic NMM functions to work. The NMM should be able to validate the modelling data: at the time of ingestion (from different sources); on model modification; and on export (according to the context profile), as well as in custom contexts if desired by the utility.

This validation functionality is multi-layered so the NMM validation must cover different levels of abstraction: syntax and format validation, basic schema validation and consistency checks, as well as higher-level analyses such as topological analyses and power flows. These validation functionalities should allow automation, triggered execution as well as execution on demand. Error handling and logging as well as the definition of amendment processes when the model update is invalid, so the modelling cycle is faster and controlled is needed as well. Additional rules should be enforceable by the utility to go beyond the rules prescribed by the CIM format validity – internal rules by the TSO may imply tighter restrictions. Finally, the NMM model validation should be a customizable feature. For a CIM-based NMM, handling of the ENTSO-E application artifacts updates is important as well. For instance, upgrades to the NMM validation rules once updated SHACL validation rules are issued by ENTSO-E should be possible.

Integration with multiple internal and external data sources (“partial sources of truth”)

This is a set of features as illustrated in Figure 20 – the NMM should be able to import and export data from different data sources. A CIM-based NMM should base these interfaces primarily on the importing and exporting of IEC CIM-data, CIM Profiles and/or extensions to CIM Profiles into the NMM. This feature should allow automation so that it integrates easily into existing business processes without adding additional manual steps.

Integration with additional internal and external network model consumers

As a central point of reference for network data, the NMM must allow adding new model data consumers to its configuration easily. Adding the additional business context integrations should not be a complicated process as this way the NMM maintains its viability and avoids the “temporary workaround” traps that would reduce its adoption.

Other user interface and enterprise features

Given that one of crucial challenges in NMM implementation is its adoption, this class of requirements is self-explanatory as the NMM should offer good performance, usability, documentation, security features, integration with enterprise features such as directories and other features required from all software solutions installed in modern enterprise environments.

4.3 CIM-based NMM Use Cases

For a CIM-based NMM, the high-level requirements indicated in the previous chapter can be formulated as a set of hands-on CIM-based test use cases – a series of typical tasks required from the NMM. There are six generic NMM requirements, described in more detail in C Annex C. A use case (UC) consists of initial preconditions, the tasks to be performed and the validation criteria. Here, these use cases are matched with the requirements above. In addition to the six use cases

coming from the document ‘Gilde TSO’ in Annex C, several additional use cases are provided, addressing more specific requirements coming from practical experiences of adopting the NMM.

The reader of this document deciding on NMM implementation is advised to base the tendering process on similar use cases, adopted to their own company context.

4.3.1 UC1: CIM import and export roundtrip

This use case verifies that the NMM tool preserves all the properties of a CIM model. It starts from a node-breaker test model, either one assembled by the utility or sourced by ENTSO-E from the conformance tests, and performs a round-trip:

- Importing of the supplied CIM EQ (equipment) file.
- Importing of the supplied CIM SSH (steady state hypothesis) file.
- Browsing the imported results.
- Computing the topology and load flow from the imported model.
- Exporting the model’s EQ, SSH, TP and SV profiles.
- Comparison of the results using an external tool such as CIMDesk.

This confirms that the NMM is capable of properly handling the CIM model and that it doesn’t introduce unwanted alterations. Special care should be paid to the mRID identifiers that are designed to be immutable – for this reason the list of high-level requirements stipulates that the NMM should pay specific care to universally unique IDs of network objects. The model structure should remain the same as in the input files, and the EQ and SSH model parameters should also remain the same as in the inputs.

A derivative of this use case is making the input data deliberately invalid to test the capability of the NMM to detect errors - at the syntax level and at higher abstraction levels. This use case (and similar derivatives) confirm that the NMM handles the CIM data properly.

4.3.2 UC2: CIM import of an increment and model assembly

This use case confirms that the NMM tool can import a diff-file of an incremental model update. Specific attention should be paid to an update that causes conflicts and requires intervention in order for the model to become valid.

This use case confirms that the NMM performs the roles of model composition properly, and that it can identify the model updates that cause the model to become invalid.

The preconditions are an originally valid and already imported and established test model within the NMM, and a change that makes the model invalid and invalidates its diagrams. The user should be able to inspect the changes and identify the “merge conflicts” and update the affected diagrams. Subsequently, on successful diff merge, the complete target model should be composed

by the NMM and all the validation rules specified should complete successfully, generating a successful report.

Finally, as in the case above, the assembled model could be exported as a collection of CIM profiles and the validity of the process verified by using an external tool. An additional verification step is to confirm in the NMM interface that the history of modifications is kept, and that all the changes are properly tracked. A close derivative is to create user roles so that the validation step requires approval. The user importing must request approval from another user that can approve the modification.

Within this use case, the NMM capability of interpreting the difference model as well as handling and maintaining the model consistency when facing changes and updates is tested.

4.3.3 UC3: CIM incremental and historical model export

In this use-case the NMM capability of handling incremental exports and producing an audit trail is tested. This case builds on the previous use case, and it is testing the concurrent editing and approval capabilities directly.

The precondition is to have a model A as the initial model state, model changesets B and C to be applied subsequently so model A becomes model B, and then model B change to become model C and finally the model D that should be applied to model A.

The changes should not be applied in order: the use case requires the users to apply and authorize the change by applying the diff models D, B and C, so that the NMM creates all the implicit versions of the model.

This UC is validated if all the incremental exports are easy to identify and make, and if all the changes are properly traceable. All steps of the historical model should be possible to be exported. Additionally, one should check if the correct version of the model is available between the import and the approval – if the approval step is not done, the NMM should show the previous approved state.

4.3.4 UC4: Power System Projects and exporting the future network model

In this use case, the idea is to test the aggregation of model changes into projects and the ability of the NMM solution to handle gradual approval and status change of the projects.

A project such as network upgrades, installation of new equipment or connection of new power plants to the grid is related to the establishment of new entities and altering the parameters of multiple network model elements. This use case tests the NMM capability of handling the aggregation of model changes into projects and the approval at the same time, and finally the NMM capability of handling future model validity as indicated in previous chapter. The planned project only becomes valid in the planning business context, while in the operational context (or model “branches”) it should only become a part of the model post the commissioning date.

This use case requires at least three users: two authorized modelers and a model manager with rights to approve the model changes.

Prior to the approval, the project and the corresponding model updates are only in the draft status and should be visible only to the original modeler user. Post-approval and prior to commissioning, the changes should also be visible only in the planning context. Post-commissioning, the changes should be visible both in planning and operational contexts. The project status and the corresponding model updates should be easily manageable for the involved users. More details are provided in Appendix C.

4.3.5 UC5: Creating and managing power system hierarchies

Power system projects are generally organized in a hierarchical manner. The NMM should support this and handle the project dependencies well and create the interim models correctly and trigger the requisite approvals properly.

This use case is an extrapolation of the prior use case with several power system projects depending on each other. The provided example considers the projects P1: a new line and a substation with arbitrary bay configuration, P2 a wind farm to be commissioned at a later stage, and finally the project P3 as another line to be commissioned later.

This use case tests the capability of the NMM to handle different contexts, project splitting, and how the project approvals affect the actual changesets of the network model, including alterations to the commissioning dates which is a common occurrence with power system projects. The NMM should correctly identify all the interim steps, and handle the element IDs properly, as well as revoke the previous changes.

The NMM passes the test if it is possible to split a project into separate groups of projects, if it is easy to handle commissioning stages of the project, if the approval interface is clear and intuitive, and if the dependencies of the projects (and underlying model changes) are correctly identified and the users alerted correctly. An additional feature to be tested in these steps is whether the NMM supports the assignment of groups of roles to be alerted for each of the mentioned steps.

4.3.6 UC6: Revisiting and expanding the model and its boundary parts

In this use case, the capability of the NMM tool of handling multiple models in parallel and adjusting the boundaries between related models is tested. The involved procedure calls for four models: a model to be expanded, one to be reduced, a boundary model and a reduced model corresponding to the model being reduced.

This use case requires the users to load the original models into the same workspace and once the elements are moved, the NMM should perform validation and object attribute updates correctly. The approval procedure triggers should work correctly as well for both of the original models involved, including model update rejections. An NMM tool is validated if it allows moving the model elements from one model to another and can manage the boundary model updates correctly and smoothly. Approval, rejection and resubmission procedure is also tested in this UC.

Up to UC6, the use cases follow the specification of principal use cases as defined in the Gilde document quoted in Annex C. The following use cases elaborate additional use cases, coming from lessons learned in the implementations of NMM.

4.3.7 UC7: Multi-User Role-Based Access Control, Concurrent Editing, Partial Locking and Model Tampering Protection

This use case singles out the case common for any larger production NMM environment where multiple users with different responsibilities must be able to operate on the model simultaneously.

The NMM must enforce clear boundaries defining what each role is permitted to do and prevent any unauthorized modifications. Similarly, the NMM should implement the approval process for model changes to be committed by the user having appropriate level rights. Additionally, concurrent edits by different users operating on the same model should not produce any conflicts, data loss or untracked changes. For this case, a read only user with no write access, two authorized modelling users working on separate model segments are required, and the model administrator tasked with approvals. To pass the use case, the users only having read-only access should be blocked from making edits with explicit and clear feedback messages. Both users that can write to the model should be able to perform their edits on their separate model extents simultaneously, however the model segment checked out for editing by one user should not be available for editing to another users. The edits pending approval should also not be visible for any consuming processes prior to administrator approval.

In full production mode, the NMM should implement measures to avoid tampering with the internal database structures, avoiding the normal NMM-sanctioned procedures. There are several underlying technological possibilities for this – here the critical feature being tested with this use case is that the NMM can be trusted to maintain and verify the integrity of its underlying data and detect unwanted tampering, ensuring thus the validity of its audit trail. The NMM should deny access explicitly and log unauthorized access, allow partial model editing and lock the model segments correctly, and the audit trail should capture all user actions with corresponding timestamps and finally the NMM should implement a mechanism to detect “out of band” tampering (such as direct edits to the database).

4.3.8 UC8: Object Registry and traceable mRID consistency on merge

There are multiple sources of partial data for the same physical assets. Either the same physical object may appear under different identifiers (false duplication), or the same identifier may carry conflicting attribute values in different data sources. The NMM must implement a resolution process for such situations and not silently accept and corrupt the model with ambiguous data.

In this use case, conflict detection capabilities are tested using two independently maintained and sourced model datasets with three deliberately introduced conflict scenarios: a physical asset appearing in both models under different mRIDs, the same mRID appearing in both models with conflicting attribute values, and assets present only in one model (with no conflict). The test examines whether the NMM surfaces all three conflict types upon import, presents sufficient information for the user to make an informed resolution decision, correctly persists the resolved

mRID mappings as aliases in the object registry, and propagates those mappings consistently across all associations and terminal connections in the merged model.

The NMM passes this test if all conflicts are detected automatically without manual comparison, if each conflict is presented with the identifiers, attribute values, and source datasets in dispute, if the merged model export contains exactly one object per physical asset with no duplicate mRIDs, and if the conflict resolution log is fully auditable and exportable. Non-conflicting assets from both source models must be preserved without modification.

4.3.9 UC9: Validation rule customization, layered validation execution and model fixing process

The NMM should be able to handle additional modelling rules beyond CIM and CGMES standards. The utility installing the NMM could implement additional conventions and restrictions on model data validity such as naming schemes, parameter plausibility thresholds, approval flags, and business-specific constraints.

The NMM should support all steps of the validation stack (from base file syntax to power flow calculations) and allow utilities to implement and enforce additional model handling rules at all levels of the stack, tightening the validation ruleset beyond the standard validation set. Additionally, the NMM should report the errors detected in a user-friendly format. To allow fixing the modelling errors, the model datasets containing errors should be exportable to external tools. When explicitly requested – so, not in the normal validation path, exporting the model data should be possible even for the models containing errors.

In this use case, the NMM's validation framework is tested using two prepared datasets. The first dataset passes all standard CGMES syntax and schema validation but violates two utility-defined rules: a specific naming convention on substation identifiers, and a parameter plausibility rule on transformer rated power. The second dataset additionally contains a topological inconsistency and a power flow divergence condition. The test examines the NMM behaviour in the model validation, whether custom rule violations are reported correctly and with sufficient detail to locate and correct the affected objects. The NMM should allow activation and deactivation of custom rules depending on the business context - individual rules can be activated and deactivated, without affecting the behaviour of other layers or the standard validation stack.

The NMM passes this test if custom rules can be defined through a supported configuration interface without vendor involvement, if each layer reports its results independently, if the validation report is exportable as a structured machine-readable artefact, and if the behaviour is consistent and predictable. The topology and power flow layers must provide diagnostic information specific enough to locate the source of each failure.

4.3.10 UC10: Handling of temporal model changes: post-correction and future validity

The required model changes do not necessarily coincide in time with the moment they are introduced. This is valid for modelling errors where an erroneous property may be incorrectly stored for a long period before being detected, but it is also valid for planned changes. In other

words, the NMM must handle changes coinciding with the model edit time and the changes having effect both earlier and later in time.

In case of retroactive (post-festum) change, the NMM should be able to assign a model correction to the correct historical timestamp and avoid “rewriting the history”. During a certain period, there has been an incorrect version of the model, but the error has not surfaced yet – and this should remain in the model audit trail of the NMM. This way, the integrity of applied changes is kept: it is correctly identifiable that the correction has been committed at a later stage.

To test this capability, by using three different model versions (V1 – with error, V2 – corrected, and V3), it should be tested whether the NMM allows navigation to the relevant historical model states, without overwriting the current (most up to date state). It should be tested whether a retroactive changeset can be assigned to a past effective timestamp and whether the NMM correctly identifies all downstream changes – at the time of retroactive change insertion. It is also tested whether the reconstructed history correctly reflects the actual history, for all three model versions, while leaving all other committed changes untouched. The test is passed if the retroactive changeset is cleanly identifiable and separable from regular changesets in the audit trail, if the impact identification is automatic and complete, if the reconstructed model versions produce consistent power flow results confirming parametric integrity, and if corrected exports are clearly flagged as revisions of previously issued datasets with references to the originals. The audit trail must be unambiguous about what was corrected, when, by whom, and which versions were affected. The analogous test can be performed for the changes committed to become valid at a later stage.

4.3.11 UC11: Export to a non-CIM consuming process / profile-constrained export

Different consuming processes require different subsets of the master model: a Regional Coordination Centre (RCC) needs transmission-level equipment in NCP format, a planning tool needs a bus-branch reduced model, and an internal analysis process needs dynamic and short-circuit data for a defined sub-area. This use case tests further the NMM capability of extracting a subset of the master model. The NMM must be able to assemble and serialize the subsets by applying the correct profile constraints, voltage or geographical filters, and model reduction logic, without requiring manual curation of the exported model subsets.

Testing for this use case will depend on the actual consuming processes in the TSO, but an example is provided here with three distinct consumer configurations. The first is a Regional Coordination Centre requiring NCP EQ and ER profiles filtered to equipment above 220 kV. The second is a planning tool requiring a bus-branch reduced model in CGMES EQ and SSH format covering the full network. The third is an internal offline process requiring EQ, DY, and SC profiles, but only for a defined geographical sub-area.

Each export should contain *exactly* the data defined by the consumer's profile, no more and no less so the consumer process doesn't need to filter the data additionally. The exports should maintain the referential integrity in the exported subset. Boundary objects at voltage thresholds and geographical perimeters should be correctly represented. Finally, the bus-branch reduction should produce a model electrically equivalent to the full node-breaker representation. The reduction logic for the bus-branch conversion can be tested in external tools consuming the export

– by comparing the results to the ones of a full model power flow, it can be verified whether the difference is within tolerance. The NMM passes this test if profile filters are applied consistently and automatically without manual intervention, if all exports are logged in the audit trail with consumer identity, profile version, and source model version, if mRIDs of exported objects are stable and identifiable in the master model, and if the bus-branch conversion satisfies the conversion criteria. An additional capability that can be tested within this use case is whether an incremental re-export after a master model change is delivering only the delta with a reference to the earlier full export.

5 NMM Implementation Governance

This chapter provides guidance for governing the implementation methodology of an NMM as proposed in this document, where a methodology for the pre-NMM phase implementation, while the post-NMM is set for incorporation in a newer version. The reader should refer to ISO/IEC 38500 [31] for a sophisticated framework for such IT implementation.

The approval workflow constraints under IEC 62443, the federated master/replica model and who holds write authority, the change management at the OT/IT boundary, and the separation of NMM configuration governance from OT cyber security operations are considered out of this document's scope.

Figure 24 depicts a high-level process for governing the process of realising a CIM-Based NMM as a single-source for network models, serving internal as well as external consuming processes. This process governs the NMM development across two distinguished phases; pre- and post-NMM:

- **The pre-NMM Phase:** refers to the process during which a TSO is establishing a ready to go-live CIM-based NMM ecosystem. This high-level phase reflects the process described in more details in chapter 3 *An Approach for Realising CIM-Based NMM*. This high-level phase reflects the process described in chapter 3 *An Approach for Realising CIM-Based NMM*, which is treated here as the pre-NMM phase. This phase can be seen as an iterative process for the realisation of a mature “enough” NMM ecosystem for serving consuming processes. The criteria for the maturity level of the NMM ecosystem, and thus the CIM-based data provided to consumers, is defined by the TSO for its internal needs. E.g., a TSO might define “mature enough” for its outage planning process to be as good as an EQ (Equipment) data load together with basic extensions, while pan-European Outage Planning Coordination defines stricter criteria for quality of the exchanged data. Furthermore, the feedback loops in each step of the cycle are to incorporate improvement points as they arise. In addition, a new cycle of the pre-NMM phase could be triggered by the availability of a new version of network model(s), containing new valuable information that is required as part of the CIM-based Totally Harmonised Model (THM).
- **The post-NMM Phase:** refers to the process during which a TSO starts serving CIM-Based data consumers after having established a CIM-based NMM. Servicing consumers undergoes a Shadowrun phase during which the integration of the NMM in the business process is being tested and accepted. The successful demonstration of the NMM Shadowrun results in a full transition from the As-Is to the To-Be situation, where NMM is the single-source of network models for the corresponding process. Note that the post-NMM phase can start for internal processes as soon as the maturity level defined by the TSO is reached. Finally, although this phase is out-of-scope of this document, it is described here to give TSOs some directions for the post-NMM phase.

Furthermore, as seen in Figure 24, the post-NMM phase could be realised in the following order – or any order of the TSO choice for that matter:

- **Serving Internal Offline CIM-Based Data Consumers:** refers to the TSO internal consuming process in the Information Technology (IT) domain, and thus none-critical Operational Technology (OT) systems. This group of consumers entitles the least risks compared to the other groups defined here.
- **Serving Internal Online CIM-Based Data Consumers:** refers to the TSO internal consuming process of critical systems inside the Operational Technology (OT) domain. This group of consumers entitles the highest risks compared to the other groups defined here.
- **Serving External CIM-Based Data Consumers:** refers to the TSO external consuming process, which involves national (e.g., DSOs, SGUs, NRA, etc.) and international (e.g., RCCs, TYNDP, ENTSO-E/ACER, etc.) data exchange.

Moreover, Table 5 lists some internal/external processes and requirements towards NMM.

Table 5: Network Model Data Consuming Processes and Requirements in terms of Structural Data in an NMM Context.

Note: While this table focuses on structural data, most operational and planning processes require the combination of structural (EQ, DY) and scenario-dependent data (SSH, TP) to produce valid simulation results.

Network Model Data Consuming Processes and Requirements	Required Network Model Data	Supporting Network Model Data
Outage Planning (Incl. any TSO specific extensions of internal process)	EQ, ER, AE, RA, CO, SSH, TP	DL, GL, OR, MR
Static Security Analysis (Incl. any TSO specific extensions of internal process)	EQ, ER, AE, RA, CO, SSH, TP	SC, DL, GL, OR, MR, SV
Dynamic Security Analysis (Incl. any TSO specific extensions of internal process)	EQ, DY, ER, AE, RA, CO, SSH, TP	SC, DL, GL, OR, MR, SV
EMS / SCADA (Incl. any TSO specific extensions of internal process)	EQ, ER, AE, RA, CO, TP, SV	SC, DL, GL, OR, MR

Regional Coordination Processes (RCP) [16, 22] (Incl. CSA, OPC, CCC, and STA)	EQ, DY, ER, AE, RA, CO, SSH (critical for e.g. CSA, CCC), TP	SC, DL, GL, OR, MR, SV
Ten-Year Network Development Plan [16, 27]	EQ, DY (conditional only for stability studies), ER, AE, RA, CO, SSH	SC, DL, GL, OR, MR, SV (optional)

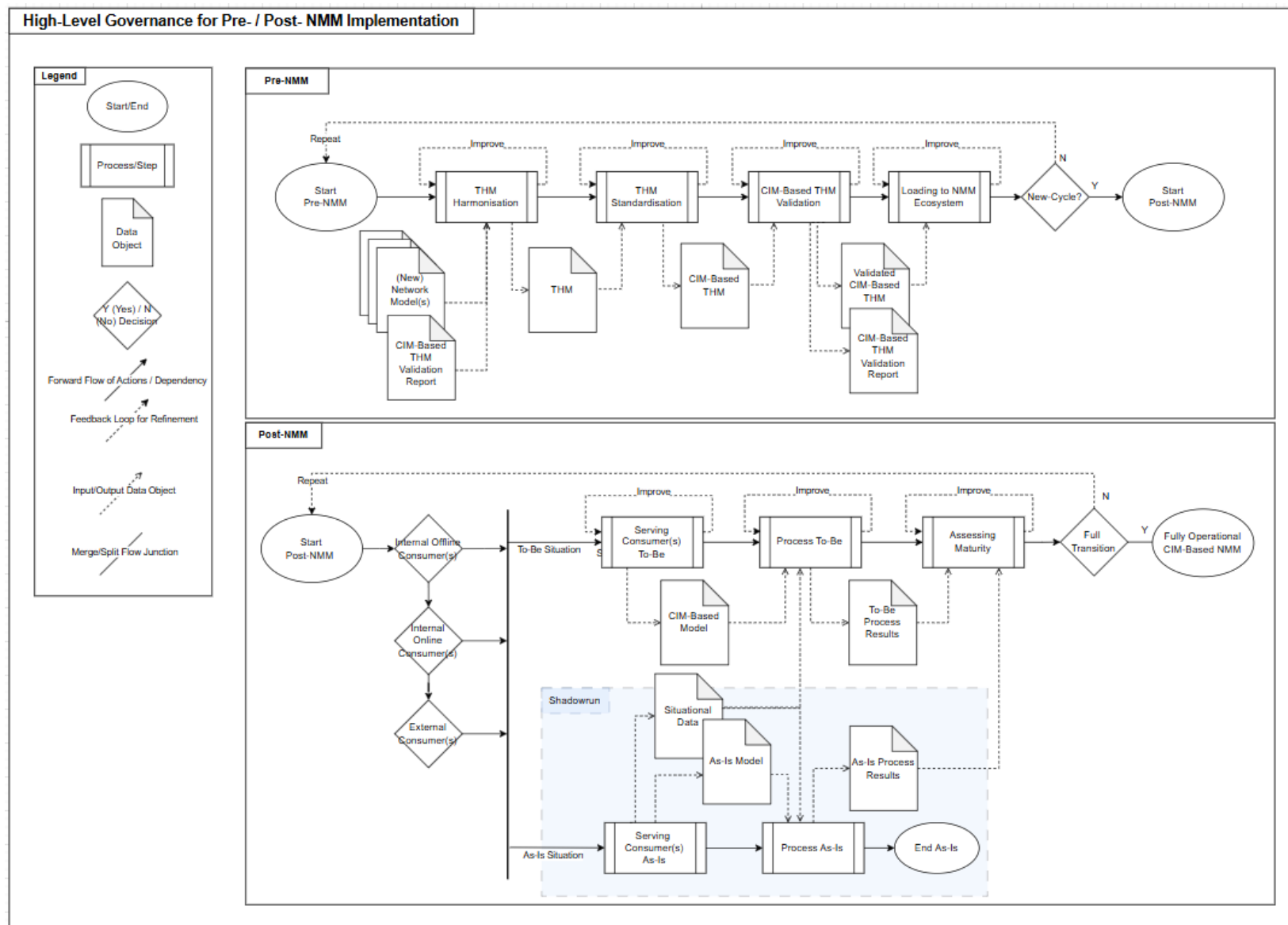


Figure 24: High-Level Governance for pre-/post- NMM Implementation.

6 Challenges, Best Practices, and Future Trends

This section outlines key challenges, best practices, and forward-looking considerations for the evolution of CIM-based Network Model Management (NMM) within the European transmission landscape. As TSOs progress from implementation toward operational maturity, NMM becomes a strategic enabler for interoperability, data quality, and coordinated system operation across organisational and geographical boundaries. At the same time, ongoing digitalisation and the transition toward Control Room of the Future paradigms are reshaping requirements for network model management, including increased integration with real-time data, advanced analytics, and modular system architectures. The perspectives presented in this section aim to support TSOs in aligning their NMM initiatives with these developments while ensuring robustness, scalability, and long-term value.

6.1 Procedural Readiness and Organisational Change Management

Chapter 2 explains the NMM vision and introduces the possible benefits of NMM, briefly indicating one of the key drivers for these benefits: the NMM *adoption* within the TSO organization and at the exchange points with external organizations. This requires a degree of procedural readiness and organizational commitment so that the implemented NMM can take over the designated roles. This (inevitably) requires the changes to the currently established procedures in the core business, and changes of that type and extent are always challenging. One of seminal works on this topic is [32].

In other words, implementing an NMM is not simply a technical incremental upgrade to the systems in the system operator. It is a change and an organisational transformation: a new system becomes *the* source of crucial network data and inevitably organisational change is required. This requires the stakeholders such as decision-makers and perspective users to be onboarded, and it is well known that most change initiatives fail for reasons well beyond the technical scope, *“underestimating the difficulty of driving people from their comfort zones”* [32].

This calls for a systematic approach to the NMM implementation – not only in technical terms but also in organisational and stakeholders’ terms. Clearly establishing the risks for not implementing a CIM-based NMM and ownership is required, as well as a group of internal and/or external stakeholders with authority and credibility to manage the change and inspire the vision is essential. *“When these individuals are not new leaders, great leaders, or change champions, phase one can be a huge challenge”* [32].

Furthermore, the procedural readiness and organisational change management also applicable to whether leveraging CGMES/NCP in the TSO approach is preferred to (Canonical) CIM approach or vice-versa as discussed in section 3.1. These kinds of decisions have a major impact on the stakeholders’ acceptance and support of the adaptation and implementation of a CIM-based NMM. In terms of change management, establishing well-crafted, early wins is important not only

for excited stakeholders who want to achieve envisioned positive results, but also for reluctant stakeholders.

6.2 Best Practices for Meeting European Transmission Network Data Requirements

Studies such as [3] provides insights about the challenges and best practices for European TSOs for meeting the network data requirements of ENTSO-E.

- Fragmentation is the root cause of inconsistencies, errors, delays, and rework in the context of network model data management across EMC, planning, etc. A CIM-based NMM overcome these challenges and lowers the long-term cost of compliance as ENTSO-E and CIM requirements are continuously evolving.
- TSOs who treat CIM/CGMES only as an external export requirement, missing the opportunity to simplify and automate other applicable processes. The paper emphasises that CIM concepts—Models, Change Models, Assemblies, Frames, Boundaries—should shape internal model management, not only for e.g. the required IGM exchange. The CGMES are the best available representation of Europe-wide conditions and therefore re-using them increases consistency and accuracy.
- Automation is emphasised because human-driven case construction and/or building future cases from memory or spreadsheets leads to inconsistencies and missed information.
- Without territorial modularisation (e.g., internal frameworks for (extra-)high-voltage grid, distribution grid, etc.), dependencies across teams become a major bottleneck. To avoid this, design frames and boundaries so that model ownership is clear, modular, and cleanly composable.
- Without automated provenance, debugging incorrect submissions is extremely difficult. Auditability is mandatory—especially with regulatory scrutiny and e.g. IGM timelines.

6.3 The challenge of calculated parameters

Building an internal CIM-based Network Model Management (NMM) ecosystem is a major strategic step for any TSO. While the approach promises long-term efficiency and improved model quality, the journey is complex—particularly when computed or derived parameters must be generated automatically. TSOs that choose to self-develop rather than procure a mature NMM solution can face several categories of challenges. Many required physical parameters—such as line impedances, thermal ratings, etc.—must be computed from raw engineering data rather than directly provided. These transformations are non-trivial and require specialized algorithms and validation workflows.

6.4 NMM in the context of digitalisation the (near-) future control centres

The evolution toward a futureproof control centre, and generally TSOs operations, is supported by a CIM-based NMM that provides a critical foundation for enabling these capabilities.

- Integration of digital twin technologies combining structural and real-time data.
- Increased use of PMU/WAMS data for enhanced observability.
- Adoption of AI-assisted operational decision support.
- Transition toward modular, interoperable system architectures.
- Increased requirements for cybersecurity and resilience.

6.5 NMM within Modular and Vendor-Agnostic Control Room Ecosystems

Control centre environments are expected to transition toward modular ecosystems composed of interoperable services rather than monolithic systems. The Vendor Agnostic Solution for Next-Generation Control Room Eco-Systems [33] emphasises principles such as modularity, transparency, standardisation, integration, and separation of concerns, enabling flexible and vendor-independent system evolution. In this context, NMM shall evolve into:

- A core platform service providing authoritative network model data.
- A shared semantic layer enabling interoperability across modules.
- A stable integration point for both legacy systems and new digital services.

This positioning requires:

- Clear interface definitions (“modular contracts”).
- Standardised data exchange mechanisms (a.o. CIM-based).
- Decoupling of data management from application logic.

A CIM-based NMM can support innovation while maintaining stability and interoperability across the control room ecosystem.

6.6 Other challenges and expected evolution of NMM

The actual path towards full adoption of NMM as the authority for handling the network model data will differ, depending on the state of the TSO at the time of implementation. One of the issues is which system will a TSO start to build the NMM from.

The NMM vision considers the NMM becoming *the* source of the modelling data. Many of the consuming processes may not be ready for this fully. For example, considering the ID consistency, with adopting the mRIDs as key identifiers there might be implementation challenges in mapping the mRIDs to the SCADA point addresses. In this step, additional consuming processes may have to be developed to ensure mappings are kept correctly and consistently – these consuming processes would ensure that the NMM remains the authority for the network model. This is

especially relevant for the EMS functionalities, closely tied to the modern SCADA systems. In other words, to ensure correct adoption of the NMM within the business processes so that the benefits of clear data trail can be utilized, a considerable technical integration work may be required.

The system operators advance towards increasingly digitalized and interconnected system operation. Different business aspects previously considered separate interact more often. In line with ENTSO-E approach towards Vendor Agnostic Solutions, the future control room environments will be characterized by modular, interoperable and resilient architectures going well beyond monolithic EMS/SCADA systems. The Digital Twin paradigm considers the structural network models should be continuously aligned with real-time system state, thus enabling predictive, coordinated, and data-driven system operation [34].

The role of the NMM in such systems is multifaceted and structurally important. The NMM evolves into a core platform service providing authoritative network data, a semantic layer enabling interoperability across systems, and a stable and standards-based integration point for legacy systems and new services. Clear interface definitions (based on standards) and decoupling data management from the application are required for this.

As it becomes the designated handler of the network structural data, the NMM is a structural backbone for the system digital twin as well. In the (near) future, this means the NMM is the glue between, e.g. the operational telemetry identifiers and the system assets that these data refer to. The NMM then becomes *the* provider of authoritative and validated network model data to SCADA and Energy Management System (EMS) environments, by providing the system-structural information that adds network model semantical interpretation to the acquired data. A signal then becomes tied to the relevant context via the structural information given by the NMM. This structural information then provides semantic alignment between telemetry and network topology, and the data enriched with additional semantic information can be used in contexts such as predictive maintenance.

Once NMM becomes the glue to bind the network signals to the underlying network elements and network model, it also enables validation to detect inconsistencies introduced by errors in signal mapping and topology changes. In the subsequent steps, the NMM also becomes the point of reference for signal lifecycle management and ensures accurate bindings of the signals to the network model.

The evolution of CIM-based NMM toward a modular, interoperable, and operationally integrated capability is a key enabler for the future European power system and by aligning with ENTSO-E's Vendor Agnostic Solutions principles and the development of federated Digital Twin ecosystems, NMM can become a central component of next-generation control room environments.

This Implementation Guide is designed to help the adoption of the NMM, and the future iterations of this Implementation Guide will build upon this version and progressively address the topics indicated above, supporting TSOs in transitioning from foundational implementation toward fully integrated, resilient, and data-driven system operation.

7 List of acronyms and abbreviations

AE – Asset Equipment

AM – Asset Management

API – Application Programming Interface

CCC – Coordinated Capacity Calculation

CGM – Common Grid Model

CGMES – Common Grid Model Exchange Standard

CIM – Common Information Model

CIM WG – CIM Working Group

CIMXML – CIM XML serialisation

CO – Connectivity

CSA – Coordinated Security Analysis

CSV – Comma-Separated Values

DL – Diagram Layout

DSA – Dynamic Security Assessment

DSO – Distribution System Operator

DY – Dynamics

EA – Enterprise Architect

EMS – Energy Management System

ENTSO-E – European Network of Transmission System Operators for Electricity

EQ – Equipment

ER – Equipment Relations

ETL – Extract-Transform-Load

GL – Geographical Location

IEC – International Electrotechnical Commission

IGM – Individual Grid Model

JSON-LD – JavaScript Object Notation for Linked Data

mRID – Master Resource Identifier

N/A – Not Applicable

NCP – Network Code Profile

NMM – Network Model Management

OPC – Outage Planning Coordination

OR – Object Registry

OT – Operational Technology

PSP – Power System Project

QoCDC – Quality of CGMES Datasets and Calculations

RA – Remedial Action

RCC – Regional Coordination Centre

RCP – Regional Coordination Process

RDF – Resource Description Framework

RDFS – RDF Schema

SC – Short-Circuit

SCADA – Supervisory Control and Data Acquisition

SHACL – Shapes Constraint Language

SSH – Steady-State Hypothesis

STA – Short-Term Adequacy

SV – State Variables

THM – Totally Harmonised Model

TP – Topology

TSO – Transmission System Operator

TYNDP – Ten-Year Network Development Plan

UC – Use Case

UCAIug – Utility Communication Architecture International Users Group

UML – Unified Modeling Language

VAS – Vendor Agnostic Solutions

XML – Extensible Markup Language

8 Glossary

The following glossary defines terms and concepts brought in this Implementation Guide. The descriptions provided combine ENTSO-E, EPRI and IEC definitions.

Term	Definition
Asset Management	Business function concerned with the lifecycle management of physical assets, including planning, operation, maintenance, and decommissioning, and their representation in information systems.
Boundary Data	Power system elements considered common border objects between adjacent modelling authorities and jointly governed to ensure consistent assembly of interconnected network models.
Canonical Common Information Model	The complete, technology-neutral information model defined by the IEC CIM standards, serving as the authoritative semantic reference from which application profiles are derived.
Common Data	Structured data that is shared and jointly governed by all participants within a given exchange context, defined once with unique identifiers and agreed attribute values to prevent duplication and inconsistency.
Common Grid Model	Union-wide power system dataset assembled from individual network models and used for coordinated operational, security, adequacy, and planning studies.
Data Loading	The process of making validated datasets available to a consuming system or process, including storage, registration, and optional triggering of downstream processing or validation.
Data Product	A packaged delivery of network model data intended for a specific consuming process, typically consisting of instance data, schemas, validation rules, and validation results.
Data Serialisation	Transformation of a conceptual or logical data model into a concrete machine-readable dataset using a defined syntax and format, such as CIMXML.
Data Validation	Process of verifying that a dataset complies with defined syntactic, structural, semantic, and business rules before use or exchange.
Extract-Transform-Load	Data processing pattern in which data is extracted from source systems, transformed into a standardised representation, and loaded into a target system or repository.
Individual Grid Model	Network model representing the electrical system of a single Transmission System Operator, serving as an input to coordinated or aggregated models.

Master Resource Identifier	Globally unique and immutable identifier assigned to an object in a CIM-based system, ensuring persistent identity across versions, contexts, and exchanges.
Network Model	Digital representation of the power system network, including topology, equipment, electrical parameters, and operational attributes, used for planning, analysis, and operation.
Network Model Management	Set of coordinated processes and capabilities for creating, maintaining, harmonising, validating, versioning, and governing network models throughout their lifecycle.
Power System Project	Logical container used to represent planned or future modifications to the power system, including staged evolution and time-dependent validity.
Profile (CIM Profile)	Uniquely defined subset of the Common Information Model specifying the classes, attributes, associations, and constraints required for a specific business context or data exchange.
Reference Data	Relatively stable, shared information defining standardised concepts reused across multiple processes and data exchanges.
Source of Truth	Authoritative system or dataset responsible for maintaining the most reliable and up-to-date representation of specific information within an organisation.
Totally Harmonised Model	Harmonised representation of network model data consolidating multiple source models and contexts, forming the basis for a CIM-based Network Model Management implementation.
Validation Rule	Formal constraint used to assess correctness, completeness, and plausibility of data, expressed using schemas, rule languages, or tooling mechanisms.
Versioning	Controlled handling of successive states of a network model or dataset, enabling traceability, reproducibility, and historical analysis.

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Appendices

A. Appendix A: Leveraging Validation Rules from the QoCDC

This appendix outlines how the validation framework defined in the QoCDC (Quality of the CGMES Dataset and Calculation) v4.1.4 [35] can be leveraged for a CIM-based NMM (Network Model Management) implementation. The QoCDC provides a structured set of validation rules — spanning processability, consistency, interoperability, and plausibility — which set out quality rules and validation guidelines for data exchanged in the context of the CGM-BP (Common Grid Model – Build(-ing) Process) for pan-European system-operations purposes (CSA, CCC, OPC, STA, after-the-fact event analysis, etc.). Applying these rules for the validation of the THM (Totally Harmonised Model) enables a uniform quality assessment layer.

This description is concerned with leveraging the QoCDC validation rules for the THM as the Single-Source-of-Truth for the network model. It is assumed that the derived data products (e.g., EQ) from the THM shall be validated against the QoCDC internally in the NMM ecosystem, hence, the NMM ecosystem should be equipped with the necessary validation module(s) to ensure the quality of the CGMES datasets and calculations before serving consuming processes.

QoCDC Validation Levels (1-8)

The QoCDC defines eight validation levels with defined constraints based on either business requirements or standards (IEC TS 61970-600-1/-2:2017 reflecting CGMES v2.4.15, and IEC TS 61970-600-1/-2:2021 reflecting CGMES v3.0). These validation rules have associated constraints' name, severity (i.e., ERROR, WARNING, or, INFO), and messages displayed upon failing validation rules. Below is a brief description of the QoCDC validation levels, followed by a more detailed explanation of the validation rules that are relevant to NMM-THM in the next section.

Level 1 — FILENAME

“Covers meta data in file names and packaging of CIMXML files. [35]”

Ensures that each file can be technically opened and parsed. Checks include XML well-formedness, valid UTF-8 encoding, correct RDF/XML syntax, and the presence of mandatory namespaces. This is the lowest gate ensuring the file is not corrupt or unreadable.

Level 2 — Schema & Profile Conformance

“Covers the structure and syntax of the individual CIMXML files as well as the meta data header [35]”

Validates that the CIMXML files follow the CGMES profiles (EQ, SSH, TP, SV, etc.), including class validity, mandatory attributes, datatypes, and RDF/XML structural correctness.

Level 3 — Internal Consistency

“Covers constraints that can be evaluated within the scope of the CIMXML files. [35]”

Ensures internal coherence of each dataset: unique rdf:IDs, valid references between objects, correct containment hierarchies (Substation → VoltageLevel → Bay), and complete connectivity structures.

Level 4 — Cross-Profile Consistency

“Covers issues that can be detected during model assembly. [35]”

Checks alignment between CGMES profiles:

- EQ ↔ SSH (operational values)
- EQ ↔ TP (equipment parameters)
- EQ ↔ SV (state variables and topology)

Level 5 — Interoperability & Linking Consistency

“Covers cross profile consistency of data. [35]”

Validates cross-model interoperability rules: global uniqueness of mRIDs, correctness of inter-profile links, naming conventions, and alignment with ENTSO-E’s profile-interaction rules.

Level 6 — Topological & Logical Plausibility

“Covers diagnostic information that may help solve convergence issues by identifying modelling issues that seem troublesome. [35]”

Ensures topological and electrical logic: valid connectivity, realistic switching states, transformer end numbering, and basic network structure plausibility (e.g., KCL-related checks).

Level 7 — Electrical Parameter Plausibility

“Focuses on coordination of IGMs in terms of neighbouring TSOs and reference values. [35]”

Checks physical reasonableness of electrical parameters and model behaviour: rated voltages, impedances, tap positions, generator/load sign conventions, and realistic operational ranges.

Level 8 — Calculation Plausibility (EMF Gate)

“Focuses on convergence behaviour of IGMs and CGMs and on the plausibility of the CGM. [35]”

Assesses plausibility of calculation outputs (e.g., TP or SV results): voltage/angle magnitudes, power flows, currents, tap positions, and consistency with network configuration and constraints.

B. Appendix B: RDF Schema Explained as Example

This appendix provides a general description about the Resource Description Framework Schema (RDFS) based on the IEC 61970-501 [9] and its protentional use for validation. This appendix illustrates how the RDF schemas can support partial validation of CIM datasets, but it does not prescribe the internal data storage or processing architecture of the NMM tool.

As shown in the figure below, an RDFS is generated from a profile that is derived from CIM Canonical Model. The RDFS is then the reference to which the dataset (or instance data) must confirm. For the purpose of this example, the Geographical Location (GL) RDFS is considered as a specific example for simplicity. The example shows how the IEC61968::Common is used to derive the IEC-61968-Part 13 (i.e., the GL profile), which is then used for generating an RDF schema for the GL dataset.

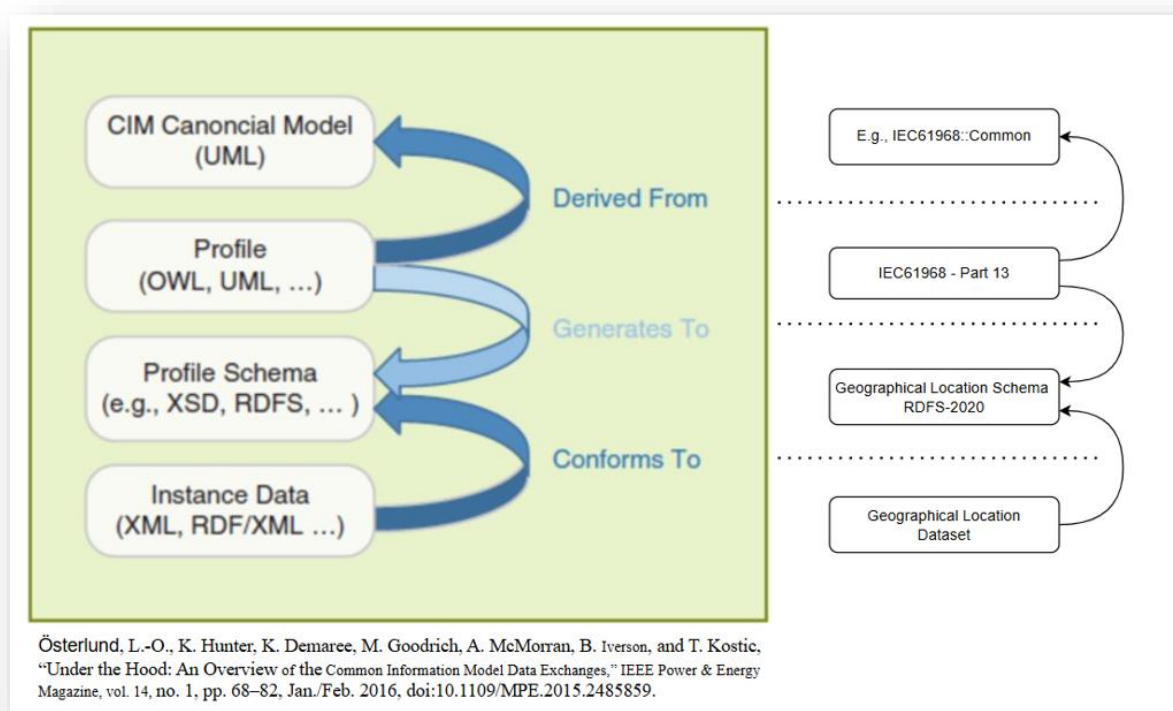


Figure 25: Adapted from [11] for the purpose of illustration in this appendix.

Starting with **CIM Canonical Model**, the example considered is from the IEC 61968 (Common package). Below is a part of the AssetEnvironment UML model, where only Location and PositionPoint classes are shown for the purpose of this appendix. As can be seen, these classes have an association which is stating that PositionPoints must have (“1”) an association to a Location, while a Location may or may not have (“0..*”) an association to PositionPoints. Notice that this association does not specify which object is pointing at the other, hence no direction is defined. However, it is common practice to have the “majority” (PositionPoints with “0..*” in this case) pointing at the “minority” (Location with “1” in this case) [13]. Moreover, the figure below

shows the attributes of each class (e.g., groupNumber in the PositionPoint class). It can be seen that attributes have zero-to-one (i.e., [0..1]) cardinalities, indicating that these attributes are optional ("0") and can have only one value ("1") at maximum.

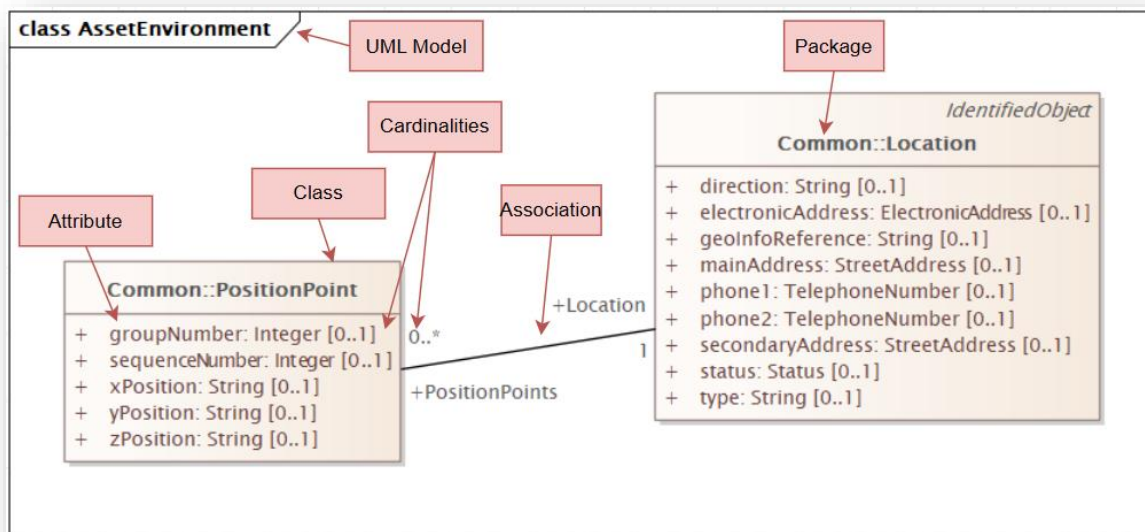


Figure 26: PositionPoint and Location from AssetEnvironment UML Model as an Example. Source: IEC 61968 CIM, adapted by ENTSO-E.

Based on the definition of (among others) the class Location and PositionPoint in CIM Canonical Model, a **Geographical Location (GL) profile** under the **IEC 61968-Part 13** [23] is derived in the context of CGMES (Common Grid Model Exchange Standard). Below is a part of the GL profile where it can be seen how “profiling” is done. In the GL profile, the direction of the association is decided to be from PositionPoint to Location although it is against the common practices. In the GL profile, it can also be seen that only a limited number of attributes is selected relative to those available in CIM Canonical Model. It is also visible that some of the selected attributes in the GL profile are mandatory (e.g., xPosition). Note that mandatory attributes have the cardinality [1..1], but this (mandatory) cardinality is not shown in the UML model. The red-boxes and numbers are used/referred to in the next paragraph.

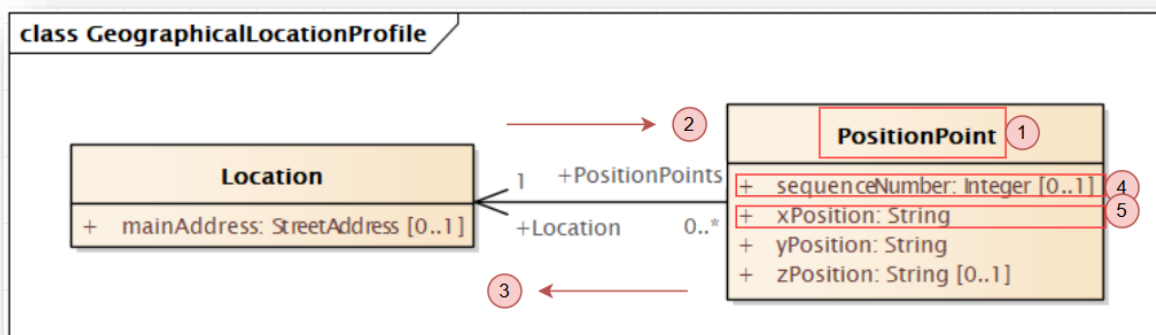


Figure 27: Location and PositionPoint from GeographicalLocationProfile UML Model as an Example. Source: IEC 61968 CIM, adapted by ENTSO-E.

Based on the GL profile, an RDFS can be generated as can be seen in Figure 28. The red boxes are numbered to easily link them back to the GL profile in Figure 27. In this part of GL RDFS which is given as an example, it can be seen that:

- The class PositionPoint (#1) is defined and given label, comment, stereotype, etc.
- The association between Location.PositionPoints (#2) is defined and declared as *not* used.
- The association between PositionPoint.Location (#3) is defined and declared to be used, hence the arrow direction in the GL profile.
- The attribute sequenceNumber (#4) is defined as a property (or attribute) of the class PositionPoint, which is an integer with [0..1] cardinality.
- The attribute xPosition (#5) is defined as another property (or attribute) of the class PositionPoint and given [1..1] cardinality, meaning that this attribute is mandatory in a GL dataset - unlike canonical CIM.

Given this high-level background information, it can be concluded that an RDFS includes valuable information that could be used for validation purposes such as the case in the GL, where:

- cims:AssociationUsed for verifying “pointing” direction of data objects (e.g., Location points at PositionPoint).
- cims:stereotype for verifying the existing of particular data objects (e.g., PositionPoint is concrete, hence must exist).
- cims:multiplicity for verifying the the min/max number of a particular information (e.g., xPosition [1..1] must exists in all PositionPoint data objects).
- cims:datatype for verifying the datatype used (e.g., string, integer, etc.)

On the other hand, what cannot be validated are:

- Multipliers, for example, whether a numerical value is in the unit Volt [V] or kilo Volt [kV].

```

1 <rdf:Description rdf:about="#PositionPoint">
  <rdfs:label xml:lang="en">PositionPoint</rdfs:label>
  <rdfs:comment rdf:datatype="http://www.w3.org/2001/XMLSchema#string">Set of spatial coordinates that
    determine a point, defined in the coordinate system specified in &apos;Location.CoordinateSystem&apos;;.
    Use a single position point instance to describe a point-oriented location. Use a sequence of position
    points to describe a line-oriented object (physical location of non-point oriented objects like cables
    or lines), or area of an object (like a substation or a geographical zone - in this case, have first
    and last position point with the same values).</rdfs:comment>
  <cims:belongsToCategory rdf:resource="http://iec.ch/TC57/ns/CIM/GeographicalLocation-EU#Package_GeographicalLocationProfile"/>
  <cims:stereotype rdf:resource="http://iec.ch/TC57/NonStandard/UML#concrete"/>
  <rdf:type rdf:resource="http://www.w3.org/2000/01/rdf-schema#Class"/>
</rdf:Description>

2 <rdf:Description rdf:about="#Location.PositionPoints">
  <rdfs:label xml:lang="en">PositionPoints</rdfs:label>
  <rdfs:comment rdf:datatype="http://www.w3.org/2001/XMLSchema#string">Sequence of position points describing
    this location, expressed in coordinate system &apos;Location.CoordinateSystem&apos;;.</rdfs:comment>
  <rdfs:domain rdf:resource="#Location"/>
  <rdfs:range rdf:resource="#PositionPoint"/>
  <cims:inverseRoleName rdf:resource="#PositionPoint.Location"/>
  <cims:multiplicity rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#M:0..n"/>
  <cims:AssociationUsed>No</cims:AssociationUsed>
  <rdf:type rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
</rdf:Description>

3 <rdf:Description rdf:about="#PositionPoint.Location">
  <rdfs:label xml:lang="en">Location</rdfs:label>
  <rdfs:comment rdf:parseType="Literal">Location described by this position point.</rdfs:comment>
  <rdfs:domain rdf:resource="#PositionPoint"/>
  <rdfs:range rdf:resource="#Location"/>
  <cims:inverseRoleName rdf:resource="#Location.PositionPoints"/>
  <cims:multiplicity rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#M:1"/>
  <cims:AssociationUsed>Yes</cims:AssociationUsed>
  <rdf:type rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
</rdf:Description>

4 <rdf:Description rdf:about="#PositionPoint.sequenceNumber">
  <cims:stereotype rdf:resource="http://iec.ch/TC57/NonStandard/UML#attribute"/>
  <rdfs:label xml:lang="en">sequenceNumber</rdfs:label>
  <rdfs:domain rdf:resource="#PositionPoint"/>
  <cims:datatype rdf:resource="#Integer"/>
  <cims:multiplicity rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#M:0..1" />
  <rdfs:comment rdf:parseType="Literal">Zero-relative sequence number of this point within a series of points.</rdfs:comment>
  <rdf:type rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
</rdf:Description>

5 <rdf:Description rdf:about="#PositionPoint.xPosition">
  <cims:stereotype rdf:resource="http://iec.ch/TC57/NonStandard/UML#attribute"/>
  <rdfs:label xml:lang="en">xPosition</rdfs:label>
  <rdfs:domain rdf:resource="#PositionPoint"/>
  <cims:datatype rdf:resource="#String"/>
  <cims:multiplicity rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#M:1..1" />
  <rdfs:comment rdf:parseType="Literal">X axis position.</rdfs:comment>
  <rdf:type rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
</rdf:Description>

```

Figure 28: RDF Schema of Part of the GL profile as an Example. Source: IEC 61968 CIM, adapted by ENTSO-E.

This appendix is derived from the Requirements for Network Model Management (NMM) Solutions enabled by CIM / CGMES developed by Swissgrid, Statnett and other TSOs within the Gilde TSOs initiative.

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C. Appendix C: A multi-TSO proposal for Requirements for Network Model Management (NMM) Solutions enabled by CIM / CGMES [1]

C.1 Introduction

In 2018, Swissgrid and Statnett launched an initiative on collecting requirements on Network Modelling and Model Management (NMM) tools. The perspective was clearly from a Transmission System Operation point of view, to support our internal processes, as well as business-to-business processes required for implementation of European Network Codes – for network model exchanges among TSOs (at Pan-European level, through our Regional Security Coordinators) and between TSOs and their DSOs.

The development of NMM owes much to the work by EPRI. In fact, the collection of requirements published here adopted many of the requirements from the EPRI's 2014 report titled “Network Model Manager Technical Market Requirements” as part of the starting point. Another important contribution was made by the findings of the 2016 CGMES Interoperability tests organised by ENTSO-E as part of its Common Grid Model efforts.

After a physical workshop in August 2018, where we exchanged on different perspectives, planned and actual projects and scope of collaboration, SVK colleagues kindly shared their existing requirements list as a starting point. Through over 60 hours of web conferences, accompanied with some off-line work, we managed to produce a list of requirements for such a tool that we believe is worth sharing with the community.

The resulting list of requirements assumes CIM / CGMES support as the primary vector of semantic and data exchange and does not include assumptions about support for other tools or formats. The reasoning behind this is to encourage vendors to provide users with:

- essential features enabled by use of standards,
- incorporate features into product baselines,
- make it easier to undergo conformance testing, and,
- have faster time to market for a large community of users.

Our small team has agreed to already now share the list of requirements with our TSO, DSO, RSC, ENTSO-E colleagues and vendors, as well as standards development organisations (e.g., IEC TC57 WG13, WG14, WG16 and WG19), user groups (e.g., CIMug) and the wider community. The list is certainly not perfect, but it is developed to a stage that allows soliciting feedback and encourage others to contribute.

C.2 Reading guide

In chapter 3 we provide explanation on the (Excel) list with requirements. We distinguished various functional as well as non-functional product features which an NMM solution should have. They are introduced at the start of the chapter. We then explain how we mapped these features to the requirements in the list and we introduce the priority classification. The list of requirements includes a glossary of terms and a list of relevant standards and documents, which are introduced at the end of the chapter.

We would expect that the list of requirements is used in assessments of products. In chapter 4 we provide a suggestion on how compliance with the requirements could be assessed.

In chapter 5 we provide a non-exhaustive set of use cases (UC), i.e. a series of tasks a system user is likely to perform when using the product (or is doing using the existing product). The UC's can be used by readers as context for the requirements. They could equally be used as a proof-of-concept for vendors or for evaluation of tenders.

Finally, chapter 6 contains (a reference to) the Requirements List.

C.3 Main objectives and product features

This section describes in section C.3.1 the main objectives that the product is to fulfil and the expected product features. This is complemented by section C.3.2 that addresses the non-functional features of the product. The product features are mapped to the requirements listed in the Requirements List (see chapter C.6) which could be used as a basis for product functional specification. Section C.3.3 clarifies how features are mapped to requirements and explains the priority classification. Sections C.3.4 and C.3.5 introduce the glossary of terms and list of reference documents that are also included in the Requirements List.

C.3.1 Expected product features

To address the management of objectives related to network model data and network model data management, the NMM system would support various roles allowing for model users, model developers, model managers etc. The system would also enable the electrical network owner to fulfill its responsibility for consolidating model data within its Master Authority MAS. It will also allow electrical Network owner to exchange model data with neighboring DSOs, TSOs and/or regional security coordinators using IEC CIM² /ENTSO-E's CGMES³ format. To achieve this a NMM tool is foreseen to have the following features:

² IEC 61970: Common Information Model (CIM) / Energy Management, www.iec.ch/smartgrid/standards/

³ ENTSO-E's Common Grid Model Exchange Standard, see www.entsoe.eu

NP - Network and Project Management

Network management is central to coordinating model update from various stakeholders. This feature covers functionality and roles for managing network models and related model parts for various planning and operational needs. This includes functionality for creating, approving, revising and assembling network models (in line with various user access rights). Furthermore, Network Model Management shall also support a well-defined model for model data versioning and modification tracking.

The Project Management feature covers functionality and roles to manage and capture the evolution of various preliminary and proposed power system projects. This includes the capability to detail such project over time and to compose different future models and scenarios⁴ based on the combination of several such projects with the models representing the current state of the grid.

MV - Modelling and Validation

Electrical Network owner has adopted and implemented certain modeling practices. Modelling practices are constrained by the tools currently used within their companies and there is a strong need for the NMM to assist in enforcing these constraints. Furthermore, this feature would also support exposing universally unique IDs of objects in the NMM to external applications. Finally, the NMM would also support implementing name and identifier mapping between those exposed by the NMM and external applications.

Model validation functionality is central for supporting testing and validation of the network model data on import, modification and export. Validation functionality includes syntax and format validation, consistency checks, topology analysis and power flow. Furthermore, this functionality shall support automation and definition of custom validation processes that can be executed in batch or on demand.

IF - Interfaces: Import, Export and Automation

This feature covers a set of functionalities that addresses the importing and exporting of IEC CIM-data, CIM Profiles and/or extensions to CIM Profiles into the NMM. This feature does also include the possibility to use as services through a standard API.

The automation feature specifies basic functionality that would provide users with an ability to automate actions and processes in the NMM, for instance using scripts. This feature would also allow users to extend the functionality of the system.

WS – Workspace: Model Assembly, User Interface

⁴ A scenario could be a combination of load, production, and topology

This feature specifies basic functionality that would provide users with the state-of-the-art intuitive user interface and workspace to browse visualize and manage network model data. This includes multiple workspaces that support modelling operations in parallel, visualization of electrical grid models and topology loaded from store or generated on demand.

C.3.2 Non-functional features

The product has to offer the users with a work environment that makes it possible to carry out tasks in a pleasant and efficient way. This requires that the product has features such as for instance user management, comprehensive documentation and features related to data security. To achieve this a NMM tool is foreseen to have the following features:

NF – Non-Functional

This feature covers a set of functionality issues that relate to the performance of the product in terms of capacity and speed, incident and change management, integration in the IT environment, usability, documentation, etc.

SEC – Security

This feature specifies the product features related to data security. This includes both data security as well as user access management.

C.3.3 Mapping in the Requirements list

The features listed in C.3.1 and C.3.2 are mapped to the entries on the 'Requirements' tab of the Requirements list (see chapter C.6). The 'ID' column displays the feature classifier followed by a sequential number. The entry under 'Topic' identifies the topic addressed by the requirement. Besides the features, the list also includes the key assumptions accompanying use of the list.

Requirement abbreviation	ID	Requirement category
NP		Network and Project Management
MV		Modelling and Validation
IF		Interfaces: Import, Export and Automation
WS		Workspace: Model Assembly, User Interface

NF	Non-functional
SEC	Security
ASM	Assumptions

Each requirement has been given a priority that reflects its weight and that can be used e.g. as a basis in a tender response evaluation.

The following priorities have been used:

S – Shall

B5 – Should, High priority

B3 - Should, Medium priority

B1 - Should, Low priority

N/A – Not applicable, used for assumptions.

C.3.4 Glossary of Terms

A glossary of terms can be found on the the 'Glossary of Terms' tab of the Requirements List (see chapter C.6).

C.3.5 Standard References

The 'Standard References' tab of the Requirements List (see chapter C.6) contains an overview of reference documents per date of this document.

C.4 Suggested use in evaluation of requirements

It is not the intention of this document to make binding statements on how requirements should be weighed and hence vendor proposals should be evaluated. This is entirely a task for and responsibility of the user. However, we do offer as a convenience a suggestion in this section.

Section 2.1 provides a suggestion for the assessment of requirements marked as 'shall'. In section 2.2 a method is suggested on how to assess the requirements that are classified as 'should'.

A. Evaluation of shall-requirements

The shall-requirements (S) have no weight or maximum score. The tender is rejected if tenderer's responses do not meet the shall-requirements. A tender that does not meet the shall-requirements is not evaluated further.

C.4.1 Evaluation of should-requirements

Should requirements (B1 – B5) are assessed using a weighted scoring. Each requirement is first scored a number of points to express the degree of compliance of the product that is offered e.g. by a tender evaluation and assessment team. The points achieved for a requirement are multiplied by a prioritization factor resulting in a score.

C.4.1.1 Points

2 points-Fulfilled: The should-requirement is met without any remarks. The tenderer has submitted a response stating that they have understood the requirement. The tenderer clearly describes how the requirement is met. The tenderer has described in the tender that it's possible to meet the requirement in a good way.

1 point- partly fulfilled: The response to the should-requirement is approved and acceptable but lacks a clear overall feasible impression. The tenderer has not described in the tender that it's possible to meet the requirement to the desired level.

0 points- not fulfilled: The should-requirement is not fulfilled.

C.4.1.2 Priority factor

The requirements are prioritized using a three-tier scale B1 – B3 – B5. Each tier is given a weight e.g. as follows:

times 5 - B5 - High priority

times 3 - B3 - Medium priority

times 1 - B1 - Low priority (1)

C.4.1.3 Examples

The scoring of should-requirements as described in the previous two sections is clarified by the following examples.

- a "High priority" requirement gets 0 points à $0 \times 5 = 0$ score
- a "Medium priority" requirement gets 2 points à $2 \times 3 = 6$ score
- a "Low priority" requirement gets 1 point à $1 \times 1 = 1$ score

C.5 Generic NMM use cases

Assessment of requirements can benefit greatly from executing hands-on tests using the proposed product. Such tests are typically defined as use cases (UC), i.e. a series of tasks a system user is likely to perform when using the product (or is doing using the existing product). To support the community in developing use cases, a set of 6 examples is included.

Use cases supplied with this document are not exhaustive, but they should be read first before digging down into the list of requirements. They could equally be used as a proof-of-concept for vendors or for evaluation of tenders.

C.5.1 UC 1: CIM import and export roundtrip

Purpose:

To verify that the tool preserves all properties in a CIM model going through import and export.

Preconditions

- Use a smaller node-breaker test model provided by the utility or available through other channels (e.g. ENTSO-E conformance test models, or EPRI Interoperability test models).

Description of operation:

- Import the supplied CIM EQ file.
- Import the supplied CIM SSH file.
- Browse the model in the workspace.
- Compute a solution (topology and load flow).
- Export to CIM EQ, SSH, TP and SV.
- Compare the imported and exported model using an external tool, e.g. CIMdesk. This step can be performed by the receiving customer.

Validation Criteria:

- The exported model shall include at least the same number of objects as the import.
- The mRID (rdf:ID) identifiers used in the exported file shall be identical to those in the imported file.
- The model structure shall be the same in the exported model as in the imported model. This includes connectivity, used CIM classes etc.
- The model parameters shall be the same in the exported model as in the imported model. (For EQ and SSH.)

C.5.2 UC 2: CIM import of an increment and model assembly

Purpose:

To verify that the tool shows the incremental updates (import of a diff-file) on actual model and also how the tool support the user to keep track of what actions that are needed to get the affected network model valid.

Preconditions

- Use a node-breaker test model and prepare a changes and updates, as a CIM incremental file, that causes conflicts in the network model and need of updates in autolayout diagrams and overview diagrams.
- The prepared test Model shall consist of at least two model parts that can be assembled to get a complete target model.

Description of operation:

- Import the incremental CIM EQ file.
- Show how the user can overview all changes imported.
- Solve the Network Model conflicts, and update affected diagrams
- Assemble the model parts to get a complete target model
- Browse the model in the workspace. Changes can be found.
- Run the validation rules (partly or all validation rules).
- Compute a solution (topology and load flow).
- Export the assembled model to CIM EQ, SSH, TP and SV.
- Import the exported model using an external tool, e.g. CIMdesk.
This step can be performed by the receiving customer.

Validation Criteria:

- The imported increment shall be possible to identify.
- The system helps you to solve the model conflicts and diagrams are correct updated and preserved.
- Assembly procedure is consistent and user friendly.
- The model parameters shall be the same in the exported model as in the imported model.
(For EQ and SSH.)

C.5.3 UC 3: CIM incremental and historical model export

Purpose: The purpose of this UC is to demonstrate the flexibility of the CIM incremental export and the relation with the audit trail, especially in case of the export between various versions of a certain model. The UC case also includes a demonstration of export of historical model (i.e. previous version of a model).

Preconditions/ background:

Model A is a model definition / model part in the Product.

Model B and model C are change sets or incremental models that have already been applied to model A, and these are shown in the audit trail for model A. There is at least some time between the application of model B and model C.

Model D is an incremental model to apply on Model A and is available in a directory accessible to the Product's Workspace.

Description of operation:

- The authorized user loads model A into the workspace.
- The authorized user loads the incremental model D into the workspace, where a change set is created and submitted.
- Another authorized user applies the change set so that model D is now part of Model A.
- An authorized user browses the audit trail, showing the change set equivalent to the Model B, C and D.
- The authorized user then export, three incremental models and two full models as as shown in Figure 3 below:
 - Incremental model export between version 2 before the application of model D and the version between the application of model B and model C.
 - Incremental export between V3, after the application of model D and V1 before model B and C.
 - Full model export of Model A, version 1
 - Full model export of Model A, version 3
 - Incremental export from V3, after the application of model D and any date/time before the application of Model C.

Validation Criteria:

- intuitive process for incremental export
- flexible incremental export capabilities.

- granularity of audit trail.
- historical model export

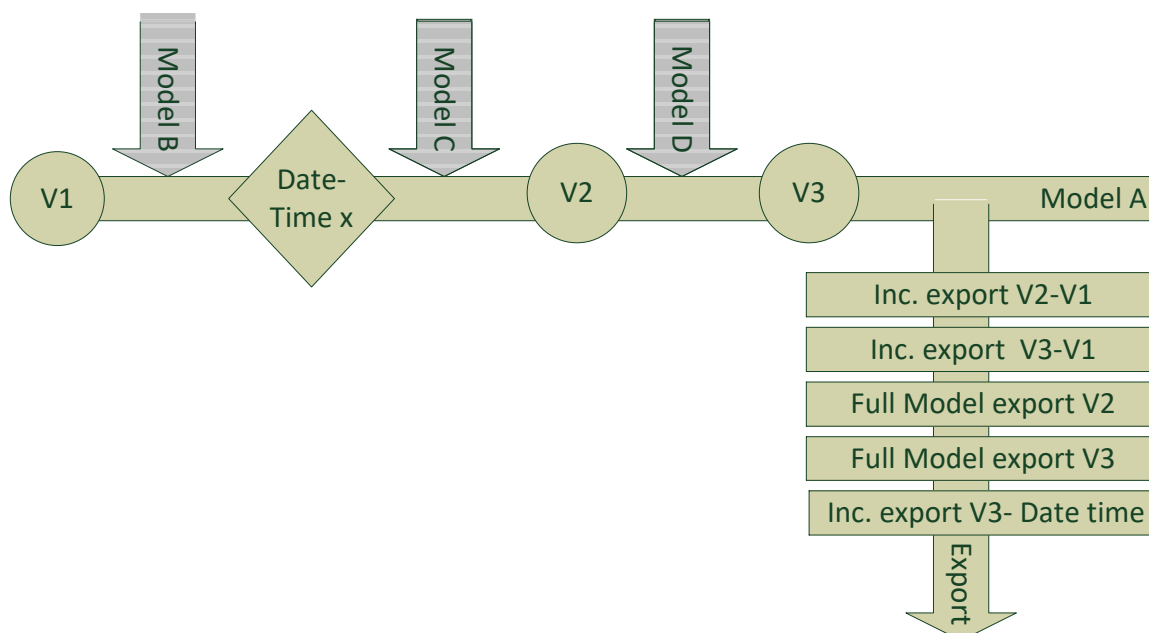


Figure 3: Model A exports in reference to various versions of the same model

C.5.4 UC 4: Create power system projects (PSP) and export future network model

Purpose: The purpose of this use case is to test the tool’s capability to create, verify and modify a future power system project (PSP), and then create and export the network model for a future target date. The use case will also examine the tool’s capability of roles and PSPs status attributes.

Preconditions/background: At least three users are involved in this use case. User 1 is an authorized modelling user that will carry out the initial modelling operation for PSP. User 2 is a model administrator or equivalent, whose role is to approve PSPs for production in the shared future network model. User 3 is another modelling user with equivalent authorization as user 1.

Model A includes the core target model (i.e. an assembled model) that the company is interested in expanding with a future PSP named “Future Project 300180”, which includes a new line planned to be commissioned January 1th 2025. The project has previously been assigned the project ID 300180. See Figure 4 below.

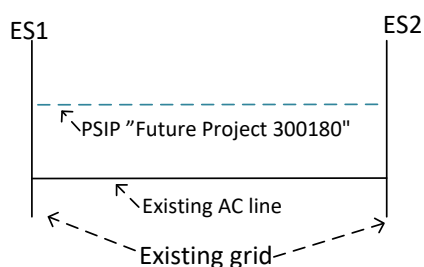


Figure 4: Creation of power system projects

Model B is Model A with the addition of user selected PSPs.

Model C is Model A with all candidate PSPs until end of 2024.

Model D is Model A with all candidate PSPs until end of 2025.

Model E is Model A with all “Approved” PSPs until end of 2025.

Description of operation:

- User 1 creates a new Private (or equivalent status) PSP with the planned commissioning date and the project ID. The PSP is only visible for user 1.
- User 1 models the PSP as a new parallel AC line (line XA2) to an existing line in Model A. XA2 has identical electrical parameters as the existing parallel AC line. The line XA2 is connected bus-to-bus (bus-branch model).
- User 1 validate that the topology in the system model with the PSP included is still valid. User 1 runs a power flow in the tool and checks that it converges.
- User 1 export Model B, which compared to Model A also includes the modelled PSP “Future Project 300180”.
- User 1 changes the status of the PSP from Private to Candidate and Stores the PSP. The PSP is now visible for user 2 and user 3.

New information is available from the ongoing power system project.

- User 3 opens the PSP and edits the line reactance of the XA2 to be commissioned. The line reactance is increased with 5%.
- User 3 models the substation equipment in detail in the PSP by adding a circuit breaker at each line end of line XA2. User 3 runs a load flow, verifies convergence, and stores the PSP changes.

- User 3 export Model C from the NMM. Verify/show that the equipment in PSP “Future Project 300180” is not included.
- User 3 export Model D from the NMM. Verify/show that the equipment in PSP “Future Project 300180” is included.
- User 3 export Model E from the NMM. Verify/show that the PSP “Future Project 300180” is not included.

The PSP has been approved by the executive board at the company. The Project administrator now changes the status of the PSP in order to put it into the Planning model production.

- User 2 “Approves” the PSP “Future Project 300180”.
- User 3 export Model E from the NMM. Verify/show that the equipment in PSP “Future Project 300180” is included.

The PSP has been cancelled by the executive board at the company.

- User 2 cancels and archives the PSP “Future Project 300180”. Show how the PSP is archived and the actions to re-activate it if the PSP would be re-considered.
- User 3 export Model E from the NMM. Verify/show that the PSP “Future Project 300180” is not included.

Validation Criteria:

- Ease of creating a PSP with equipment, an effective date, ID and project name. The PSP creation with equipment data is expected to be intuitive and effective.
- Ease of executing a model export which only includes user selected PSPs, or all PSPs up to a certain target date.
- Ease of setting up a first initial Private PSPs and then sharing the model as Candidate.
- Ease of creating different user roles with various authorizations in the planning process.
- Ease of approving PSPs which are put into production in the official future network model with Approved PSPs.
- Ease of changing existing equipment data in shared PSPs.
- Ease of running verification with topology processor and power flow when modelling a PSP.
- Ease of cancelling and archiving PSPs.

C.5.5 UC 5: Creating power system project hierarchies

Purpose: The purpose of this use case is to test the tool’s capability to support hierarchies of PSPs and stages of implementation, reflecting how projects are managed during its lifecycle. The purpose is also to evaluate how dependencies are handled between several PSPs, and PSP dependency with core model.

Preconditions/background: At least four users are involved in this use case. User 1 and User 3 are authorized modelling users that will carry out the initial modelling operation for PSPs. User 2 is a model administrator, or equivalent, whose role is to approve PSPs for production in the shared future network model. User 4 has equivalent authorization as user 2 but is also able to update the PSP commission dates.

The “Future Project 300180” (from UC Case3) is now split into three PSPs, P1 , P2 and P3. See Figure below. The new substation is located in the middle of the previously planned line, and line parameters in P1 and P3 are set accordingly.

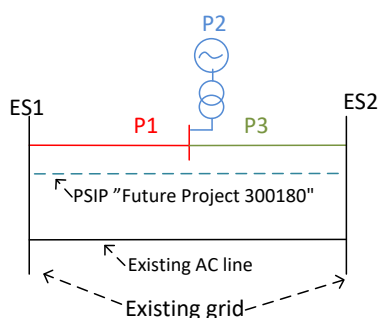
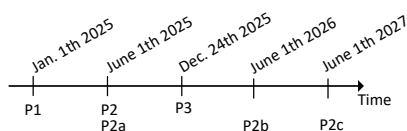


Figure 5: Creating projects hierarchies

PSP P1 includes a new line and a substation with an arbitrary bay configuration. The project ID for P1 is 300180 with the planned commissioning January 1th 2025. PSP P2 includes the wind farm PSP (or other generating facility), with project ID 400200, and commissioned date June 1th 2025. PSP P3, with ID 400300, includes the new line and is planned for commission in December 24th 2025. The following timeline summarizes the commissioning dates for the PSPs (P2a-P2c is explained later in the UC).



Model A includes the core target model that the company is interested in expanding with a future PSP. Model B includes Model A with all Approved PSPs to end of 2026. Model C includes Model A with all Approved PSPs to end of 2030.

Description of operation:

- User 1 splits “Future Project 300180” into two new Private projects (or sub-projects or equivalent) P1 and P3. User 1 also modifies PSP P2 to be connected to the new substation in P1.
- User 1 verifies the system model including PSP P1-P3 with a load flow and then changes the status of the three PSPs from Private to Candidate.
- User 2 verifies the system model including PSP P1-P3 with a load flow. If successful verification (e.g. load flow converges), user 2 changes the attribute of the three PSPs (P1-P3) to Approved.
- User 3 exports Model B. Show that all relevant equipment data for the three PSPs is included.

At a later stage in the planning process it is decided to commission the wind farm (P2) in three stages of 30 MW with one year interval. Dates are shown in the previous timeline and the stages are here named P2a, P2b and P2c for clarification. The generator model for the stages can be arbitrary modelled (e.g. one generator or three), however there will only be one power transformer.

- User 3 models the three stages (or sub-projects or equivalent) of this new PSP version of P2, and sets its status to Candidate.
- User 2 changes the attribute of the new version of PSP P2 to Approved.
- User 3 exports Model B. Show that all relevant equipment data for the three PSPs is included, except for the last stage in PSP P2, P2c. Only 60 MW should have been commissioned.
- User 4 updates and approves the commissioning date for PSP P1 to January 1th 2026.
- User 2 exports Model C. Show how the time dependencies errors between PSP P1, P2 and P3 can be handled by the tool.
- User 4 prolongs the commission date for PSP P2 and P3 with one year to solve the time dependency issues.
- User 4 renews the ES1 substation (new mRID), with a change set, making the line connection in PSP P1 invalid.
- User 2 exports Model C. Show how the topology dependencies errors between ES1 and PSP P1 can be handled by the tool.
- User 2 updates and approves changes in PSP P1 in order to take care of the errors. User 2 then export Model C. Show that all relevant equipment data for the three PSPs is included.

Validation Criteria:

- Ease of splitting up one PSP into new PSPs.
- Ease of using different commissioning stages in PSPs.
- Intuitive and effective work flows between different roles in the planning process.

- Capability to report issues of time and topology dependencies between PSPs and existing grid model. Changes with impact on dependencies could be alerted directly, or detected in a cyclic verification process, or at export. All relevant users attached to the PSPs or change sets could be notified in order to take action.

C.5.6 UC 6: Revising & expanding model and boundary parts.

Purpose: This UC case aims to demonstrate the tool's capability to manage multiple models and to adjust the boundary between related models. The UC case will also bring to light the process the tool/vendor proposes for achieving, this, as well as the functionality provided to manage and audit such changes.

Preconditions/ background: At least three users would be involved in this use case (but it can be performed on one instance). User 1 and User 2 are authorized modelling users that will carry out the modification operations in the NMM. User 4 is a model administrator or equivalent, whose role is to approve such the changes carried out by user 1. Available to the users are 3 models.

- Model A: The core target model part that user 1 is interested in expanding
- Model B: The source model part from which modelling elements will be moved.
- Model C: A boundary model between model A and model B
- Model D: A reduced model part representing the same network as model B.

Description of operation:

- User 1 loads both model A and model B into a workspace or equivalent transfers some modelling objects from model A to model B, performing the necessary object attribute updates and addition of new modelling elements as necessary.
- The NMM performs a validation on model A and model B to verify that constructs are well defined, executing topology analysis, power flow etc.
- User 1 updates model C changing the boundary point between model A and B.
- User 1 validates in the workspace the resulting model (can modify corresponding base for the merged model of A & C) and submits the changes to the model manager on success.
- User 2, continues and updates the model D.
- User 2, merges and validates Model C and D and submits changes to the model manager on success or failure
- User 3, loads all changes in the data base for the time period under which user 1 and user 2 has carried out operations or by selecting changes by user 1 and user 2 or equivalent.
- User 3 approves changes to model A, B & C but rejects changes to model D.

Validation Criteria:

- Ease of managing multiple models in the workspace and moving objects between models,
- Ease and practicality of expanding model parts
- Capability to update boundary models
- Ease and smoothness of validation of resulting models after changes
- Ease and practicality of approving and rejecting some created change sets.

C.6 Requirements List

The Requirements List is contained in the file NMMSolutionsRequirements.xlsx available on the ENTSO-E CGMES Library (<https://www.entsoe.eu/data/cim/cim-for-grid-models-exchange/>).